
Project
Management

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1 Basic Concepts and Project Life Cycle

Project

- A project is concluded when a certain set of interrelated activities are performed within a fixed period with certain specifications and within certain costs.
- As per the ISO-21500, "A project is a unique process consisting of a set of coordinated and controlled activities with start and finish dates, undertaken to achieve an objective conforming to specific requirements, including constraints of time, cost, and resources."
- According to Little and Mirrless, "A project is any scheme or a part of a scheme for investing resources which can be reasonably analysed and evaluated as an independent unit."
- As per Harison, "A project can be defined as a non-routine, non-repetitive, one-off undertaking, normally with discrete time, financial and technical performance goal."
- As per the Project Management Institutes (PMBOK Guide), A Guide to Project Management Body of Knowledge, "A project is a temporary endeavour undertaken to create a unique product, service, or result." This definition, which has been given by PMBOK, is widely accepted and uncontested.
- The basic classification of projects is temporary and unique.
- A temporary project is a routine operation from beginning to finish, having a defined scope and resources.
- A unique project does not have routine operations but a specific set of operations designed to accomplish a singular goal.

Classification of Projects

Projects can be classified under different heads, which are as follows.

Classification Based on Type of Work

- Industrial projects
- Non-industrial projects

Classification Based on Project Completion Time

- Long duration projects (over 5 years)
- Medium duration projects (3–5 years)
- Short duration projects (1–3 years)
- Special short duration (less than 1 year)

Classification Based on Project Value

- Mega value projects (> ₹1000 crore)
- Large value projects (₹100–1000 crore)
- Medium value projects (₹1–100 crore)
- Small value projects (< ₹1 crore)

Classification Based on Ownership

- Private sector projects
- Public sector projects
- Joint sector project
- Government sector projects

Classification Based on Execution of Work

- Fast track projects
- Normal track projects

Characteristics/Features of a Project

Stakeholders

- A stakeholder is a person, or group of individuals, or organisation that is positively or negatively impacted by the project and/or anyone who can exert influence over the project's objective and results.
- The project has a range of stakeholders, who have particular needs and expectations that have to be identified and addressed. Generally, stakeholders for a project are as follows.

Project Sponsor

A project sponsor is an individual or a group of people who provides for the project. The sponsor (client) develops a business case to address certain corporate requirements and corporate strategies (vision and values).

Project Manager

A project is managed by a project manager as the single point of responsibility. This implies that the project manager is responsible and answerable for all project deliverables.

Project Team

- A project is administrated by a project team working in the project management office, abbreviated as PMO. The project manager leads and manages the project team, the suppliers, contractors, incharge, and consultants.
- The project team is in charge of the execution and delivery of the project objectives.

Project Charter

A project charter outlines the scope of work of the project and how the stated objective will be delivered. Along with this, it gives the project manager official authority to make use of the company's resources.

Project Life Cycle

- A project is subdivided into several stages which are represented in order along the project timeline.
- Each project has several stages of development. The managers and executives are required to have a clear understanding of all these stages to have better control over the project.
- Life cycle stages/phases include all stages/phases from the start to end of a project. This number of stages, i.e., the life cycle of the project, may vary from project to project.
- The project life cycle consists of some important elements: operations/activities, resources, constraints, and external conditions. The activities/operations are to be performed in a sequence. Resources include material, money, industrial machinery, manpower, etc.
- The important stages/phases of the life cycle are depicted below.

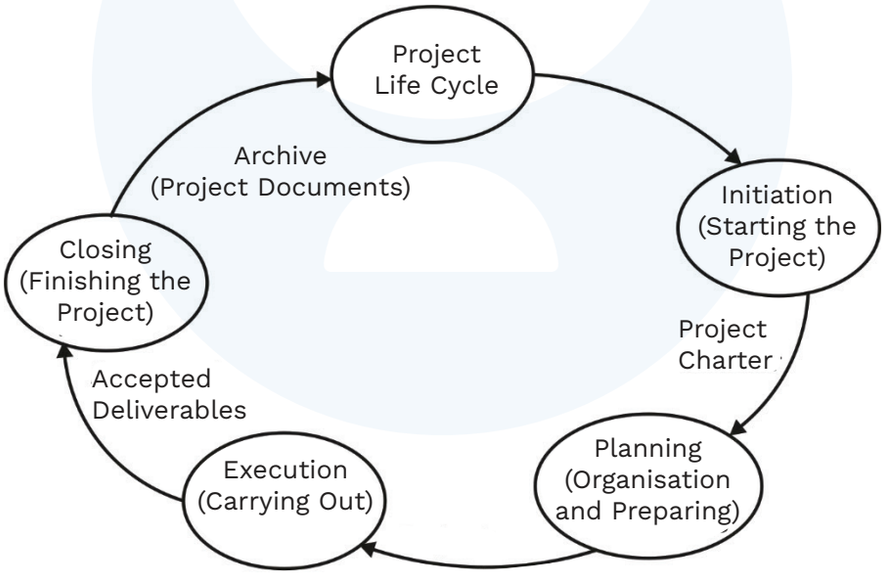


Fig. 1.1 Project Management Plan

Phase I: Initiation/Feasibility/Conception Stage

- During the initial phase, a business case is assessed to check how the project can be implemented and delivered. If there exist numerous viable business case solutions for a client's needs, the study will evaluate these cases according to how well they fulfil the client's requirements.
- It is during the initial stage that the problem is identified and possible solutions are put forward. Once the feasibility study gets wrapped up and the goals are clearly chalked out, a thorough analysis of the proposed

solutions is done, considering the benefit, financial, and risk analysis.

- After a thorough analysis, if the proposed solutions seem viable, they are given a green signal.
- Therefore, the conception stage of a project is summed up as identifying the requirements of the client, establishing the fundamental practical approach of the project aim, along with identifying the alternate possible solutions, and analysing them on the basis of benefits, finances, and risks involved.

Phase II: Design/Planning/Scheduling Stage

- The scheduling stage makes use of the variables from the feasibility study to plan the project, set up a working process, and bring out certain schedules known as the outline plans for all areas that are required to accomplish the project.
- Following the capital decision, the planning stage of the project begins. During this stage, original ideas are expanded to give an outline for the next stage. The blueprint gets ready for execution by the end of the design phase. The technical parameters are fixed and the initial design is complete. Along with this, the specifications of the equipment are finalised and the cost estimation is done in detail. A schedule for executing the project is planned. The 'detailed project report' comprises the deliverables of the planning phase.
- Additionally, the 'detailed project report' is assessed further by the relevant organisation. The magnitude of the intensity of operations grows from the first to the second stage of a project's life cycle.

Phase III: Execution/Production Stage

- During the production stage, the project plan is executed. The progress of the project is repeatedly monitored and kept under check. Whenever there are deviations from the original plan, suitable action is taken to get it back on track. Every output of the production phase is reviewed for quality along with measuring it against the acceptance criteria. Once all deliverables are produced and are ready for acceptance by the client, the project managers move to the next stage, which is known as the termination phase.
- The execution phase is the most important. The severity of activities is further enhanced and is at its peak in the production phase. The responsibilities of the project manager are also at peak. Once the execution phase approaches completion, the severity of activities starts to fall again.

Phase IV: Termination/Commissioning/ Handover/Closing Stage

- Before handing over the project to the client, the final inspection is done in the project commissioning and handover phase, which reaffirms that the project is made according to the accepted design.
- Termination, as the name suggests, is the last stage of the project cycle. Trials are carried out on constructed solutions, and problems, if any, are solved. After the successful completion of the trials, the commissioning gets completed.
- The commissioning stage may include the training of operating personnel as well. The intensity of activities is at its minimum at the end of the commissioning stage.

Scope of Work

- The scope of work defines both the work that is included and the work that is not included in the project.
- The two frameworks known as 'product breakdown structure,' abbreviated as PBS, and work breakdown structure, that is, WBS, can be used to define the scope of work in detail. A PBS subdivides the project into various deliverables, whereas a WBS subdivides work into different work packages, and these work packages correspond to different deliverables.

Project Schedule

- A project's schedule employs the critical path method (CPM) to compute the start and end dates of all activities, as well as, the list of critical activities whose duration determines

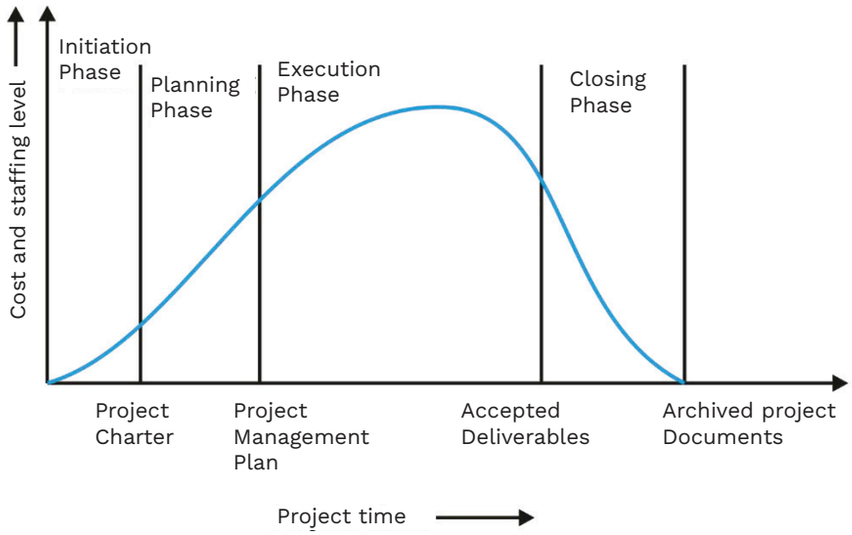


Fig. 1.2

the project's duration. The Gantt chart is one of the most important schedule-detailing papers.

Budget

A project is assigned a budget to complete the work, which the WBS subdivides into a budget per work package. The project cash flow statement helps the project manager to balance income and expenditure.

Resources

A project uses resources (equipment and workforce). A resource histogram helps the project manager to balance the resource requirements and available resources.

Procurement

The materials and components used in a project are procured from outside suppliers. This is typically accomplished through the procurement process and schedule.

Quality

A project has to achieve predetermined quality requirements. This is achieved through the quality assurance plan and the quality control plan.

Risks

A project is subjected to unknown risks that threaten to prevent the project from achieving its objectives. These risks are addressed by the risk management plan.

Management

- Management is the act of getting people together and using the available resources efficiently and effectively to accomplish a task or goal.
- The project management process consists of a linear sequence of steps: initiation, planning, execution, monitoring and control, and closure.
- There are various type of resources in management. They are human resources, financial resources, time resources, technological resources, mechanical resources, and natural resources.

Different Types of Management

General Management

A successful project manager must be competent in a wide range of general management skills. These include personnel and human resources, sales and marketing, accounts and salaries, computer systems, and legal contracts.

Portfolio Management

- Portfolio management is effective, centralised management, which includes identifying, prioritising, authorising, and controlling

a collection of projects or programmes, and other work grouped to meet strategic business objectives.

- A portfolio is a set of projects and/or programmes that are not necessarily related but are brought together for the sake of control, coordination, and optimisation.
- Portfolio management is mainly applied to groups of projects and programmes that might not be related in the business sense but draw on a common pool of scarce resources. A portfolio manager coordinates all projects and possible programmes within an organisation during the process of evaluation, selection, monitoring and controlling, reprioritisation, and closure. The division of a complex project into sub-projects is part of normal project management.

Programme Management

Programme management is defined as centralised and synchronised management to accomplish its defined goals and objectives, i.e., in programme management, interrelated, and interdependent projects are managed under one head to achieve the desired goal.

Production Management

Although projects are deemed to be unique, in reality, they usually consist of several similar and repetitive tasks that could be grouped and completed using a production line approach.

Process Management

Process management is used by companies that manage a product that typically flows from one process to another.

Project Management

- Project management, generally abbreviated as PM, is the discipline of putting into use the established principles, knowledge, procedures, tools, and policies to attain the project goals and objectives. The key idea here is the optimised use of resources to accomplish these objectives within a specified timeframe.
- According to the PMBOK Guide, 'Project Management is an application of knowledge, skills, tools, and techniques to project activities to meet project requirements.'

Features of Project Management

Project Charter

A project charter, as stated above, is a PM document that outlines the scope of work of the project and translates the client's business problem into project goals and objectives.

Tools and Techniques

There are various tools and techniques that management makes use of along with the knowledge and processes that have been developed by research over the years to execute a project. Some of these tools include the Gantt chart, CPM analysis, histogram, activity crashing methodologies, matrix organisation structure, and earned value graphs and tables.

Project Plan

It is a document which defines the implementation and life stages of the project, including the benefits, financial, and risk analysis of the proposed solution.

Project Management Processes

These involve a set of techniques used to manage a linear sequence of processes associated with each other to churn out a project or service with certain goals and objectives. The project management process includes conception, design, production, and termination stages.

Project Organisation Structure

With the help of management technique known as project organisation structure, a project leader can form temporary teams or organisation structures that can accomplish the objectives of the project within the specified financial and time constraints.

Project Methodology

With the help of the project life cycle technique, the project can be subdivided into distinct phases where each phase will produce a unique deliverable. The sequence of distinct phases forms the foundation of this methodology and

links all subjects at intra-phase and inter-phase levels.

Level of Effort (Intensity of Activities)

- Any parameter that goes along the project and can be quantified or measured comes under the level of effort. The most commonly used level of effort is expenses/costs or man-hours. The project manager uses the level of effort to estimate the amount of work that is to be performed or completed within the different stages of the project. The line graph of 'cumulative expenditure' and/or a line graph of 'rate of expenditure' is typically used to represent the intensity of activities or the level of effort.
- The sum of effort to date is known as the classic curve profile. This 'S curve' profile is like the one that is used in the calculation of earned value. The level of effort is an important feature that can be used by project managers as similar projects usually have similar intensities of activity profiles.

Product Life Cycle

The length of time between a product's introduction to its customers and its removal from the market is known as a product's life cycle. There are four stages in a product life cycle—introduction, growth, maturity, and decline.

Introduction

- This stage of the product life cycle involves building a market strategy, typically through investing in advertising and marketing to inform consumers of the product's advantages.
- As demand is created at this point, sales tend to be slow. Depending on the product's complexity, how fresh and original it is, how well it meets clients' needs, and the level of market rivalry, this phase can be time-consuming. There is ample evidence that products can fail at this stage, meaning that stage two is never achieved, even when new product development is tailored to client needs. Due to this, many businesses opt to improve an existing product and release their version, following the inventive pioneer's footsteps.

Growth

- If a product successfully completes the market introduction phase, it is prepared to begin the growth phase of its life cycle. This should result in increase in production and wider distribution of the product due to growing demand.
- As the product begins to get traction in the market, its gradual growth during the debut and development phase suddenly accelerates. At this juncture, rivals may enter the market with their versions of goods, either exact copies or variants with some modifications. When the consumer has the option to go elsewhere, maintaining the product's market position becomes dependent on branding. In the face of intensifying competition, the pricing and availability of products in the market become crucial for maintaining sales momentum. At this moment, the life cycle moves to the third phase, which is market maturity.

Maturity

- At this stage, a product is well-established in the market. As a result, the costs associated with its production and promotion will decrease.
- As the product life cycle reaches this stage of maturity, market saturation begins to develop. By this time, numerous consumers will purchase the product, and new competitors will emerge.
- Consequently, branding, price, and product distinction will be even more crucial for preserving the market share. Instead of attempting to market their goods, as they may have in stage one, retailers will become stockists and order takers.

Decline

- Eventually, as competition continues to increase and other companies attempt to replicate your success by adding product

features or lowering the prices, the product's life cycle will deteriorate. New developments that supersede your product can also contribute to its decline.

- As a result of market saturation, many businesses will begin to pursue new opportunities. Some companies will survive the fall and may continue to offer the product, but production is likely to be scaled back and prices and profit margins may decline. Consumers may also abandon a product in favour of a newer option. But in certain situations, this trend can be reversed as styles and fads return to revitalise interest in an earlier product.

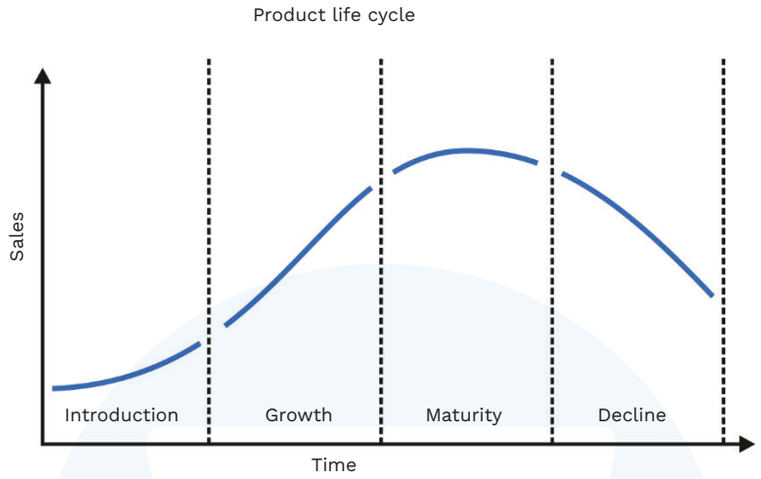


Fig. 1.3

Concurrent Engineering

- Concurrent engineering, or simultaneous engineering, or integrated product development or agile, is a work methodology where work in various areas (like research and development, design, manufacturing, testing, etc.) required to bring a product from conceptualisation to sales is conducted in parallel as opposed to step-by-step. Decreased time to market leads to increased sales and market capture.
- The main challenges in this work methodology are the requirement for increased training for all team members, coordination, and high levels of communication between the concurrent work areas. The absence of this will result in a failed product.

Halo Effect

- The halo effect is the assumption that because the person is good at technology/ technical skills, s/he will be a good project manager. It describes error in thinking, which would help one to draw an inference about the person.

Practice Exercise

1. The steps or phases taken by the project from beginning to end are collectively called its _____.

(A) hierarchy of steps to be followed on the project
(B) the manager's method of completing the project
(C) the project life cycle
(D) a collaborative approach between the manager and the team to complete work on the project

2. The project management process groups are _____.

(A) initiate, plan, expedite, and control
(B) plan, organise, develop, and control

(C) plan, do, observe, and commit
(D) initiate, plan, execute, monitor, control, and close

3. Ideally, a project manager should be selected and assigned at which point in the project life cycle?
(A) During the initiating process or conception stage
(B) During the project planning process
(C) At the end of the design phase of the project life cycle
(D) Before the beginning of the design phase of the project life cycle

4. The second-largest employment generating industry in the country is _____.
(A) agriculture
(B) petrochemical industry
(C) construction industry
(D) leather industry

5. The _____ recognises that a project or phase should begin and commits the organisation to do so.
(A) initiating process
(B) solicitation process
(C) scoping process
(D) planning process

6. Projects are initiated in the life of an organisation as a result of _____.
(A) an immediate financial need
(B) keeping employees busy and productive
(C) achieving pre-set strategic objectives
(D) making products available for sale in market

7. A collection of generally sequential project phases whose name and number are determined by the control needs of the organisation or organisations involved in the project is called _____.
(A) project life cycle
(B) product life cycle
(C) portfolio management
(D) programme management

8. In which of the following project management process groups, most of the time and money are typically spent?
(A) Initiating
(B) Planning
(C) Executing
(D) Controlling

9. All of the following are characteristics of a project, except _____.
(A) temporary
(B) definite beginning and end
(C) interrelated activities
(D) repeats itself every month

10. What do knowledge, skills, tools, and techniques stand for when applied to the activities of a project if it also entails effective management of a process?
(A) Human resource management
(B) Project management
(C) Financial management
(D) Project management process

11. A project is defined as _____.
(A) a process of considerable scope that implements a plan
(B) an endeavour, which is planned, executed, and controlled; performed by people; and constrained by limited resources
(C) a temporary endeavour undertaken to create a unique product or service
(D) an objective based effort of temporary nature

12. Which of the following statements is correct?
(A) Process groups are the same as project phases.
(B) Each process group has a mapped project phase.
(C) Process groups are not project phases.
(D) Monitoring and control process group applies only in the monitoring and control phase.

13. In case of a project being given by an external client to a company, the client must have provided all information, except _____.
(A) project statement of work
(B) agreements

(C) business case
(D) organisational process assets

14. In general, a project charter covers which of the following?
(A) Project justification, background, scope, objectives, available resources, a project schedule, estimates, procurement schedule, and any limitations that top management wants to apply
(B) Project justification, background, scope, objectives, available resources, organisational responsibilities, estimates, detailed procurement schedules responsibilities, and any limitations that top management wants to apply
(C) Project justification, description, details of sponsor and project manager, budget, assumptions and constraints, and boundaries to apply
(D) Project scope, objectives, team responsibilities, a project schedule, estimates, procurement schedules, and any limitations that top management wants to apply

15. Consider the following statements regarding project management
I. It is the process of attaining project objectives in a stipulated time to produce quantified and qualified deliverables.
II. It is the art of bringing together the responsibilities, resources, and people necessary to accomplish the business goals and objectives within the specified time limit and the financial grant.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I, nor II

16. Consider the following phases of project management
I. Identification
II. Formulation
III. Appraisal
IV. Implementation
Which of the above phases are relevant, sequentially?
(A) I, II, and III only
(B) I, II, and IV only

Answer Key

1. (C) **2.** (D) **3.** (A) **4.** (C) **5.** (A) **6.** (C) **7.** (A) **8.** (C) **9.** (D) **10.** (B)
11. (C) **12.** (C) **13.** (D) **14.** (C) **15.** (C) **16.** (D)

2 Project Initiation

Project Identification and Development

A typical project would go through a development process with the following three phases.

- Pre-investment
- Investment
- Operation

Pre-investment Phase

It is related to the authorisation of investment decisions for a particular project idea under prevailing conditions. It involves the following stages.

Identification of Project Idea/Conceptualisation of Project Idea/Opportunity Studies

- The primary goal of project identification is to generate ideas about the possible project that can be undertaken or implemented. Primary ideas can be discovered from various internal and external sources.
- Some techniques for project identification are strengths, weaknesses, opportunities, and threats (SWOT) analysis, brainstorming, etc.

SWOT Analysis

- SWOT provides a framework for identifying strengths, weaknesses, opportunities, and threats expected in a project.
- Strengths are positive internal factors that a company can use to accomplish its mission, goals, and objectives.
- Weaknesses are negative internal factors that prohibit a company's ability to accomplish its mission, goal, and objectives.
- Opportunities are positive external factors that a firm can exploit to accomplish its mission, goals, and objectives.
- Threats are negative external factors that inhibit a company's ability to achieve its mission, goals, and objectives.

Brainstorming

- The top management, along with the concerned people, brainstorm for idea generation and project identification. It is the most widely used technique for generating ideas. During this process, ideas generated are listed without suppressing any idea at source. A list of ideas is then subjected to pre-feasibility studies.
- No investor, whether an individual or a company, desires to miss out on lucrative prospects to enter a new attractive business, or to expand an existing one. By their very nature, opportunity studies are indicative rather than specific; hence, they rely on macro parameters and imprecise estimates. These studies may also be conducted on a broader scale to find opportunities in one of the three following areas.
  - A given geographical area (area studies)
  - A specific sub-sector of economy or industry
  - The exploitation of renewable and non-renewable natural, agricultural, or industrial products
- Identifying a new project is complex. It involves a careful study from different perspectives.
- Tracking the performance of competitors (existing industries): It involves tracking the performance of the existing industries in the sector.
- Price trend: If price levels increase, demand levels also increase, which creates an opportunity for a new project.
- Domestic and international prices: A product with a higher international price than domestic price can give higher returns as it provides an opportunity to export, whereas a product with a lower international price than domestic price poses a risk as it can be imported.

- Location: It is an important factor based on the ease of procuring raw material in a particular area, government policies existing there, market size, proximity, and infrastructure available like mode of transportation, electricity, and power.
- Government policies: If the government promotes a certain sector, it provides an opportunity to businesses to explore and establish themselves as the government provides extended support for infrastructure and other facilities for its growth. Numerous industry groups have been set up in the field of infrastructure as the current government provides support in that sector.
- Financial positioning: The approximate cost incurred during this phase of the project is also estimated. Once the availability of the funds is confirmed, it proceeds to the next step.

Pre-feasibility Studies

- This is also called prima facie analysis or screening of ideas or preliminary filtration. This is a crucial step as 99 per cent of projects are dropped during this step. It involves careful study from different perspectives. The projects which cross the hurdle of the pre-investment phase reach their destination for implementation.
- The pre-feasibility study helps to filter out project ideas once the identification stage is completed. In this stage, a more elaborate study is required than the opportunity study stage. This stage lies between feasibility and opportunity study.
- This stage is recommended when project formulation and techno-economic feasibility studies are costly and time-consuming. This stage usually should not take more than three months. This study formulates the base for an investor whether:
  - the project has to be accepted or rejected.
  - the project requires a detailed analysis, i.e., feasibility study.
  - any part of the project needs to be particularly scrutinised.

Feasibility Studies

- Projects which have crossed the hurdle of 'project screening' are studied in detail to have a look at the viability of an idea with emphasis on identifying potential problems.
- A feasibility study is an analysis of how successfully a project can be completed. In feasibility analysis, we study technical feasibility, market demand, financial feasibility, economic feasibility, social feasibility, and environmental impact assessment (EIA).
- The feasibility study requires in-depth research involving higher-order skills for analysing the techno-economic criteria of the project.
- Even though the basic framework is similar but it is based on more reliable data gathered through survey, research, and pilot plant study.
- Even if a feasibility study report has covered all aspects, it may still lead to an incorrect project decision if:
  - project formulation is based on inadequate data.
  - the data are unrealistic.
  - the report has been deliberately tailored to fail acceptance criteria.
  - unknown or unpredictable factors occur.

Project Appraisals

Technical Analysis/Technical Feasibility

- Technical analysis includes the selection of appropriate technology, a source for the selected technology, capacity planning, location selection, raw material identification, etc.
- Technical feasibility is different from other feasibility studies (financial, commercial, and managerial feasibility studies). Some parameters to be analysed for this study are as follows. These parameters may differ from project to project.

Location and Site

- Location is one of the most important parameters for the feasibility of the project. It should align with the goals and objectives

of the organisation. It should be decided strategically so that the raw materials, labour, and infrastructure required are readily available in the nearby market.

- It starts with identifying as many locations as possible, and then that location is chosen that satisfies all needs and requirements.
- The government policies and benefits should be taken into consideration when choosing a location as these affect the project. The Location Cost Index (LCI) should be used for different locations by taking into consideration the capital loans, tax concessions, and subsidies offered by the government.

Calculating the Location Cost Index

Location Cost Index (LCI) for location A is given by

$$\frac{Ca}{Cb} \times 100$$

Where, Ca represents the cost of establishment of the plant at a location 'A', and Cb represents the cost of establishment of a plant at location 'B.'

Check the value of LCI. If LCI is less than 100, the project location is considered suitable.

Plant Size

- Plant size can have an impact on the profitability of a business. It may play a part in deciding when the business will reach break-even. The plant size needs to be optimised as it can directly impact the economics of the business.
- Technology is one of the most important aspects of a project. It should be carefully selected as inappropriate technology can lead to an unsuccessful project.
- A technology is considered to be appropriate, if it is assessed satisfactorily under a specific project situation.

Design, Layout, and Machinery

- One of the most important aspects of machinery and plant layout is that they should meet the required industrial safety standards. The safety standards are non-negotiable, and anything that does not comply with these should be discarded. Design should be done in such a way that it is practical and should operate at minimum costs and sustainability.
- The feasibility study should provide a layout of various facilities and major equipment required, with key specifications and available sources of supply.

Inputs

- These are related to the operation phase. However, they need to be identified at the feasibility stage.
- The main inputs to be examined are raw material, processed material, components and sub-assemblies, spare parts, and water, gas, and fuel supply.

Infrastructure

Organisations should plan infrastructure requirements, including water availability, transportation, electricity, etc., for example, it is beneficial for industries to be situated near water bodies that require a large amount of water for operating.

Manpower

- Organisations should determine what type of labour pool they require and form their requirements based on that.
- Sometimes a company may require highly skilled workers. In that case, the company should include the desired educational qualification. On the other hand, some companies may require a large number of workers rather than highly skilled labourers.

Environmental Impact Assessment

- Environmental Impact Assessment (EIA) is a management tool or process used for evaluating the likely socio-economic, cultural, and physical or health impacts of a proposed project or development.
- The main aspects of EIA are as follows.
  - Appraisal of prevailing environmental conditions

- Appraisal of production methods both existing and proposed
- Methodologies related to EIA
- The possible impact of projects on the environment—both existing and proposed
- Development of the techniques of conservation of environment by modifying and improving the existing production technology

- The main steps for conducting EIA are as follows.
  - Identification of project
  - Screening
  - Scoping and considering alternatives
  - Impact prediction
  - Mitigation
  - Reporting to a decision-making body
  - Public hearing
  - Review (EIA report)
  - Decision-making
  - Post-project monitoring and environmental clearance

Market and Demand Analysis

- One of the first steps in market feasibility is to estimate the potential size of the market for the proposed product (service planned to be offered) to be manufactured and get an idea about the market share that is likely to be captured.
- To assess a market, appropriate pricing, a well-established distribution plan, and other marketing ideas, including digital marketing, should be taken into consideration, along with the estimation of demand for the particular product.
- The market analysis can be done using a top–down, or bottom–up approach. The top–down approach is beneficial in some cases, whereas the latter is suited for others. Along with the above technique, the demand–supply gap is also considered to be one of the finest techniques for reckoning the market potential. If the demand–supply gap is positive, it favours the market potential; if it is negative, it does not.
- How does the process take place? It starts with evaluating the demand in future. The calculated futuristic demand is then compared with the present supplies and possible new suppliers.
- To determine the future demands, a demand forecast or demand prediction can be used. A demand prediction considers an estimate of future trends and events subjectively without considering any past data. On the other hand, a demand forecast combines the historical data, and then, estimates the future trends and events.

Demand Forecasting

- Demand forecasting combines historical data and estimates future expected trends and events.
- It should also take into consideration the export possibilities and domestic potential.
- A good forecast is judged on accuracy, plausibility (reasonable and consistent), user-friendly, availability, durability, and flexibility.

Demand Forecasting Methods

Many methods are used to compute demand forecasts; some of them are listed below. The demand forecast is used to assist a suitable and authentic investment proposal.

Qualitative Techniques

Sales Force Composite Method or Collective Opinion Survey Method

- The collection opinion survey method forms the basis of market analysis and forecasting the demand. This technique is subjective and is based on a market survey involving the opinions of sales and marketing professionals.
- Salespeople are closest to customers and have the finest understanding of the market. As a result, their opinions can be used to predict future demand or assess the consumer's responses to the product. In this strategy, salespeople must predict the anticipated demand in their respective cities, states, or regions.
- The total expected demand is determined by combining the estimates of individual salespeople. Then, the optimism or pessimism bias is removed from these values.

- They are then re-evaluated to account for changes in pricing, product design, advertising budget, anticipated change in competition, purchasing power, income distribution, employment, population, and so on. After considering these criteria, the final projection is generated.
- This strategy capitalises on the collective knowledge of salespeople, departmental leaders, such as production managers, sales managers, marketing managers, management economists, and senior executives.
- This strategy is straightforward and useful for forecasting the sales of new products. This strategy can only be utilised for short-term forecasting because salespeople only consider the current trend. In addition, they may lack the vision required for predicting the future.

Consumer Survey Method or Survey of Customer Intention Method

- It is a useful method when the volume of sales supplied by producers is high. It is a direct method that helps to evaluate future demand in the short run. The method works by directly asking the customers about their plans to buy a product in the imminent period, generally a year.
- The consumer survey method is directly based on 'what the customer plans or is expected to buy.' The information can be collected by using personal contacts or referrals or by getting information through questionnaires or emails.
- The important point to note here is that the onus of forecasting gets shifted on the customer. In this method, customers may tend to exaggerate their requirements. As the number of customers is high, this method is laborious, impracticable, and costly.

Delphi Technique

- In this method, a panel of experts is asked sequential questions, in which the response to one question is used to produce the next question. It is a step-by-step procedure, in which information available to some experts is made available to others, and the final forecast is obtained only from the common opinion of all experts. It is useful to analyse the economic conditions by organising opinion polls.
- The following steps are used to analyse the demand forecast.
    - Step 1: The supervisor, coordinator, or manager lists a set of questions and distributes it to the panel of experts. The experts are expected to write a brief prediction.
    - Step 2: Once all experts write their predictions, they are collected, edited, and recapitulated.
    - Step 3: Based on the summary, the supervisor then sets an entirely new set of questions, and step 1 is repeated, i.e., the experts' panel is asked to write a prediction again.
    - Step 4: The predictions are again collected, edited, and recapitulated.
    - Step 5: Until a unison is reached, steps 3 and 4 are repeatedly executed by the supervisor or coordinator.
    - When the Delphi Technique is being conducted, the identity of the experts is not disclosed and direct interaction between them is also avoided.
    - With this method, interpersonal conflicts are also avoided, and the strong-willed experts can also not dominate the group.

Jury of Executive Opinion or Nominal Group Technique

- It is an improvised version of the Delphi technique. This method allows personal interaction, and a panel of seven to 10 experts discusses the ideas and suggestions, and rates or ranks the proposals in descending order. The procedure followed is mentioned below.
    - Step 1: The panel of experts sits around a table in clear view of one another. The experts are allowed to interact with each other.
    - Step 2: A copy of the questionnaire is handed over to the experts and each one of them is asked to write down their suggestions and ideas for the questions.

- Step 3: Once every expert has listed down his/her ideas, the moderator asks each expert to share one idea with the panel from his/her file. The shared idea is written on the board that is visible to everyone.
- Step 4: The experts are then asked to debate/discuss each idea given in detail. The moderator/facilitator makes sure that every idea presented is discussed in-depth. The ideas/suggestions having similarities are then clubbed together and rephrased which leads to a decrease in the number of ideas.
- Step 5: Once the discussion is complete, the experts are asked to rank the ideas written in descending order.

Jury Opinion

- Jury opinion is a method of combining the views of several executives regarding a forecast. The general practice is to bring together top executives from various fields of management, which are finance, human resources, marketing, purchasing, and production.
- They provide information, experiences, and opinions for demand forecast. Jury opinion is simple and commonly used as it is less time-consuming.

Quantitative Methods

- The two models used in quantitative forecasting are the following.
  - Time series model
  - Casual model
- Time series: In this method, past data are arranged in a chronological order as dependent variable, and time as independent variable. Based on the past data, the demand for future is projected.
- Causal method: In this method, the forecaster tries to establish the cause and effect relation between the demand of a product and any other variable upon which demand is dependent. Regression is a mathematical technique that is used to relate dependent and independent variables by the use of linear equations.

Simple Average Method

The forecast by the simple average method is given by the mean or average of actual demand data for the previously specified period.

$$\text{Simple average} = \frac{\text{Sum of demands of all periods}}{\text{Number of periods}}$$

Simple Moving Average or Rolling Average Method

The moving average method uses past data and calculates the rolling average for a constant period. The fresh average is computed at the end of each period by actual demand data for the most recent period and deleting the data for older periods.

$$\text{Moving average} = \frac{\text{Sum of demand for the chose period}}{\text{Number of chosen period}}$$

Solved Examples

1. Calculate the forecast (using the simple moving average method) for 2013, if demand were 32, 36, 40, 35, 32, 35, and 45, respectively, in each corresponding year from 2006 to 2012.

Year	Demand	4-year Moving Average Forecast
2006	32	—
2007	36	—
2008	40	—
2009	35	—
2010	32	35.75
2011	35	35.75
2012	45	35.50

Table 2.1

Solution:

- The forecast for 2013 is the arithmetic mean of demand for the last four years, viz, 2009 to 2012. Hence, the forecast for 2013 is 36.75.

Weighted Moving Average Method

- This method gives unequal weight to each demand data in such a manner that the summation of all weight is always equal to one. The most recent data are given the highest weight, and the weight assigned to the oldest data will be the lowest.
- Weighted moving average is given by the following relation.

  D1 × W1 + D2 × W2 + D3 × W3 ... Dn × Wn

  Here, D1, D2, D3 ... Dn are the demands for different periods, and the weight assigned to these periods are represented by W1, W2, W3 ... Wn.
  Note: W1 + W2 + W3 ... Wn = 1.

Solved Examples

2. Compute the forecast for 2013 using weight in the proportion of 1:2:3:4.

Year	Demand	Weight	Weight × Demand
2006	32	—	—
2007	36	—	—
2008	40	—	—
2009	35	1	35
2010	32	2	64
2011	35	3	105
2012	45	4	180
		10	384

Table 2.2

Forecast for 2013 = (W × D) W = 384/10 = 38.4.

Regression Analysis

- Regression methods involve determining the trend of consumption based on past consumption and projecting future consumption by extrapolating this trend.
  Linear relationship D = a + bT

  Where D = demand, T = year, a and b = constants,

  and $a = \frac{\sum D}{T}, b = \frac{\sum DT}{T^2}$.

- Regression is the most commonly used technique for forecasting demand in project initiation as it provides demand forecasts for many years to come.
- The earlier methods are generally applied in operations management, whereas this method is significantly useful in project management due to its long time horizon.

Solved Examples

3. Determine the forecast for 2010 and 2013 using the least square method for the data given below.

Year	2005	2006	2007	2008	2009
Demand	35	42	45	51	56

Table 2.3

Solution:

Year	Demand (D)	Year (T)	T2	TD
2005	35	−2	4	−70
2006	42	−1	1	−42
2007	45	0	0	0
2008	51	1	1	51
2009	56	2	4	112
Total	229		10	51

Table 2.4

Equation of line is D = a + bt

$a = \sum DN, b = \frac{\sum DN}{T^2}$

a = 45.8, b = 5.1, forecast for 2010 is 61.1 (using T = 3)
D = 45.8 + 3 × 5.1 = 61.1

Similarly, forecast for 2013 is 76.4 (using T = 6)
D = 45.8 + 6 × 5.1 = 76.4

- In the above example, the middle year is assigned the value of 0; if the number of data available is even, the two middle terms can be assigned values of –1 and +1, respectively, with a common difference of 2.

Errors in Demand Forecasting

- Forecast error is the difference between the actual demand and the forecasted demand.
- Any error in the demand forecast will impact the reliability of the economic appraisal. A demand forecast can have the following type of errors.
  - Measurement and data error
  - Model specification error
  - External error
- The most commonly used methods to find out the forecast errors are described below.

Mean Absolute Deviation

- It indicates the average magnitude of an error made in every period without considering any positive or negative sign with that error, i.e., it is measured in absolute terms.
- The mean absolute deviation (MAD) is given by the following expression.

  MAD = sum of absolute values of forecast errors for all periods/number of periods
- Note that the forecast errors are normally distributed, and a relation exists between the mean absolute and standard deviations. The forecast errors will be smooth when the forecast is fairly correct. This is known as smoothed mean absolute deviation (SMAD). Here,

  Standard deviation = 1.25 × SMAD.

Mean Forecast Error or Bias

- Mean forecast error (MFE) or bias measures the forecast error about the direction and shows any tendency over or under forecast. Mean forecast error is given by the following expression.

$$\text{MFE or BIAS} = \frac{\text{Algebraic sum of forecast errors of all periods}}{\text{Number of periods}}$$

- Positive bias indicates underestimated forecasting, whereas negative bias indicates overestimated forecasting. If the forecast is accurate, both MFE and MAD will be zero. Note that lowering the value of MAD will also reduce the value of MFE (bias).

Costs of Forecast Errors

- Important decisions (short, medium, and long-term) are based on forecast. When forecasting is done for analysing investment proposals that are long-term and irrevocable, high inaccuracy in forecasting can lead to expensive blunders.
- The forecast inaccuracy may be due to an error in magnitude or direction. In some conditions, errors in magnitude may be serious, while in some, errors in direction may lead to serious implications.

Costs and Accuracy of Forecasts

- It is possible to relate the percentage of inaccuracy in a forecast with its costs. This can be done by selecting, validating, continuing, and sustaining a specific forecast technique.
- Using techniques like time series will result in increased costs. In contrast, the use of qualitative methods, such as Delphi will reduce the amount of work and cost required. However, it may lead to increase in the inaccuracy of forecasting.
- The use of advanced statistical techniques makes an increment in the precision of the forecasts. The expenses or costs related to these techniques differ as specified by graphologists. The cost associated will be less when the inaccuracy is less. The cost will increase, if there is a high margin error in strategic planning.
- The total cost curve is snuggled between the complex statistical methods and simple statistical methods. There is a zone that shows the minimum total costs in this curve, which is termed the optimal zone and is placed in the middle of the total cost curve.

Financial Feasibility

- A financial appraisal of a project focuses on two main areas: figuring out how much the project will cost, and figuring out how

to pay for it. By 'way of financing,' we mean the combination of equity and debt. The right mix of debt and equity for a project depends on how much money it can make. Even though equity does not earn interest and does not have to be paid back, the debt used to finance the project has to be paid back with interest. So, the project should be able to make enough money to pay off the debt with interest. The project should also bring in more money to cover the cost of equity capital. Even though the people who started the project put in their own money, it does not come for free. An entrepreneur puts money into a project with the hope of getting returns on the money, in addition to paying off the debts.

Methods of Financial Analysis

Average Rate of Return on Investment (ROI) or Accounting Rate of Return (ARR)

- This is an accounting method. There is no agreement on the definition and several alternative methods of calculating it are available. The most common ratio is as follows.

$$\text{ARR} = \frac{\text{Average annual profit after taxes}}{\text{Average investment over the project life}} \times 100$$

- The average annual profit after taxes is calculated by adding the after-tax profits for each year of project life and dividing it by the number of years of estimated useful life (for annuities, after-tax profit is equal to 1 year's profit).
- The average investment over the project life is computed by dividing the net investment by 2 (straight line depreciation is assumed), adding the salvage value that would be received at the end of the project life (since it remains invested throughout), and the total amount of working capital required.

$$\text{Average investment over the project life} = \frac{1}{2}(\text{Initial cost} - \text{salvage value}) + \text{salvage value} + \text{working capital}$$

- Companies select projects where ARR exceeds a 'predetermined value' or rank the alternative proposals in the descending order of magnitude of ARR to select the most attractive one.
- In case of the highest ARR or ARR above the minimum expected rate of return, accept the project.
- In case of a lower value of ARR or ARR below a minimum expected rate, reject the project.

Payback Period

- It is a traditional method widely used for project evaluation. It is measured in terms of the time it will take to recover the investment cost.
- If cash inflow after tax (CFAT) per annum is uniform/even/constant, then

$$\text{PB} = \frac{\text{Initial investment}}{\text{Constant annual cash flow (CFAT)}}$$

- If cash inflow is not even or varies year-to-year, then

$$\text{PB} = A + \frac{B}{C}$$

Where, A = Last period with negative cumulative cash outflow (net invested cash flow)

B = Absolute value of cumulative cash outflow (i.e., net invested cash flow) at the end of period A

C = Cash inflow during the next period after A

- If there are two projects, accept the project with the least PB.
- The annual cash inflow, if expressed as a percentage of initial investment, is termed an 'unadjusted rate of return,' i.e.,

$$\frac{\text{Annual cash inflow}}{\text{Initial investment}} \times 100$$

Advantages and Disadvantages of Payback Period

- It is easy to calculate. It can be used to measure risk in a project.
- For companies facing liquidity problems, it provides a good ranking of projects that would return money early.
- It does not take into account the time value of money. To remove this drawback, discounted cash flow method can be used.

- It does not take into account the cash flows that occur after the payback period.

Solved Examples

4. Let a project need ₹ 80,000 as initial investment, and it will generate an even annual cash inflow of ₹ 20,000 for 10 years. The payback period will be 4 years, i.e.,

$$\text{Payback period} = \frac{\text{Initial investment}}{\text{Annual cash inflow}}$$

$$= \frac{80,000}{20,000} = 4 \text{ years}$$

Discounted Cash Flow Techniques

- Traditional methods of project evaluation do not take into account the complete benefits from the entire life cycle of a project, nor do they account for the time value of money. These strategies discount the cash flows by the cost of capital, which is a discounting factor for adjusting the time value of money. This study takes future free cash flow projections and discounts them to arrive at present value estimates, which are then used to evaluate the investment potential.
- Discounted cash flow is calculated as follows.

$$DCF = \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \cdots \frac{CF_n}{((1+r))^n}$$

CF = future cash flow, r = discounted rate 1, 2, ... *n* is period.

If DCF > Present Value (PV)/Current cost, then the project should be preferred.

Net Present Value

- Before discussing Net Present Value (NPV), we should understand Present Value (PV). PV is based on the economic theory of time value of money, which states that a rupee today is worth more than a rupee tomorrow. This occurs for three reasons.
  - Potential for earning interest (cost of finance)
  - Impact of inflation
  - Effect of risk
- NPV computes the net present value of cash inflows (CFAT) in each of the future years by discounting them suitably and then subtracting the cash outflow in each year.

  $FV = PV (1+i)^n$
- If NPV is positive, accept the proposal.
- If NPV is zero or negative, reject the proposal.

Solved Examples

5. The initial investment (cash outflow) is ₹ 240000 and cash inflow (after tax) is ₹ 80000 in the first year, ₹ 120000 in the second year, ₹ 70000 in the third year, ₹ 40000 in the fourth year, and ₹ 20000 in the fifth year. If the company's cost of capital is 9% (interest rate), calculate NPV.

Year	Cash Flow (₹)	Discounting Factor at 9%	Present Value (₹)
1	80000	0.917	73360
2	120000	0.842	101040
3	70000	0.772	54040
4	40000	0.708	28320
5	20000	0.650	13000
			Σ = 269760

Table 2.5

Present value of cash inflow, A = 269760
Present value of cash outflow, B = 240000
NPV = 269760 – 240000 = +29760
NPV > 0

Internal Rate of Return

- The Internal Rate of Return (IRR) is the rate at which the net present value of an investment becomes zero. In other words, IRR is the rate that equates the present value of the future cash inflows of an investment with the initial investment.
- The IRR method is known by many names, i.e., yield on investment method, marginal efficiency

of capital method, marginal productivity of capital method, rate of return method, time adjusted rate of return method, etc. It also takes into account the time value of money by discounting cash inflows and outflows.

- Also known as the Economic Rate of Return (ERR), IRR is a method of calculating the rate of return. The term internal means that the calculation does not incorporate environmental factors (i.e., interest rate or inflation). It is also called the discounted cash flow rate of return. Sometimes, it is also known as the effective interest rate.
- IRR is used for capital budgeting and to compare the profitability of investments.
- Greater IRR is desirable. The project should be undertaken only when IRR > cost of capital (acceptable rate of return or interest rate).
- IRR gives rate and is an indicator of efficiency/quality/yield of investment, whereas NPV is an indicator of the value of the magnitude of investment.
- IRR can be used as an indicator to evaluate whether a new plant is viable or an extension of the existing plant is viable.
- IRR can also be used in evaluating stock buyback programmes. Sometimes, a company's own stock is a better investment (higher IRR) than any other use of funds.

$$NPV = \sum_{n=0}^{N} \frac{c_n}{(1+r)^n} - \text{present cost outflow} = 0$$

n = number of periods/years

c_n = cash inflow in the nth year

Solved Examples

6. Find IRR for the tabulated data shown below.

Year (n)	Cash Flow (Cn)
0	–123400
1	+36200
2	+54800
3	+48100

Table 2.6

– is outflow, + is inflow

Solution:

NPV = present value of cash inflow – present value of cash outflow = 0

$$\frac{36,200}{(1+r)^1} + \frac{54,800}{(1+r)^2} + \frac{48,100}{(1+r)^3} - 1,23,400 = 0$$

r = 0.0596 = 5.96%.

Profitability Index (PI)

Profitability index = PV of future cash inflows/PV of future cost × 100.

- Accept the project when PI > 1
- Reject the project when PI < 1
- May or may not accept when PI = 1

Economic Feasibility or Economic Analysis

- Economic feasibility is done on a broader aspect and not just limited to the financial aspects. It is carried out from the frame of reference of a society or the overall economy.
- The economic feasibility of the projects should include whether:
  - the national priorities are aligned with it.
  - there is any contribution to the development of that sector.
  - it is rational to consume the limited resources of the country.
  - the project has to be evaluated for its economic viability.

Social Cost-Benefit Analysis

- Social Cost-Benefit Analysis (SCBA) may be defined as the systematic evaluation of a concern's social performance as distinguished against its economic performance. It considers all such activities which have a social or macroeconomic impact.
- The technique of SCBA was developed by a French Engineer named Jules Dupuit in 1844. During the early twentieth century, while working on welfare economics, AC Pigou pointed out the distinction between financial cost and benefit as against social cost and benefits.
- He said, "A project which may yield profit in monetary terms may not bring corresponding benefits to the society." Its practical application started in 1930 onwards. The

United States government used it to analyse flood control measures in 1936.

- Investment in a project is made for the main reason of generating a high feasible rate of return. Consequently, the main objective of the project's promoters is the maximisation of wealth. Thus, project promoters tend to evaluate a project based solely on its commercial viability. However, some projects may not be commercially profitable but are still pursued because they have societal ramifications. These include road, railway, bridge, and other transportation projects, and irrigation and power projects, among others, for which socio-economic concerns play a considerable role in addition to pure commercial viability. Such projects are analysed for their net socio-economic advantages. The profitability analysis of such projects is referred to as national profitability analysis, which is nothing more than the socioeconomic cost-benefit analysis performed at the national level.
- Each project puts specific costs on the nation and yields particular benefits for the nation. There are two possible types of cost: direct and indirect. Similarly, the benefits generated from every project will be of two types: direct and indirect.
- The SCBA considers the real cost of inputs, i.e., the cost of inputs if they had not been subsidised. For SCBA, the necessary modifications are made to the direct costs of inputs.
- The socioeconomic cost-benefit analysis considers indirect costs and advantages to the nation, whereas the commercial profitability analysis considers only direct costs and direct gains. While a country bears the indirect costs, its citizens reap the indirect advantages.

Methods or Approaches to SCBA

There are two approaches.

United Nations Industrial Development Organization Method

The United Nations Industrial Development Organization (UNIDO) gave SCBA for projects in developing countries in the 1960s and followed it with a detailed guideline in 1978, which suggests five stages as given below. The UNIDO method measures cost-benefit in terms of domestic prices.

- Calculation of financial profitability of the project measured at market prices
- Calculation of net benefits of projects in terms of economical or shadow prices
- Adjustment for the impact of the project on savings and investments
- Adjustment for impact of the project on income distribution
- Adjustment for impact on merit goods and demerit goods for which social values differ from monetary values

Shadow Price

- If the price does not reflect the actual value of a good or commodity or no market value for a good exit, then shadow pricing can be used.
- The shadow price is a proxy value of a good, often defined by what an individual should pay to buy one unit of a good. It may be greater, equal, or less than the market price.

Little–Mirrless (L–M) Method

- The L–M method assesses costs and benefits in terms of international price (sometimes referred to as border price), whereas the UNIDO method analyses costs and benefits in terms of domestic price. The argument in favour of adopting the international price as shadow price is that it helps to offset domestic price volatility and justifies the project economically. Additionally, it assists in increasing productivity and trading efficiency.
- L–M shadow price for traded goods: The shadow price for exported goods is the freight on board (FOB) price, whereas the shadow price for imported goods is the cost, insurance, and freight (CIF) price.
- L–M shadow price for non-tradeable goods: Non-tradeable (excluding labour) items include land, buildings, and services, such as electricity, internal transportation, etc. The shadow price for non-tradeable (excluding labour) items is derived from marginal social cost and marginal social benefit. The marginal social cost and marginal social benefit are determined using

border prices. By utilising border pricing even for non-tradeable, the L-M approach assures that both tradeable and non-tradeable are valued using the same yardstick.

- Measuring the marginal social cost and marginal social benefit of non-tradeable, especially at border prices is challenging. L–M strategy recommends separating the cost component of non-tradeable into tradeable labour and residual component, using a suitable standard conversion factor (SCF) for tradeable and residual components, and utilising standard wage rate (SWR) for the labour component. The challenge, however, lies in selecting a suitable SCF, as this entails a subjective factor.
- L–M Standard Wage Rate (SWR): L–M approach suggests the following relationship.

  $$SWR = C - \frac{1}{s}[c - m]$$

  Where, SWR = standard wage rate

  C = additional resources devoted to consumption

  1/s = social value of a unit of consumption so committed

  c′ = consumption of wage earner

  m = marginal productivity of wage earner
- This method was developed by IMD Little and JA Mirrless. The approach has the following similarities and dissimilarities with the UNIDO method.

Similarities

- Calculations of shadow prices for foreign exchange savings and unskilled labour
- Considerations for equity and use of discounting method (DCF analysis)

Dissimilarities

- L–M method measures costs and benefits in international (or border) prices, whereas UNIDO suggests measuring costs and benefits in domestic currency.
- The L–M method measures costs and benefits in terms of uncommitted social income and the UNIDO method measures them in terms of consumption.
- The UNIDO suggests a stage-by-stage analysis of efficiency prices, savings, and retribution. But, the L–M method reviews these together.
- India has been one of the pioneers in using SCBA for public sector projects which were, at one time, at the commanding heights of the economy. The government of India also followed economic analysis, which they call social profitability analysis, for all industrial projects. They also follow a simplified version of the L–M method. The key elements of analysis are detailed as follows.
  - Tradeable inputs are valued at international prices.
  - Taxes, duties, transfer costs, etc., are ignored
  - Nontradable items like power and equipment are valued at marginal cost.
  - Foreign exchange for inputs and outputs is valued at a specified premium.
  - Social wage rates are applied to unskilled and semi-skilled labourers.
  - The unit of currency in savings is rupees rather than foreign exchange.
- Unfortunately, despite pioneering work done in India, there have been serious gaps in arriving at investment decisions that are financially and economically sound, owing to political expediencies.
- Indian financial institutions use the following indicator for cost-benefit analysis.
  - Economic Rate of Return (ERR) or Internal Rate of Return (IRR)
  - Domestic Resource Cost (DRC)
  - Effective Rate of Protection (ERP)

Economic Rate of Return

- The Economic Rate of Return (ERR) quantifies the rate of return to society. When used to determine a project's financial viability, the current market pricing and costs do not reflect the project's true economic value.
- Private project promoters may benefit from this financial appraisal; social initiatives are excluded.
- The ERR is the discount rate that compares the economic cost of the project to the economic benefits accruing during the project's life. Instead of market pricing, shadow prices are utilised for inputs (costs) and outputs (benefits) for calculating ERR. Prior to selecting

one for implementation, social initiatives are compared based on ERR. A project may have a higher ERR but a lower financial rate of return, or vice versa.

Domestic Resource Cost

- It measures the domestic resource cost of manufacturing a product against the cost of importing it.
- The Domestic Resource Cost (DRC) attempts to quantify the cost of import substitution. Foreign exchange is saved, if the manufacture of a product planned by the project results in import substitution. By calculating the DRC of such a project, it is possible to estimate the number of domestic resources that must be expended to save foreign exchange.
- Additionally, a product may be exported, resulting in foreign exchange earnings. Thus, when a product generated by a project replaces an import, foreign exchange is saved owing to a reduction in imports. When a product is exported, foreign exchange is earned to the extent that exports grow. The establishment of the project and production of the product may necessitate the import of machinery and/or raw materials, resulting in foreign exchange outflow that must be accounted into the DRC calculation.

$$\text{DRC} = \frac{\text{Value added at domestic prices}}{\text{Value added at world prices}} \times \text{exchange rate}$$

- The DRC be used as a measure of the entire amount of rupees expended to conserve one unit of foreign currency. (This concept applies when the commodities produced are substitutes for imports.) Similarly, DRC can be used as a metric of the total amount of rupees required to earn one unit of foreign currency. (This term applies when goods are manufactured for export.)
- The DRC is an indicator for determining whether it is profitable to manufacture a product in the country. If the DRC is less than the exchange rate of the unit of foreign currency, it implies that manufacturing the goods domestically is recommended. This is because the product can be manufactured at a cheaper cost than the cost of foreign exchange. Thus, implementing the project is worthwhile.

Effective Rate of Protection

- It determines the level of net protection granted to domestic projects. The domestic price of a commodity is influenced by concessions, tariffs, subsidies, quantitative controls, etc., in a country. Consequently, such controls and concessions have an impact on the input costs of a product. Costs and prices, excluding concessions and regulations, are the only indicators of a product's competitive strength in the global market. In the absence of concessions and regulations, the value added to a product when assessed at domestic and international prices will be the same. If there is a discrepancy between the value addition evaluated at domestic and foreign prices, this demonstrates the level of protection.

$$\text{ERP} = \frac{\text{Value added at domestic prices} - \text{value added at world prices (free trade)}}{\text{Value added at world prices}}$$

- The ERP of zero indicates that the project does not enjoy any protection from international competition.
- A positive value of ERP indicates that the project enjoys protection from international competition. Higher the positive value of ERP, the higher the protection enjoyed by the product.
- Projects with a higher ERP require careful analysis as to whether the protection available against international competition is likely to continue. It may be possible that the project may have a higher ERP because of the subsidised cost of inputs. Suppose, in future, there is a possibility of the government withdrawing the subsidies presently allowed on inputs. In that case, the ERP will come down, and the project may get exposed to international competition.

Detailed Project Report

- Detailed Project Report (DPR) has details of all feasibility studies, i.e., technical, financial, market, etc.
- A DPR is an essential building block for projects as it serves the purpose of investment, decision-making, and approval. Hence, DPR is to be prepared carefully and with sufficient details to ensure appraisal, approval, and subsequent project implementation in a timely and efficient manner.
- As DPR contains complete details of the project, the same is submitted to banks and financial institutions for getting financial assistance. The DPR also acts as a base document for project planning and implementation.
- The contents of a DPR are mentioned are as follows.
    - Availability of raw materials and tie-up (MOU document/willingness certification)
    - Availability of land and tie-up (a lease document)/willingness certification
    - Organisation type and structure (entrepreneur/proprietary, private limited/ public limited, cooperative, NGO, etc.)
    - Brief project description
    - The tie-up with technology, equipment suppliers
    - Financial analysis and profitability study
    - Incentives and concessions expected from other government and public bodies for demonstration and future multiplications
    - Initial contribution in terms of finance, technology development, technical and equipment tie-up by the promoter and user agency (mention separately)
    - Organisations to operate and maintain the demonstration project
    - Organisation to replicate the project in a specific region or throughout India
    - Fulfilment of statutory requirements (like PCB clearance, environmental clearance/ safety, etc.)

Procedure to Prepare a DPR

Step 1: Client signs a contract with the consultant.
Step 2: Consultant receives inputs from the client and carries out necessary studies.
Step 3: The client evaluates the draft and makes comments, suggestions, and requests for modifications.
Step 4: Consultant submits the revised draft for approval.
Step 5: Consultant submits the final DPR after approval from the client with all appendices and annexure.
Step 6: Evaluation of DPR by client/owner.

Investment Phase/Project Feasibility Studies

It involves a detailed design and actual implementation. This phase involves several interdisciplinary tasks. It has the following four phases.

- Negotiating and contracting
- Detailed project design and engineering: Site investigation, design, approval of layout, drawings, and blueprints, preparation of time schedule and PERT charts, final selection of technology, equipment, and detailed estimation of costs
- Construction and erection (execution and control)
- Trial runs, commissioning, and optimisation

Operation Phase

- The post-commissioning stage is when a project is run for output or benefits.

Practice Exercise

1. The reason that the Schedule Performance Index (SPI) is shown as a ratio is to _____.

(A) enable a detailed analysis of the schedule regardless of the value of the schedule variance
(B) distinguish between critical path and non-critical path work packages
(C) provide the ability to show performance for a specified time period for trend analysis
(D) measure the actual time to complete the project

2. If the cost variance is the same as the schedule variance and both numbers are greater than zero, then _____.

(A) the cost variance is due to the schedule variance
(B) the variance is favourable to the project
(C) the schedule variance can be easily corrected
(D) labour rates have escalated since the project began

3. Preliminary project report for a road project must contain _____.

(A) a detailed estimated cost based on detailed design
(B) several alternatives to the project that have been considered
(C) the soil survey, traffic survey, concept design, and approximate cost
(D) the contract documents for inviting tenders

4. Match List-I with List-II and select the correct answer using the codes given below the lists.

List-I

A. To allocate the resources among competing projects and to determine whether or not the returns from investment are adequate
B. To see that adequate and timely funds are made available for investment, including that for operation and maintenance
C. To analyse various aspects of a project, such as a process, design, size, layout, capacity, etc.
D. To find the effects of fiscal, monetary, wage, exchange rate, policies, etc.

List-II

I. Technical analysis
II. Macroeconomic analysis
III. Financial analysis
IV. Feasibility analysis
V. Economic analysis

Codes:

	A	B	C	D
(A)	V	IV	II	I
(B)	IV	III	I	II
(C)	III	V	II	I
(D)	V	III	I	II

5. Match List-I (feasibility) with List-II (emphasised on or calls for) and select the correct answer using the codes given below the lists.

List-I

A. Social
B. Economic
C. Input
D. Coordination

List-II

I. Optimising marginal costs and benefits
II. Forward and backward linkages
III. Interfaces
IV. Improving wealth

Codes:

	A	B	C	D
(A)	I	IV	III	II
(B)	IV	I	III	II
(C)	I	IV	II	III
(D)	IV	I	II	III

6. Cost-benefit studies are essential to _____.

(A) assess the total cost of the work
(B) ascertain the relevant escalation in prices
(C) monitor the expenditure
(D) evaluate the viability and worthiness of taking up the project

7. The following are project baselines that are generally part of the project management plan, except _____.

(A) technical
(B) scope
(C) time
(D) cost

8. Projects are particularly susceptible to risk because _____.
(A) Murphy's law states that "if something can go wrong, it will"
(B) there is uncertainty in all projects
(C) project management tools are generally unavailable at the project team level
(D) there are never enough resources to do the job

9. The Delphi technique is a particularly useful method for identifying risks to _____.
(A) present a sequence of decision choices graphically to decision-makers
(B) define the probability of the occurrence of specific variables
(C) reduce the bias in the analysis and keep any one person from having undue influence on the outcome
(D) help to take into account of the attitude of the decision-maker towards the risk

10. Each one of the following statements about risk avoidance is true, except that it _____.
(A) focuses on changing the project management plan to eliminate the threat
(B) isolates the project's objectives from the risk's impact
(C) accepts the consequences of the risk event should it occur
(D) changes the project objective that is in jeopardy

11. The simplest form of quantitative risk analysis and modelling techniques is _____.
(A) probability analysis
(B) sensitivity analysis
(C) delphi technique
(D) utility theory

12. Risk mitigation involves _____.
(A) using performance and payment bonds
(B) eliminating a specific threat by eliminating the cause
(C) avoiding the schedule risk inherent in the project
(D) reducing the probability and/or impact of an adverse risk event to an acceptable threshold

13. What is the primary difference between risk audit and risk reassessment?
(A) A risk reassessment is conducted at the completion of a major phase; audits are conducted after the project is complete.
(B) Project stakeholders conduct risk audits; management conducts reassessments.
(C) Risk reassessments are regularly scheduled; risk audits are performed as defined in the project's risk management plan.
(D) There is no difference; they are virtually the same.

14. Accurate and unbiased data are essential for performing qualitative risk analysis. Which one of the following should you use to examine the extent of understanding of project risk?
(A) Data quality assessment
(B) Project assumptions testing
(C) Sensitivity analysis
(D) Influence diagrams

15. Which of the following is 'not' an objective of a risk audit?
(A) Confirming that risk management has been practised throughout the project life cycle
(B) Confirming that the project is well managed and that the risks are being controlled
(C) Evaluating the effectiveness of risk responses in dealing with identified risks
(D) Ensuring that each risk identified and deemed critical has a computed expected value

16. Which of these is not a component of the feasibility report?
(A) Profit margin
(B) Works cost
(C) Overheads
(D) Pre-commissioning expenses

17. Arrange the following elements of the project cycle in the correct order.

I. Project appraisal
II. Feasibility analysis
III. Execution
IV. Project selection

(A) I – II – III – IV
(B) II – I – III – IV
(C) II – I – IV – III
(D) I – III – IV – III

18. You are the project manager who is incharge of developing a new shipping container for Globus Ocean Transport, which needs to withstand winds of 90 knots and swells of 30 metres. In determining the dimension of the container and the materials to be used in its fabrication, you convene a group of knowledgeable professionals to gather initial requirements, which will be included in the _____.

(A) project charter
(B) bill of materials
(C) WBS
(D) project statement of work

19. Feasibility report is prepared during which of the following phase of a project?

(A) Intermediate phase
(B) Initial phase
(C) Final phase
(D) After completion of a project

20. The economic viability of a project can be assessed by which of the following methods?

(A) benefit-cost ratio
(B) payback period and net present value
(C) return on investment and internal rate of return.
(D) All of the above

21. Break-even analysis can be used in _____.

(A) choosing the best product
(B) selecting among alternatives
(C) choosing a critical method
(D) selecting the best equipment

22. The S-curve is used for _____.

(A) forecasting cash requirement
(B) forecasting only shortages
(C) forecasting only surpluses
(D) None of the above

23. Which of the following methods are considered most favourable to production organisations liable for taxation?

I. Straight line method of depreciation
II. Declining balance method of depreciation
III. Sum of the years' digits methods of depreciation
IV. Sinking fund method of depreciation.

Select the correct answer from the codes given below.

(A) I and II
(B) II and III
(C) III and IV
(D) IV and I

24. A project on software development and construction of a bridge are similar to each other in terms of _____.

(A) project life cycle
(B) project management process groups
(C) technology
(D) resources

25. Enterprise environmental factors that influence the development of a project management plan include _____.

(A) configuration management system
(B) scheduling software tool
(C) government standards
(D) All of the above

26. Some of the accepted types of project appraisals are _____.

(A) Technical, financial, economical, and formal.
(B) Technical, financial, ecological, and environmental.
(C) Technical, financial, commercial, and environmental.
(D) Commercial, technical, financial, and superficial.

27. The main purpose of conducting a project appraisal is to ensure that _____.

(A) the project has met all its objectives concerning the scope
(B) the funds assigned for the project have not exceeded the initial assessment
(C) all related areas of the project are examined before approving the execution of the project
(D) a promoter is a qualified person who can be trusted with the project

28. Technical appraisal of a project includes all of the following, except _____.
(A) availability and transport of raw material
(B) layout of the plant including employee safety
(C) project scheduling and implementation
(D) use of the latest technology on the project

29. To ensure the success of a good product mix in the output from a project, the promoter can ensure the following, except _____.
(A) flexibility in the layout of the product facilities
(B) availability of more funds in case the original plan does not work out
(C) elaborate study on the market for the product prior to finalising production plans
(D) offering a discounted initial rate for new, unusual items in the market

30. Demand forecasting techniques to ensure the success of a proposed project include which of the following?
(A) Consumer surveys, expert opinion, Delphi techniques, regression, and trend analysis
(B) Consumer surveys, expert opinion, Delphi techniques, regression, and exit poll analysis
(C) Consumer analysis, Delphi surveys, expert judgement, regression, and trend studies
(D) Consumer studies, Delphi methods, good judgement, TV, and newspaper analysis

31. Which is/are the main reason(s) for conducting a financial appraisal?
(A) Estimating the cost of the project and the ability of the promoter to manage it
(B) Obtaining funding for the project and estimating the promoter's ability to spend the money carefully
(C) Estimating the cost of the project and the break-even point
(D) Estimating the cost of the project and the ability to obtain funding for this cost through debt or equity

32. Risk planning on projects is mainly done to _____.
(A) check if all technical options on the project have been explored
(B) adopt a proactive approach to ensure that the team members are prepared for both positive and negative risk events
(C) see whom to hold responsible when things go wrong on the project
(D) figure out how to get more money to fix things once problems happen on the project

33. You have always been asked by your management to cut your project estimate by 10 per cent after you delivered it to them. The scope of your new project is unclear and there are over 30 stakeholders. The management expects a 25 per cent reduction in downtime as a result of the project. Which of the following is the best course of action in this situation?
(A) Re-plan to achieve a 35 per cent improvement in downtime
(B) Reduce the estimates and note the changes in the risk response plan
(C) Provide an accurate estimate of the actual costs and be able to support it
(D) Meet the team to identify areas of low per cent savings

34. Factors which govern the operating cost of equipment are as follows.
I. Purchase price of the equipment
II. Depreciation due to regular use
III. Cost of operation, maintenance, and repairs
Which of the above factors are correct?
(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II, and III only

35. During the life cycle of the project, a plot of the project's expected expenditures will

usually follow a characteristic 'S' shape. This indicates that _____.

(A) there is a cyclic nature to all projects
(B) problems will always occur in the execution phase
(C) there are high expenditures during closeout
(D) a bulk of the project budget will be spent in the execution phase

36. Project A has an Internal Rate of Return (IRR) of 21 per cent in 2 years. Project B has an IRR of 17 per cent in 3 years. Project C has an IRR of 31 per cent in 2.5 years. Project D has an IRR of 29 per cent in 1 year. Which of these would be the 'best' project for implementation financially?

(A) Project A
(B) Project D
(C) Project C
(D) Project B

37. Risk on a project involves which of the following?

(A) Proactively planning and preparing for positive and negative risk events as a one-time process during the planning phase
(B) Reactively ensuring that all positive and negative risk events are taken care of
(C) Proactively planning and preparing for positive and negative risks iteratively throughout the project life
(D) Reactively checking to ensure that all new risk events do not impact the project negatively

38. Risk is defined as _____.

(A) a negative event that may occur during a project
(B) a positive event that may occur during a project
(C) an event that may disturb the team on a project
(D) Both A and B

39. A system development project is nearing the closing phase when a previously unidentified risk is discovered. This could potentially affect the project's overall ability to deliver. What should be done next?

(A) Alert the project sponsor of potential impacts to cost, scope, or schedule.
(B) Measure the impact of the risk on the project.
(C) Mitigate or reduce this risk by developing a risk response plan.
(D) Develop a workaround to fix the problem immediately.

40. The 'economic life' of a building is considered to be at an end when:

I. the net income from the building fails to justify its existence.
II. the building facilities become obsolete.
III. the capitalisation rate is high.

Which of the above statements are correct?

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II, and III

41. During project execution, a project team delivers a project deliverable to the customer. However, the customer neither acknowledges the deliverable nor says if it is acceptable, although approval is required. What is the 'best' thing to do?

(A) Continue with the project
(B) Document the situation
(C) Contact the management for help
(D) Call a meeting of the team

42. A particular stakeholder has a reputation for making any changes to projects. What is the best approach a project manager can take at the beginning of the project to manage this situation?

(A) Say 'no' to the stakeholder a few times to dissuade him from submitting more changes
(B) Get the stakeholder involved in the project as early as possible
(C) Talk to the stakeholder's boss to find ways to direct the stakeholder's activities to another project
(D) Ask that the stakeholder not be included in the stakeholder listing

43. The document that officially sanctions the project is _____.

(A) project charter
(B) project plan
(C) feasibility study
(D) cost-benefit analysis

Answer Key

1. (C)	**2.** (B)	**3.** (C)	**4.** (D)	**5.** (B)	**6.** (D)	**7.** (A)	**8.** (B)	**9.** (C)	**10.** (C)
11. (B)	**12.** (D)	**13.** (C)	**14.** (A)	**15.** (D)	**16.** (A)	**17.** (C)	**18.** (A)	**19.** (B)	**20.** (D)
21. (B)	**22.** (A)	**23.** (A)	**24.** (B)	**25.** (D)	**26.** (C)	**27.** (C)	**28.** (D)	**29.** (B)	**30.** (A)
31. (D)	**32.** (B)	**33.** (C)	**34.** (C)	**35.** (D)	**36.** (C)	**37.** (C)	**38.** (D)	**39.** (B)	**40.** (B)
41. (C)	**42.** (B)	**43.** (A)							

3 Project Organisation

Introduction

- Once the project has been selected, the next step is for the senior management to choose a project manager (PM).
- A PM is an individual with authority, accountability, and responsibility for managing a project to achieve specific objectives.
- A PM, new or experienced, must be well versed in functional areas like design, human resources, finance, planning, etc.
- An organisational structure is concerned with the allocation of tasks, establishment of an authority and its responsibilities, and the relationship between members of an organisation.

Characteristics of a Project Manager

An effective PM should have the following skills or abilities.

- Technical skills
- Leadership skills
- Resource management skills
- Human resource management skills
- Communication skills
- Negotiation and influencing skills
- Conflict management skills
- Marketing, contracting, customer relationship skills
- Budgeting and costing skills
- Scheduling and time management skills
- Team building skills
- Motivation skills
- Decision-making skills
- Political and cultural awareness
- Trust building skills

Powers of a Project Manager

Meaning of Power

- Power implies the ability and authority to make something work.
- Within the disciplines of political science and business, power can be described as the capacity to influence the actions of others to achieve a predetermined objective. In companies, legitimate power is referred to as 'authority' and is typically exercised by superiors over subordinates or team members to achieve organisational goals. The proper application of authority does not require the use or threat of force (coercion).

Types of Power

- Legitimate power, formal power, or positional power: This refers to the power invested in an individual based on one's position within an organisation or project. It is typically connected with the person's title, position, or uniform, as well as, one's duties and responsibilities in the organisation, for example, the owner of a business.
- Expert power: This refers to the expertise earned through years of job experience or attainment of higher educational degrees in a particular field of work. These persons are referred to as subject matter experts or SMEs. This authority is a result of their capacity to comprehend and analyse problems and suggest a practical solution. This form of authority is typically limited to the domain of expertise possessed by an SME.
- Referent power: This is the power acquired by using interpersonal skills and personal magnetism to obtain and maintain the loyalties of others and persuade them to think as you do. Thousands of individuals joined Mahatma Gandhi's non-violent movement because his personality and way of life inspired them.
- Reward power: It is the best type of power to use, if provided appropriately. In this situation, a person leverages one's power to the advantage of employees or team members by providing them with incentives, rewards, increased tasks, promotions, etc., for doing their job successfully.
- Penalty or coercive power: This is the worst type of power as it is used to punish employees

or withhold their rewards for non-performance or disobedience. Although sometimes required, the extended or ongoing exercise of only this type of authority creates a toxic work environment within the organisation.

Strategy

- Strategy may be defined as a high-level planning that incorporates contingency planning or alternatives that accommodate for uncertainties in the original plan. Essentially, two actions are required. Firstly, the formulation or strategic thinking is required. It considers all possibilities pertinent to the scenario at hand and chooses the best decision while keeping in mind the options required in uncertain circumstances. Secondly, implementation or execution needs to be taken care of. It is the act of carrying out the selected action plan.
- As far as organisations are concerned, there are three types of strategy.
  - Business strategy defines the purpose and plan for the organisation as a whole.
  - Operational strategy defines what must be done daily to achieve the business goals.
  - Transformation strategy involves radical and drastic changes to the business and operation strategies when the company decides to change its business strategy substantially.

Problem-solving

- Problems are a regular occurrence in a project and one of the fundamental qualities of a competent manager is the ability to tackle problems as they arise. Some issues are clearly defined, while the majority are not. Without a strategy for defining, comprehending, and resolving problems, it becomes exceedingly challenging.
- The basic steps in problem solving are listed below.
  - Defining the problem
  - Generating alternatives for solutions
  - Testing these solutions against the problem at hand
  - Evaluating and selecting one or two effective solutions
  - Implementing the selected solution
  - Following up to check if the solution was adequate

The main barriers to effective problem-solving are the following.

- Beginning the problem-solving procedure with a solution already in mind—confirmation bias. This involves discovering the solution before defining or establishing the problem, and then, finding a way to arrive at this already decided solution, thereby, confirming one's initial thought incorrectly.
- Utilising approaches and solutions that have worked in the past without verifying their applicability to the current problem–mental set. Consequently, neither originality nor invention is employed to solve the problem.
- Refusing to generate numerous solutions to a problem and clinging to the most obvious or straightforward one—functional fixity. By thinking in only one way or presuming that an object has only one use, there are no fresh ideas, and one has to employ extraneous data that may mask the genuine solution.

Roles and Responsibilities of a Project Manager

- Formulation of objectives and policies for the project and operationalising them properly (scope management)
- Ensuring the availability of resources and their best utilisation
- Acquiring and motivating personnel (managing human resources)
- Making project goals trade-offs and maintaining a balanced outlook
- Preparation and approval of all operational plans, procedures, and project plans
- Delegation of duties and responsibilities to project staff
- Monitoring and controlling cost, progress, performance, and quality
- Ensuring safety standards and legal requirements
- Serving as the prime point of contact for the project
- Ensuring the establishment and operation of communication, information, and reporting system
- Analysing the potential risk and their mitigation

Types of Project Organisation

Organisational structure can be of three types. They are as follows.

- Functional organisation
- Product organisation
- Matrix organisation

Functional, or Traditional, or Classical Organisation

- A functional organisation is the simplest and most rational organisational system. It unites all personnel engaged in a single activity or multiple related tasks in a department.
- It has a lengthy communication network and a chain of command. The relevant functional head resolves project-related difficulties.
- This organisational structure is appropriate for tiny businesses with a limited product offering. As an organisation grows by expanding its product range or geographically, the inadequacy of its functional organisational structure will become obvious, as controlling will lose its effectiveness. Another downside of functional organisation is that it is difficult to establish accountability and evaluate members' performance.

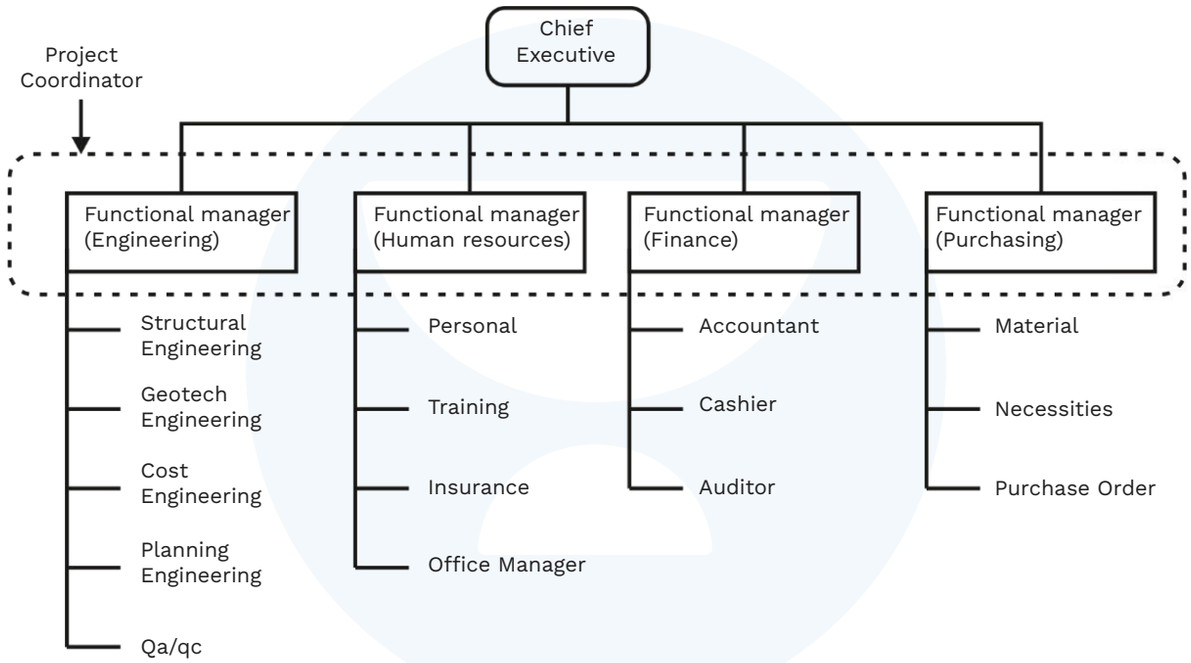


Fig. 3.1

Advantages

- Individual experts can be utilised in different projects. The functional division contains the normal path of advancement for individuals whose expertise is in the functional area. Employees have a sense of security, and thus, perform their job responsibilities diligently.
- Employees tend to remain loyal to the organisation due to the sense of job security.
- It has been observed that cost control and budgeting are uncomplicated.
- There is flexibility in the use of workforce.
- As the employees have experience in the same field, they tend to be more skilled, competent, and efficient.
- The roles and responsibilities of every employee are fixed, and there is no replication of work.
- Every employee reports to his/her functional manager where he/she is working. This is known as clear-cut hierarchy, and the outcome of this is improved coordination, cooperation, and communication.
- There is better accountability of work.

Disadvantages

- No one person is responsible for the total project. The project-oriented focus for achieving the project goals is absent.
- Decisions are influenced by stronger functional departments.
- Work of functional division usually takes precedence over the project work.
- Lack of coordinated effort tends to make the response to client needs slow.
- One of the disadvantages is that the work may become repetitive and monotonous, which might bore the employees, and they may become lazy.
- The cost involved in having highly skilled workers in the organisation is also higher.
- A self-centred mentality may develop, i.e., in this case, the functional manager only concentrates and takes charge of his/her department, neglecting or ignoring the other departments.
- Inter-department communication is poor, which results in lack of coordination between the departments. This leads to reduced pliability, innovation, and transformation.
- As the company grows bigger, managing functional departments can become a strenuous task because of their size. Each department can act like a mini-company having its own working culture, guidelines, and management style.
- A PM has little or no authority.
- The carrier path is not clearly defined for a PM. Sometimes, they may be in part-time pay roles.

Project or Product Organisation

- In a project or product organisation structure, the PM has the authority to assign priorities, apply resources, and assign work to persons involved in the project.
- The structure of a project organisation differs a lot from the structure of a functional organisation. It is opposite to the functional organisation structure. This type of organisation will either not have a functional manager, or if there is one, his/her role, responsibility, and authority is limited.
- The focus area of project organisation is only on the project work that it obtains from clients or customers.
- The PM recruits necessary personnel from both within and outside the parent company, such as admin, accounting, human resource, etc., to support the project management activities.

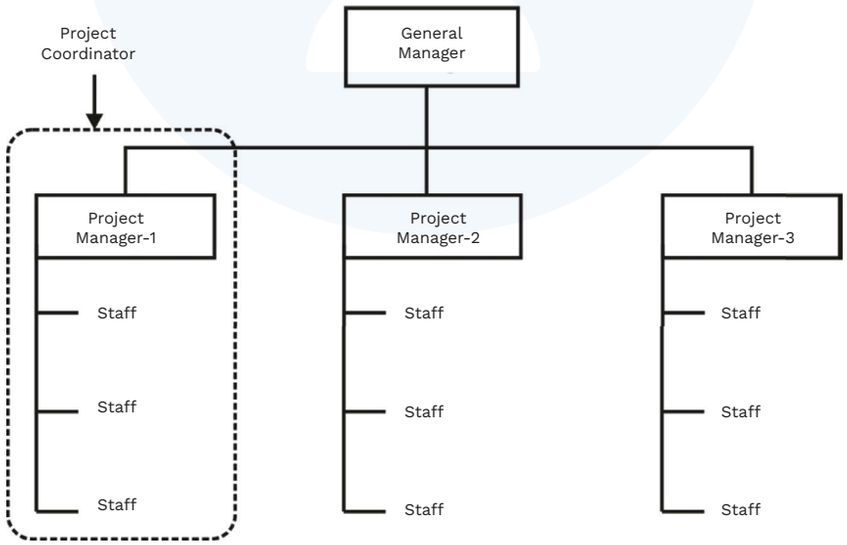


Fig. 3.2

Advantages

- It is simple to establish.
- Fast completion of the projects is ensured.
- There are no functional division heads whose approval needs to be sought.

- The PM directly communicates with the senior management, which results in shorter communication channels.
- The project team forms a strong and separate identity, which results in a high level of dedication.
- Motivation is high. Unity of command exists.
- Decisions are made at a faster pace because all members of the team report directly to the PM. As a result, there is an explicit line of command, and this decreases conflict. Hence it speeds up the decision-making process and aids in having better flexibility.
- The team members get to work on distinct projects, which help them get a multifaceted skill set and become adaptable to any type of work.
- There is robust and efficacious communication within the entire team due to the presence of an exclusive reporting system.

Disadvantages

- It creates an internal rift in the organisation. It is known as projectitis, which means a gap gets created between the project teams and other people in the parent organisation, as project members feel they are only important for the organisation.
- One of the disadvantages is that the PM may become pompous and haughty as he/she has full command and control over his/her team.
- If there are multiple ongoing projects, it can lead to below-par communication amongst them, which can, in turn, result in the replication of the resources.
- The employees may think that once the project gets completed, they might lose their jobs, and thus, have a sense of insecurity. Hence, the employees tend to be less faithful to the company.
- As specialised tools, equipment, and high skilled labour are required for a short duration, the cost associated with these can be higher.

Matrix Organisation

- As per the Project Management Institutes (PMBOK) Guide, the combination of functional and project organisations forms the basis of a matrix organisation. It takes advantage of both types of organisational structure.
- In a matrix organisation, the expertise, competence, knowledge, and skill set of an employee come handy within the functional department, as well as, the project management team.
- In the matrix organisation structure, there are two chains of command, one along functional lines and the other along project lines.
- Instead of delegating segments of a project to different units or creating a separate (or project) team, project participants report simultaneously to both functional and project managers.
- Managers, who supervise the functional part of a project, are functional managers. They determine the work that needs to be done by the juniors or assistants, and how the distribution of work is to take place.
- Managers, who have command over the managerial aspect of a project, are project managers. A project manager will decide how the work will be done, what the timeline will be, and assess the team members' performance.
- Sometimes, it can be baffling to work in a matrix organisation as the employees have to report to two managers.
- Clear instructions should be given to the employees regarding their job roles and responsibilities in a matrix organisation so that there is no doubt and the work is performed efficiently.
- This type of organisational structure is generally present in those organisations that have multiple projects and are of substantial size. The benefit of the matrix structure in these organisations is that the employees can be transferred where they are needed, according to the expertise they possess.
- Based on the power and authority of the functional and project manager, the matrix organisation structure can be categorised as follows.
  - Weak matrix structure
  - Balance matrix structure
  - Strong matrix structure

Weak Matrix Organisation Structure

- The functional manager has command over the budget in a weak matrix organisation structure. Here, the project manager will have less power or authority. His/her role will be mostly part-time and no administrative personnel will be reporting to him/her. The project manager's role in this type of structure will resemble the role of a coordinator.
- The attributes of a weak matrix organisation structure mimic the attributes of a functional organisation structure.

Balanced Matrix Organisation Structure

- A balanced matrix organisation structure shares power and authority between both the project and the functional managers.
- The role of the project manager will be full-time. But the administrative personnel reporting to him/her will be limited.
- The project budget is controlled by both the project and the functional managers.

Strong matrix Organisation Structure

- As opposed to the weak matrix organisation structure, the project manager holds maximum power and command over the project in a strong matrix organisation structure.
- Unlike the weak matrix organisation, the project manager has a full-time role, and all the administrative personnel report to him/her. The project manager controls the project budget in a strong matrix organisation structure.
- As opposed to the weak matrix organisation structure, the role of a functional manager is limited.
- The attributes of a strong matrix organisation structure mimic those of a project organisation structure.

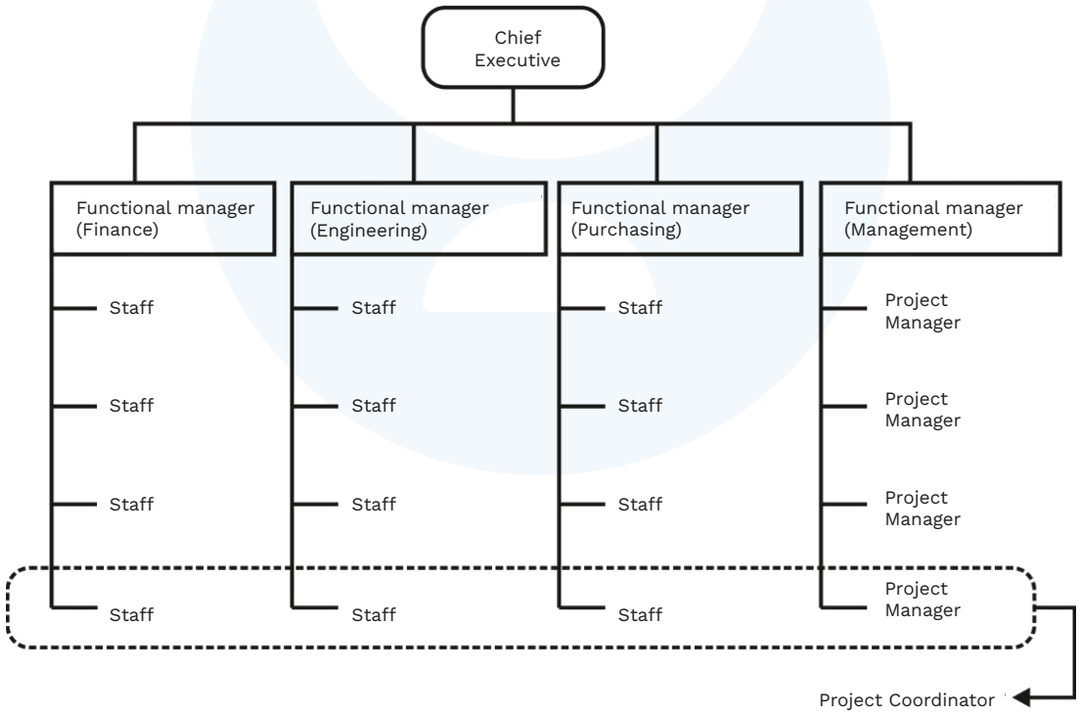


Fig. 3.3

Advantages

- The project manager is responsible for managing the project, bringing it on time, and within cost and specification.
- The response to the requirement of clients is fast. The expertise of high-skilled workers comes in handy for both functional departments and units.
- As inter-department communication between employees is better in a matrix structure, it is more dynamic than in a functional structure.

- One of the advantages is that a great working culture and cooperative environment is established, which is important to maintain harmony in an organisation.
- There are many opportunities for employees to enhance their skill set by learning and taking part part in different type of projects.
- Employees tend to remain loyal to the organisation due to their job security, and hence, the productivity increases.

Disadvantages

- Matrix management violates the management principle of unity of command.
- As stated above, it can be baffling for some employees as they have to report to two managers. A conflicting situation may arise between the decisions made by the project and the functional manager.
- Clear priorities should be given to employees regarding their job roles and responsibilities, mainly when doing the work that differs from the one they were doing earlier.
- If a resource is limited, rivalry may arise to make use of it. This may lead to a chaotic situation and can reduce the efficiency of operational tasks.
- In addition to the regular work assigned to the employees, they might have to do additional project work allocated to them, which can reduce their efficiency and lead to exhaustion.

Practice Exercise

1. Match List-I with List-II and select the correct answer using the codes given below the lists.

List-I

A. Translates policy into a method of achieving the objective set out.
B. Consists of defining the responsibilities of employees.
C. Transmits all information to the supervising staff.
D. The organisational set-up is aided to operate efficiently with the flow of information, decisions, and results in all directions.

List-II

I. Coordinating
II. Planning
III. Organising
IV. Directing

Codes:

	A	B	C	D
(A)	III	II	I	IV
(B)	II	III	I	IV
(C)	III	II	IV	I
(D)	II	III	IV	I

2. In a matrix organisation, which of the following is true?

(A) The project manager is responsible for employee skills improvement.
(B) The functional manager is responsible for employee skills improvement.
(C) The project manager is responsible for the employees' annual appraisal.
(D) The employee is responsible for the improvement of his or her own skills.

3. Match List-I (indications of terms) with List-II (terms) and select the correct answer using the codes given below the lists.

List-I

A. Used for recording instructions given by the executive engineer at the site.
B. Used widely for civil engineering construction.
C. One of the principles of organisation.
D. One of the functions of management.

List-II

I. Coordination
II. Unity of command
III. Line organisation
IV. Site order book

Codes:

	A	B	C	D
(A)	IV	III	II	I
(B)	IV	II	III	I
(C)	II	IV	I	III
(D)	IV	I	II	III

4. The 'halo effect' refers to the tendency to _____.
(A) promote from within
(B) hire the best
(C) move people to project management because they are good in their technical fields
(D) move people to project management because they have had project management training

5. The overall coordinating person in a construction team is the _____.
(A) owner
(B) architect
(C) project manager
(D) contractor

6. The functional organisation system of working was introduced by _____.
(A) FW Taylor
(B) Henry Gantt
(C) MR Walker
(D) JE Kelly

7. In a matrix organisation the ability of a project manager of influence the cost of a project _____.
(A) increases with time
(B) remains the same over time
(C) depends on the availability of resources and funds
(D) decreases over time

8. Name the type of organisation, in which the project manager has little formal authority and acts as a staff assistant to an executive who is ultimately responsible for the project.
(A) Balanced matrix
(B) Weak matrix
(C) Projectised
(D) Strong matrix

9. Who is ultimately responsible for quality management on the project?
(A) Project engineer
(B) Project manager
(C) Quality manager
(D) Team member

10. A project manager is trying to complete a software development project, but cannot get enough attention for it. The resources are focused on completing process-related work and the project manager has little authority to properly assign the resources. What form of organisation must the project manager be working in?
(A) Functional
(B) Matrix
(C) Expediter
(D) Coordinator

11. What are the major advantages of a functional type of organisation?
(A) Single point of contact for the customer
(B) Stable organisational structure
(C) Project orientation
(D) Multifunctional teams are easy to form

12. Breaking the major deliverables into smaller, more manageable components to provide better control is called _____.
(A) scope planning
(B) decomposition
(C) scope baselining
(D) bill of materials

13. In which organisational form is it most difficult to integrate project activities?
(A) Classical/traditional
(B) Projectised
(C) Strong matrix
(D) Weak matrix

14. In which organisation form would the project manager possess the greatest amount of authority?
(A) Classical/traditional
(B) Projectised
(C) Strong matrix
(D) Weak matrix

15. In which organisational form do project managers have the greatest likelihood of possessing reward power and having a wage-and-salary administration function? (The project and line manager are the same people)

(A) Classical/traditional
(B) Projectised
(C) Strong matrix
(D) Weak matrix

16. In which type of matrix structure would a project manager most likely have a command of technology?
(A) Strong matrix
(B) Balanced matrix
(C) Weak matrix
(D) Cross-cultural matrix

17. During project staffing, the primary role of line management is _____.
(A) approving the selection of the project manager
(B) approving the selection of assistant project managers
(C) assigning functional resources based on who is available
(D) assigning functional resources based on the availability of resources and the skill set needed

Answer Key

1. (D)	**2.** (B)	**3.** (A)	**4.** (C)	**5.** (C)	**6.** (A)	**7.** (D)	**8.** (B)	**9.** (B)	**10.** (A)
11. (B)	**12.** (B)	**13.** (A)	**14.** (B)	**15.** (A)	**16.** (A)	**17.** (D)			

4 Project Planning and Scheduling

Planning

- The purpose of the project planning process is to kick off a new project and establish an accurate plan and schedule.
- During this phase, the general requirements of a project are identified and broken down into discrete activities. The sequence and interdependence of these activities are also determined. The required resources are estimated, and the critical path is identified.

Scheduling

- Project scheduling is the process of determining the project completion time so that the implementation of the project is done within the stipulated timeframe.
- This stage overlaps with the planning stage. Each activity is assigned a time estimate, and the total project duration is computed. The earliest and latest beginning and closing dates for each activity are determined.

Steps Involved in Planning and Scheduling

- Determine the planning details or Work Breakdown Structure (WBS).
- Work breakdown should be converted into activities and activity interrelationship should be established.
- Develop alternative planning strategies and give standardised inputs.
- Develop resource optimisation, considering constraints and determining activity duration.
- Prepare project histogram, or bar chart, or link bar chart, or milestone chart, or network diagram.
- Make time calculations and find project duration. Review the project progress at regular intervals throughout the process of updating. Monitor the progress regularly.

Implementation and Control

- The implementation of control phase begins once the project is undertaken. Here you employ a variety of techniques to achieve the optimisation of resource usage, monitor progress and cost, forecast completion dates and overall project cost, and ensure that critical activities are completed on time.
- The earliest technique that was invented for implementation and control was known as 'project, planning, and scheduling' abbreviated as PPS. This technique was transformed to Critical Path Method abbreviated as CPM. The deterministic procedure of the CPM technique is helpful when projects are unchanging and repetitive.

Work Breakdown Structure

- The Work Breakdown Structure (WBS), also known as top–down diagram, looks like a tree formation. It accounts for the explanation of the jobs for the accomplishment of the project.
- It is used to divide the work logically by graphical representation, following a stratified and relational structure.
- The factors on which the most important jobs get segmented into several small easily controllable jobs are time, how much effort is required for the accomplishment of a job, the scope of work, etc. Work breakdown may be done into the following stages: project, sub-project, task, sub-task, and work packages.
- The components of work that are at the lowest level of the WBS are known as work packages.

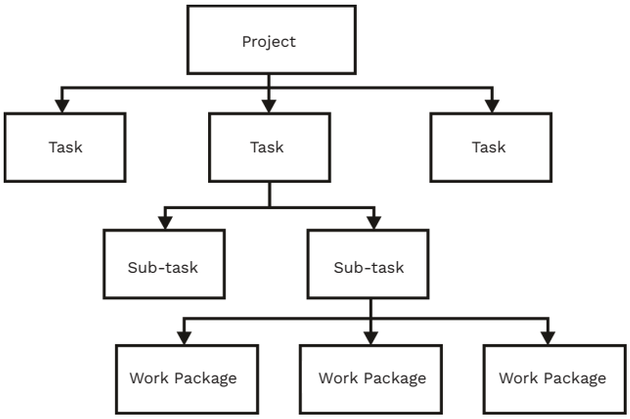


Fig. 4.1

Techniques Used for Scheduling

Bar Charts

- These are also known as Gantt charts as they were developed by Henry Gantt around the year 1900.
- It is a pictorial or graphical representation of activity versus time. The time, which has elapsed, is depicted on the horizontal or X axis, and the activities or jobs that are to be carried out are represented on the vertical or Y axis.
- The chart is useful for finding the time needed for accomplishing an activity. The starting and ending point of the bar depict the beginning and end of that specific task or activity, whereas the length of the bar depicts the total time needed for completing the activity.
- There is no particular order in which the bars have to run one after the other. The jobs may run parallelly, or may start one after the other, or may even overlap each other. So, it means that the bars may start after one gets completed, or may be parallel to each other, or may overlay each other too.
- The bar chart has a few constraints as well, which are as follows.
    - Absence of detailing of sub-activities: The sub-activities cannot be depicted separately in a bar chart. This becomes a problem, especially when the projects are enormous and only the important activities can be shown in the bar chart. If all activities of a big project are shown on the bar chart, then it may become messy. This leads to ineffective command over all tasks or activities in case of extensive projects.
    - Unable to track the progress of a project: With the use of a bar or Gantt chart, we are unable to trace the progress of a project as it does not depict the progress of activities.
    - Correlation between activities: The dependencies or correlation between the activities cannot be represented by a Gantt chart.
    - Completion time estimation for uncertain activities: In activities like research projects, where the completion time is uncertain and unsure, Gantt charts are not handy.
    - Difficult resource levelling and smoothening: One of the most important aspects of accomplishing resource levelling and smoothening is that one should be able to distinguish between critical and non-critical activities, which in case of Gantt charts, is not possible.
- To summarise, the projects where time evaluation can be done with certain conviction, such as production, construction, small-scale, etc., Gantt charts are convenient to use.

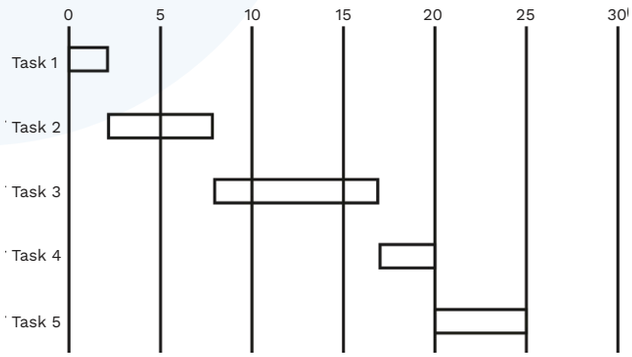


Fig. 4.2 Gantt Chart

Milestone Charts

- A milestone chart is an improvised version of Gantt chart. These charts give information about the accomplishment of sub-activities, as the milestones are the key events of the important activities depicted by a bar.

- A milestone can be represented by an arrow, a square, or a circle on the bar of that particular activity.
- It has been observed that when a particular activity represented by a bar is very long, the details of sub-activities will be lacking. If this activity is broken down into a number of sub-activities or key events, each of these can be recognised during the progress of that activity and through which control can be achieved.
- In the milestone chart, relationships between two activities are not revealed.

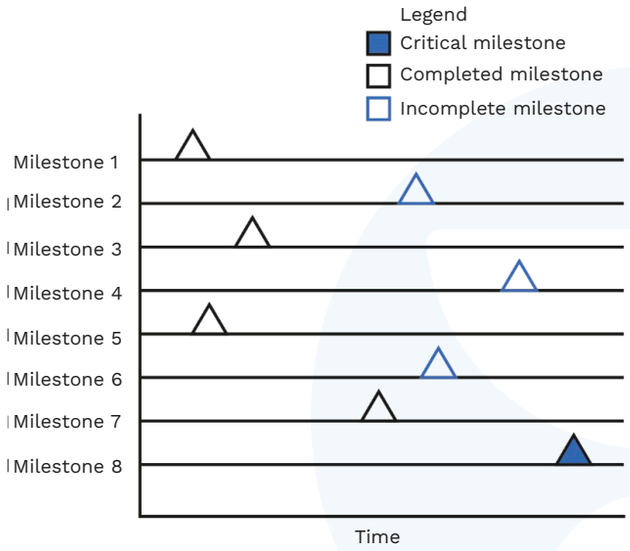


Fig. 4.3 Milestone Chart

Benefits of Network Method as Compared to Gantt and Milestone Charts

- In the network method, the correlation between events and activities of the project is distinctly evident.
- There is better control over the project with the use of this technique as there is a clear interrelation display between the sub-activities.
- As opposed to the Gantt chart, the network method is beneficial when the complexity of the project is immense and the number of activities is also high.
- It denotes the time interval between two activities during which a project's schedule can be rescheduled.
- In contrast to the Gantt chart, it includes the provision of time uncertainty. Hence, this method may be useful for projects related to research and development.
- This method serves as the foundation for all planning and forecasting activities, and assisting the management in determining the optimal allocation of resources to meet time and cost objectives.
- They increase visibility and empower the management to manage 'one-of-a-kind' initiatives.
- They assist the management in evaluating alternatives by addressing issues, such as how delays will affect project completion, where slack exists between elements, and which elements are critical to meet the project's completion date.
- They establish a framework for acquiring facts necessary for decision-making.
- They employ a technique known as time network analysis to ascertain manpower, material, and capital requirements, as well as, to monitor progress.
- They establish the fundamental framework for data reporting. They demonstrate the interdependence of activities.
- They enable 'what if' scenarios to be explored. They assist with risk analysis during scheduling.

Line of Balance Technique

- The line of balance (LOB) technique is an improvised version of milestone chart. A linear-shaped or 'S'-shaped curve depicts the graphical plan of action for project execution. These graphs, representing the work cycles, are referred to as 'cyclographs' or 'cyclograms.'
- On the horizontal or X-axis of a cyclograph, the unit of time in days or weeks is shown, whereas on the vertical or Y-axis, the total number of matching work units of a project is shown.
- The unit of time represented on the horizontal axis can additionally be split into weeks or months after evaluating the availability of operating days for each working month.

Network Method of Scheduling

It is an outcome of advancements in milestone charts and is referred to by several names, such as CPM, Programme Evaluation and Review Technique (PERT), LESS, UNETICS, SCANS, and TOPS. The written methods of scheduling are established from the following elementary systems of networks.

- PERT
- CPM

Network Diagram

- It is the logical and sequential flow of a diagram composed of activities and events.
- Network diagrams are of two types.
  - Activity-on-Arrow network (A-O-A)
  - Activity-on-Node network (A-O-N)

Elements of a Network Event

- It denotes the point of time or accomplishment occurring at a moment and is used to denote the starting and ending point of an activity.
- The event neither consumes any time nor resources for its completion.
- Some of the shapes that are globally used for representing an event are depicted below. In a network, events are the ones depicted by nodes. It may have any of the following shapes. The most commonly adopted shape for events is circular.



Fig. 4.4

- The numbering of the events follows the Fulkerson's rule (tail event number must be less than head event number, or succeeding event has a higher number than preceding event).

Starting or Tail Event

The starting or tail event marks the commencement of an activity. In a special case, where it is the first and foremost event of a project, it is also referred to 'as the starting or initial event'. This event only has an outgoing arrow.

Head or Final Event

As the tail event marks the beginning of an activity, the head event marks the termination of an activity. In a special case, where it is the last event in a project, it is known as the 'ending or finish event.'

Dual Role Event

A dual role event is one that has both incoming and outgoing arrows. Leaving the first and the terminating event, all events of a project are characterised as dual role events.

Successor Event

A successor event refers to an event that immediately follows a previous event without any intervention. Hence, an event that comes immediately after a particular event is referred to as the successor event.

Predecessor Event

A predecessor event is defined an event that comes before a particular event. Hence, such an event will be referred to as predecessor event for the referred event, for example, in the figure given below, event 1 is the tail event, event 3 is the head event, and event 2 is the dual role event.

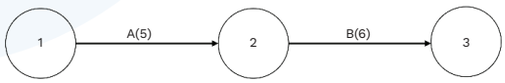


Fig. 4.5

Merge Event

A merge event is one when two or more activities turn up from an event.

Burst Event

A burst event is one when more than one activity leaves an event.

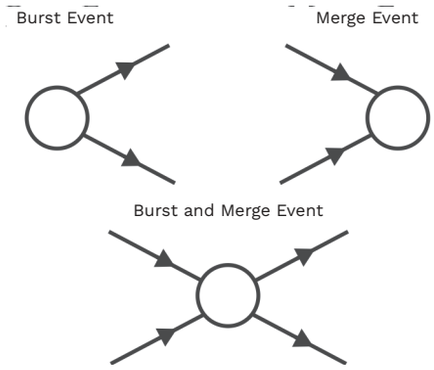


Fig. 4.6

Activity

- A project can be broken down into various jobs in the form of tasks necessary for its completion. It is a recognisable part of a project, which consumes time and resources for completion, and may involve physical and mental work.
- These activities can be classified into three categories known as parallel, successor, and predecessor activities.
- Parallel activities are the ones that take place at the same time. In Fig. 4.7, parallel activities are denoted by A, B, and C.
- Similarly, predecessor activities are the ones that are carried out before an activity or precede that activity. In Fig. 4.7, activity A is the predecessor to activity D.
- The third category is the successor activity. Successor activities are those that are carried out after a particular activity, and hence, succeed that activity. In Fig. 4.7, activity F is the successor to activity C.

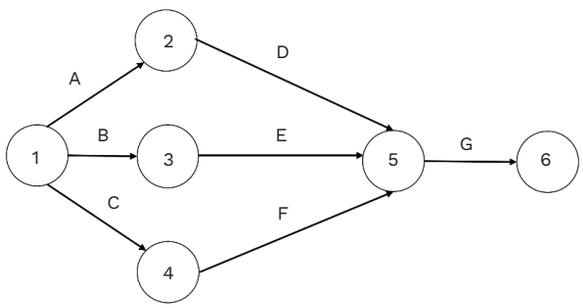


Fig 4.7

- The difference between A-O-A and A-O-N network is that in A-O-A network. The activity is shown by arrows between the events, whereas in A-O-N network, the activity is depicted by nodes.



Fig. 4.8

Dummy

- An activity that is used to show logic, dependency, or relationship between one activity over the other, but does not consume any time or resource for completion, is called a dummy activity. Unnecessary dummies should be avoided.
- Dummy activity should only be used when necessary. But there is no restriction on the number of dummy activities that may be used.

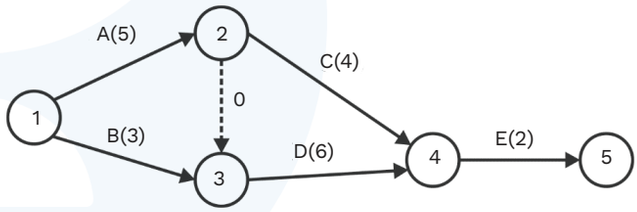


Fig 4.9

- The length and direction of the arrow are indicative only, and time flows from left to right in the network diagram. There should be no looping and dangling in the diagram, for example, the dummy activity shown by a dashed line in Fig. 4.9 between nodes 2 and 3 indicates that activity D can only start once activities A and B are completed.

Rules for Network Construction

- It is a graphical representation of a logical sequence, in which different activities are interrelated while completing a project.

- If the number of activities in a project is 'n', then the number of arrows in the network will also be '*n*.'

  n activities in the project = *n* arrows in the network
- If the activity is started, then it must be completed, i.e., the activity must have two events.



Fig. 4.10

- An activity can only be started when all activities prior to it are completed. No two or more activities may have the same head and tail events.
- Fig 4.11 shows that the fifth event can occur only after the completion of events C, D, and E.

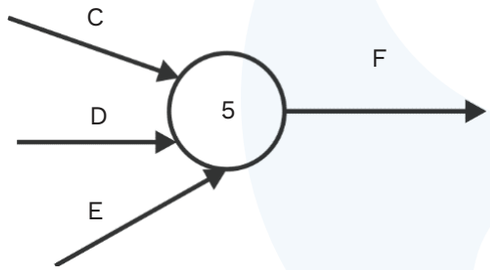
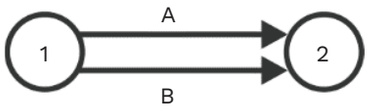


Fig 4.11

- As stated earlier, two or more activities cannot have the same starting and termination point, i.e., they cannot have the same head and tail event.



Wrong

Fig. 4.12

- The length of the arrows is just an indicator and does not depict any dimension. Also, it is recommended to use a straight arrow that connects the two nodes.
- As shown in the following figure, the arrows should not cut or cross each other. But if a situation arises, where it is impossible to draw the network without crossing the arrows, then an arrow can be bridged over the other.

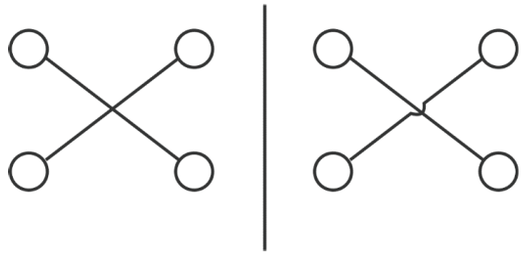


Fig. 4.13

Fulkerson's Rule for Numbering Events

- The numbering of the events follows the Fulkerson's rule (i.e., the tail event number must be less than the head event number, or the succeeding event must have a higher number than the preceding event).
- The Fulkerson's rule helps in avoiding looping errors (loops should not be formed).

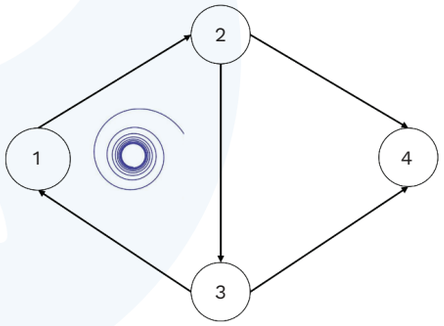


Fig. 4.14 Looping

- Dangling: When an activity other than the final activity of a project does not have any successor activity, then this situation is called dangling. Such activity should be connected directly to the last event of the network to avoid dangling errors.

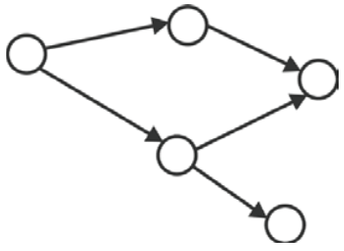


Fig. 4.15

Network Scheduling Techniques

- Programme Evaluation and Review Technique (PERT)
- Critical Path Method (CPM)
- Graphical Evaluation and Review Techniques (GERT)
- Precedence Diagram Method (PDM)

Critical Path Method

- The critical path method is activity oriented and is associated with deterministic activities.
- It is based on a single time to complete an activity and is used for repetitive jobs where one has prior experience in handling similar projects.
- It gives importance to cost analysis and crashing is done to minimise the cost of the CPM project. This technique is mainly used for construction projects.

Critical Path

It is the maximum time-consuming path from the first event to the last in a network diagram.

- The time taken along the critical path is termed as 'expected project completion'. The activities along the critical path are called 'critical activity', and are represented by double line arrows.

Time Computation of Events

Earliest Expected Occurrence Time

- The time at which an event is expected to occur the earliest. It is generally denoted by E.

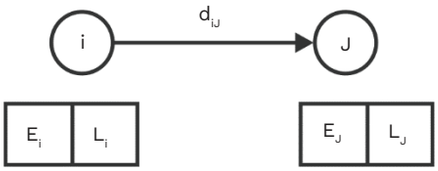


Fig. 4.16

- In the above figure, Ei is the basis, and Ej is calculated as:

  Ej = Ei + Duration of the activity (Dij)

  Here, Ei is the EOT of event i, Ej is the earliest expected occurrence time (EOT) of event j, and Dij is the duration of the activity I – j.

- The forward pass technique is used to calculate EOT.

  Starting node/event = First event
  Ending event = Last event in the network

- Starting event value Ei = 0, Ending event Ej = Project completion time
- When there is more than one path, then

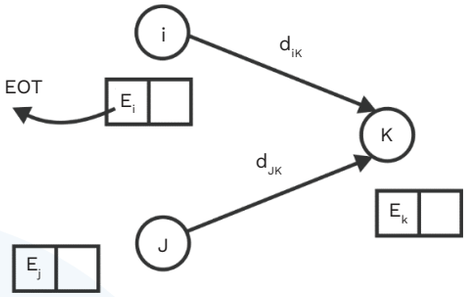


Fig. 4.17

Ek = Maximum value of (Ei + Dik, Ej + Djk)

Latest Occurrence Time

It denotes the latest allowable time at which an event must occur to keep the project on schedule. It is generally denoted by L

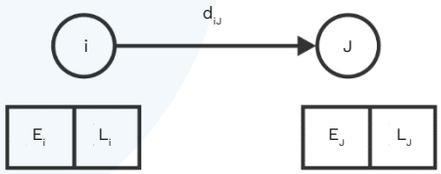


Fig. 4.18

- Lj is the basis and Li is calculated as

  Li = Lj – Duration of the activity (Dij)

  Here, Li is the latest occurrence time (LOT) of event i, Lj is the LOT of event j, and Dij is the duration of the activity I – j.

- The backward pass technique is used to calculate LOT.

  Starting node/event = Last event in the network
  Ending event = First event

- Starting event value Li = 0, ending event Lj = project completion time
- When there is more than one path, then

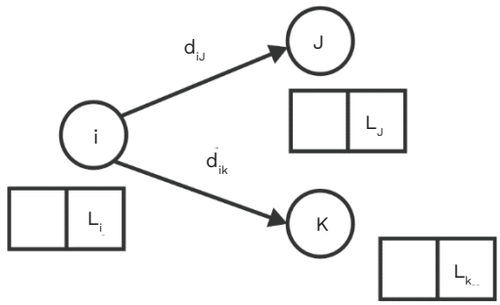


Fig. 4.19

Li = Minimum value of (Lj – Dij, Lk – Dik)

Slack or Event Float

Slack denotes the amount of time by which a particular event can be delayed without delaying the project schedule. It denotes the surplus time available without delaying the project.

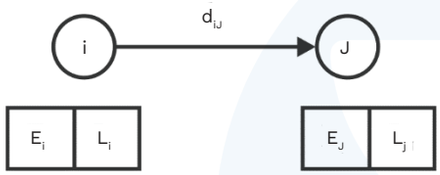


Fig. 4.20

Slack for the event j = Lj – Ej

Slack for the event i = Li – Ei

Float

It is the range within which the start or finish time of activity may fluctuate without affecting the project completion time. Floats are of the following types.

Total Float

- It denotes the amount of time by which an activity may be delayed without delaying the project schedule. Hence, it is the extra time available for an activity without delaying the project schedule.
- If the total float (TF) is negative, it implies that the resources are not sufficient and the activity may not be completed on time.
- If the TF value is zero, it implies that the resources are sufficient to complete the activity on time.
- If the TF value is positive, this means that the resources are surplus and may be allocated for other activities.

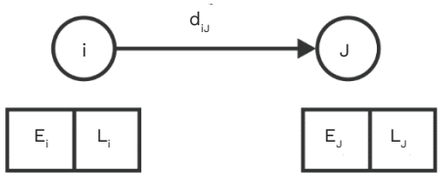


Fig. 4.21

- The expression for TF is equal to the maximum available time – duration of the activity.

  Total float (TF) = (Lj – Ei) – Dij
- The TF of an activity affects the TF of succeeding and preceding activities.

Free Float

- It is that part of the total float, which can be used without affecting the float of the succeeding activity.
- It is the extra time by which an activity may be delayed so that the succeeding activity may be started at the earliest start time.
- Free float (FF) is given by – slack at head event

  FF = (Lj – Ei) – Dij – (Lj – Ej)

  = Ej – Ei – Dij

Independent Float

- It is a permissible delay period, in which the activity may be delayed without affecting the occurrence of preceding or succeeding activity. It is the amount of time that can be used without affecting either the head or tail event.
- Independent float (IF) is given by

  Total float – slack at head event – slack at tail event

  Independent Float (IF)

  = Free Float – slack at tail event

  = Ej – Ei – Dij – (Li – Ei)

  = Ej – Li – Dij

Interference Float

Interference float (InF) is much like head event slack.

Interference float (InF) = total float – free float

Critical Path in Critical Path Method

If the total float is zero or minimum along a path, then that path is referred to as critical path. On the critical path, all activities are critical. There

is a possibility that there can be more than one critical path.

- If the total float for an activity is greater than zero, then that activity is known as a sub-critical activity. The path joining the sub-critical activities is known as the sub-critical path.
- If the total float for an activity is less than zero, then that activity is known as a super critical activity. The path joining the super critical activities is known as the super critical path.

Solved Examples

1. Construct the network diagram. Find the critical path, and slack on non-critical events.

Activity	Predecessor	Duration (days)
A	–	2
B	–	2
C	A	7
D	A	12
E	B	10
F	D, E	3
G	C, F	4

Table 4.1

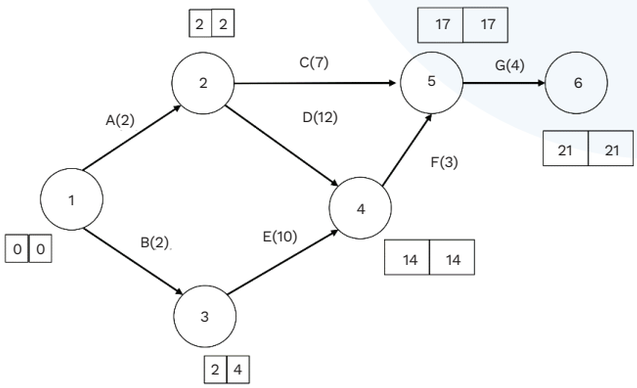


Fig. 4.22

Critical path: 1 – 2 – 4 – 5 – 6 = 21 days
Slack on event 3 is 2

2. Construct the network diagram. Find the critical path, and slack on non-critical events.

Activity	Predecessor	Duration (days)
P	–	3
Q	–	4
R	P	5
S	Q	5
T	R, S	7
U	R, S	5
V	T	2
W	U	10

Table 4.2

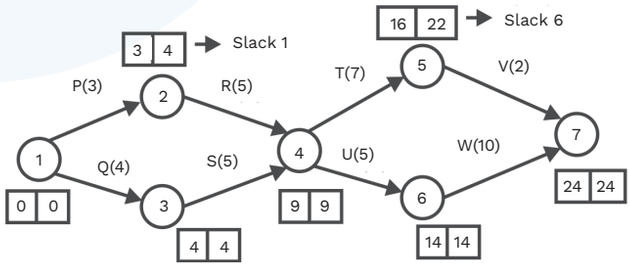


Fig. 4.23

Critical path: 1 – 3 – 4 – 6 – 7 = 24 days

3. Construct the network diagram. Find the critical path, and slack on non-critical events.

Activity	Predecessor	Duration (days)
A	–	3
B	A	4
C	A	5
D	A	4
E	B	2
F	D	9
G	C, E	6
H	F, G	2

Table 4.3

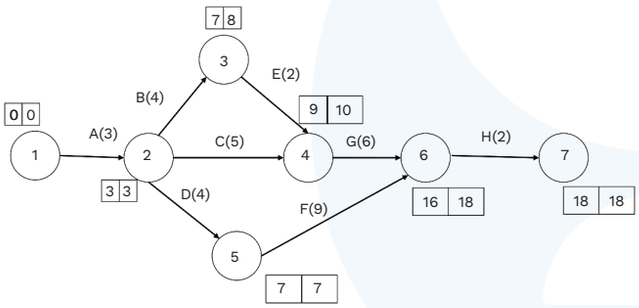


Fig. 4.24

Critical path: 1 – 2 – 5 – 6 – 7 = 18 days, slack on event 3 is 1, and event 6 is 2

4. Construct the network diagram. Find the critical path, and slack on non-critical events.

Job	Predecessor	Duration (days)
A	–	3
B	–	4
C	A	6
D	A	8
E	B, C	6
F	B, C	4
G	F	2
H	F	8
I	D, E, G	6

Table 4.4

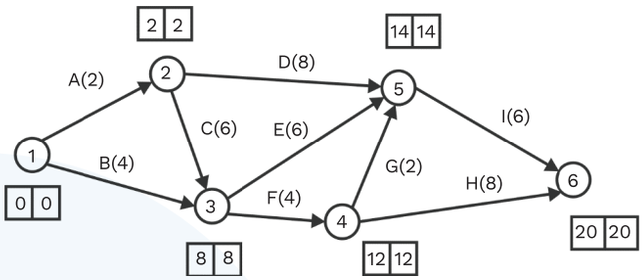


Fig. 4.25

Critical path: 1 – 2 – 3 – 4 – 5 – 6 = 20 days
1 – 2 – 3 – 4 – 6 = 20 days

5. Construct the network diagram. Find the critical path, and slack on non-critical events.

Job	Predecessor	Duration (days)
A	–	5
B	–	3
C	A	4
D	A, B	6
E	C, B	2

Table 4.5

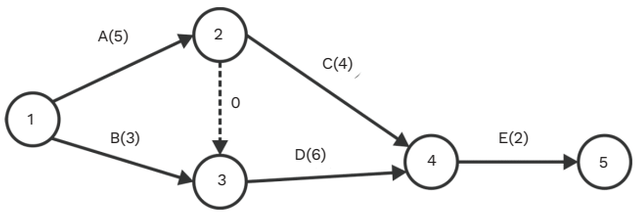


Fig. 4.26

Critical path: 1 – 2 – 3 – 4 – 5 = 13 days

6. Construct the network diagram.

Activity	Predecessor
A	–
B	–
C	–
D	A
E	A, B
F	A, B, C
G	D, E, F

Table 4.6

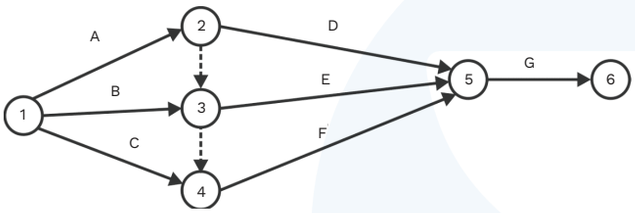


Fig. 4.27

7. Construct the network diagram. Find the critical path, slack on non-critical events, and total float of activities E and F.

Job	Predecessor	Duration (days)
A	–	2
B	–	4
C	A	2
D	B	3
E	C	2
F	C	4
G	D, E	5

Table 4.7

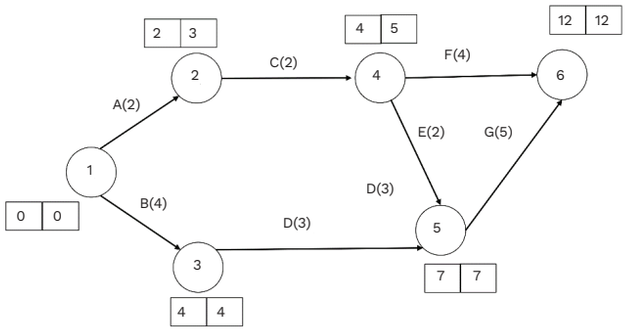


Fig. 4.28

Critical path: 1 – 3 – 5 – 6 = 12 days
Slack on event 2 is 1 and event 4 is 1
Total float of activity E (Dij – 2)

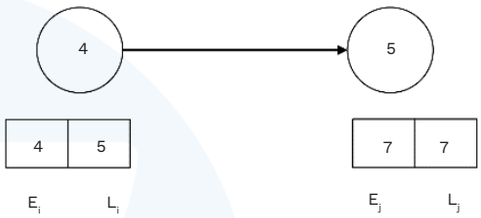


Fig. 4.29

Total float (TF) = (Lj – Ei) – Dij
(7 – 4) – 2 = 1
Total float of activity F (Dij – 4)

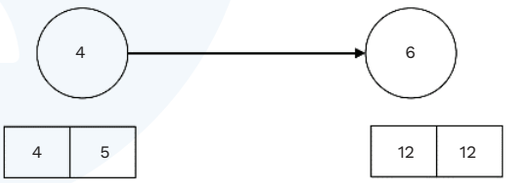


Fig. 4.30

Total float (TF) = (Lj – Ei) – Dij
(12 – 4) – 4 = 4

8. Construct the network diagram. Find the critical path, and free float of Activity 4–6.

Activity	Duration
1–2	3
1–3	2
1–4	2

Activity	Duration
2–5	2
3–6	5
4–6	4
5–6	3
6–7	5

Table 4.8

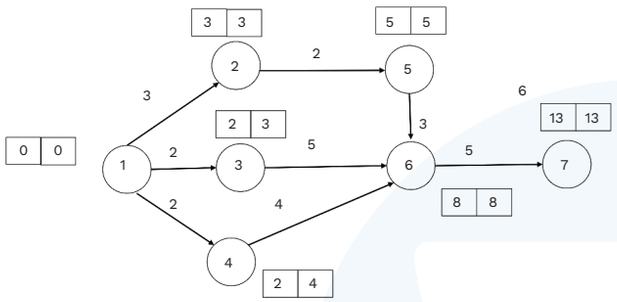


Fig. 4.31

Critical path: 1 – 2 – 5 – 6 – 7 = 13 days
Free float of activity 4 – 6

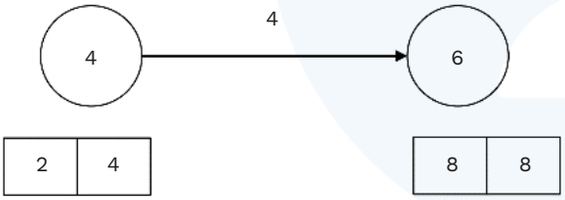


Fig. 4.32

Free float = Ej – Ei – Dij
8 – 2 – 4 = 2

9. Construct the network diagram. Find the critical path. Find the relationship between Total Float, Free Float, Independent Float, and Interference Float of Activity D.

Activity	Predecessor	Duration
A	–	3
B	–	5
C	A, B	5

Activity	Predecessor	Duration
D	A	4
E	C, D	4
F	A	4
G	A, B	6

Table 4.9

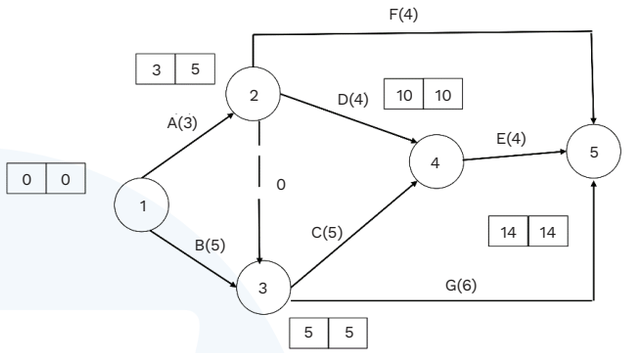


Fig. 4.33

Critical path: 1 – 3 – 4 – 5 = 14 days

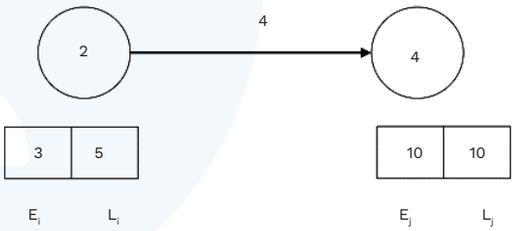


Fig. 4.34

Total float (TF) = (Lj – Ei) – Dij
(10 – 3) – 4 = 3
Free float = Ej – Ei – Dij
10 – 3 – 4 = 3
Independent float (IF) = Ej – Li – Dij
10 – 5 – 4 = 1
Interference float (InF) = Total float – Free float
3 – 3 = 0
Total float = free float > independent float > interference float

10. In the network shown in the following figure, find the total float for Activity 3–5 (the numbers on the arrow show the duration of the activity).

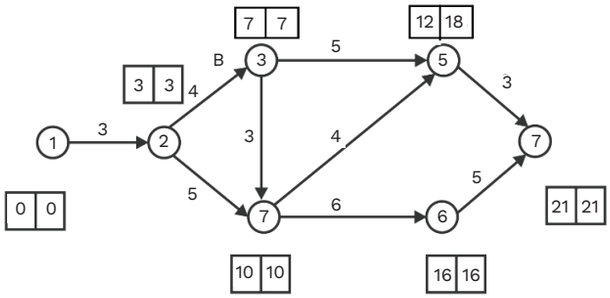


Fig. 4.35

Critical path: 1 – 2 – 3 – 4 – 6 – 7 = 21

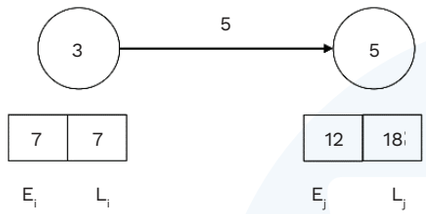


Fig. 4.36

Total float (TF) = (Lj – Ei) – Dij
(18 – 7) – 5 = 6

Programme Evaluation and Review Technique

- Programme Evaluation and Review Technique (PERT) was developed by the United States Navy during the late 1950s to accelerate the development of the Polaris Fleet Ballistic Missile
- PERT is event-oriented and is based on a three-time estimate to complete an activity.
- It is used where the time required to complete various activities is not certain. It usually does not consider cost analysis and is used mainly for projects involving research and development.
- An activity has uncertain time phases and the expected duration of activity follows a random distribution, also called beta (β)-distribution.
- Overall project duration follows normal distribution.

Time Estimates Used in PERT

- Optimistic time (To): It is the minimum time required to complete an activity when everything goes according to the plan.
- Pessimistic time (Tp): It is the maximum time required to complete an activity when everything goes against the plan.
- Most likely time (Tm): It is the time required to complete an activity when executed under normal working conditions.

Mean, or Expected, or Average Time

- A fundamental assumption in PERT is that the three-time estimates form the end point of the distribution curve and activity is assumed to follow the β-distribution.
- It is also assumed that the probability of completing the activity in time 'a' and 'b' is equal and that of completing the activity in time 'm' is four times that of either 'a' or 'b.'
- The average or expected time to complete an activity is given by:

$$T_e = \frac{a + 4m + b}{6} = \frac{T_0 + 4T_m + T_p}{6}$$

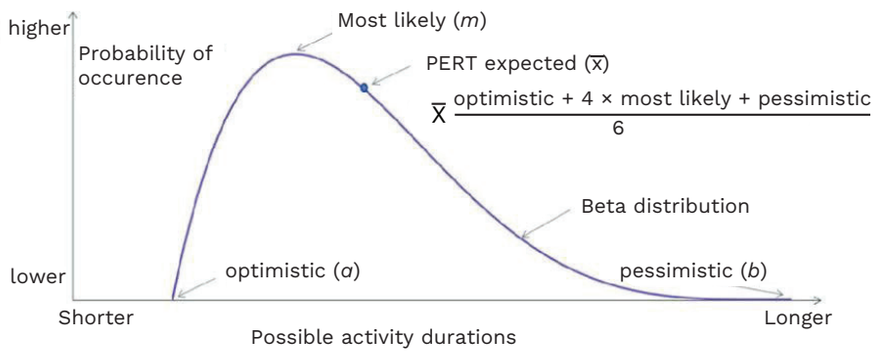


Fig. 4.37

Standard Deviation of an Activity (σ)

- This is the measurement of uncertainty, which is approximately one-sixth of the time range, i.e.,

$$\sigma = \frac{T_p - T_0}{6}$$

- It can be seen above that σ is affected by relative distance from the most optimistic estimate to the most pessimistic estimate.
- Therefore, a wide range of time estimates represents greater uncertainty.

Variance of an Activity (σ^2)

- The square of standard deviation is the variance of an activity.
- Variance gives the measure of uncertainty of activity completion.
- The higher the value of variance, the greater the uncertainty will be.

Central Limit Theorem

- The theorem states that a project consists of a large number of activities, where each activity has its own meantime (Te), standard deviation (σ), variance (σ^2), and β-distribution curve. The distribution of time for the project as a whole will approximately be a normal distribution, i.e., mean time, or expected time of a project is Te = Te1 + Te2+... along the critical path, and the variance is:

$$\sigma^2 = \sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \ldots$$

- Hence, the standard deviation of the project as a whole along the critical path will be:

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \ldots}$$

Probability of a Project to be Completed in Scheduled Time (Ts)

- The probability of project completion in expected time (Te) is just 50 per cent. The probability of completion increases or decreases with an increase or decrease in scheduled time, respectively.
- The probability distribution curve for the entire project is a normal distribution curve.
- Probability can be calculated by using the probability factor (Z) which is defined as:

$$Z = \frac{T_s - T_e}{\sigma}$$

Notes

- At Z = 0, Ts = Te, and probability = 50 per cent
- Z < 0, Ts < Te probability < 50 per cent and vice versa
- For an intermediate value of Z, interpolation is required.
- The time and probability distribution for a project as a whole is normal distribution. The area under the curve is equal to unity.
- A total of 68 per cent of the area lies within the 'Te – σ' and ' Te + σ' range. This is called the probable range.
- The area bound by Te + 2 σ and Te – 2 σ is about 95 per cent.
- The area bound by Te – 3 σ and Te + 3 σ is about 99.7 per cent, which is approximately equal to an area bound by To and Tp, which is the maximum probable range.

$$\sigma = \frac{T_p - T_0}{6}$$

Solved Examples

11. If To = 2, Tm = 5, and Tp = 8, then find the expected time (Te).

$$T_e = \frac{T_0 + 4T_m + T_p}{6}$$

$$T_e = \frac{2 + 4 \times 5 + 8}{6}$$

Te = 5

When To, Tm, and Tp are in arithmetic progression, then Te = Tm

12. Find the critical path, standard deviation. variance of critical path, and project duration.

Task	Predecessor	Mean Duration (t_E)	Variance
a	–	30	25
b	a	40	64
c	b	60	81
d	c	25	9
e	b, c	45	36
f	d, e	20	9

Table 4.10

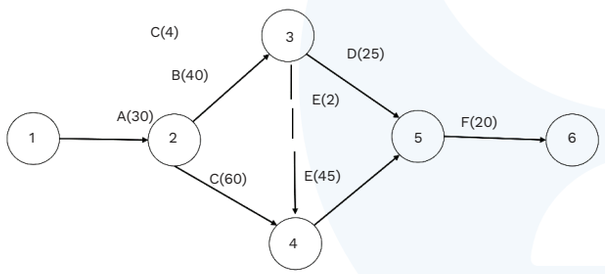


Fig. 4.38

Critical path: 1 – 2 – 4 – 5 – 6 = 155

$\sigma_{cp} = \sqrt{V_{c_P}} = \sqrt{25 + 81 + 36 + 9} = \sqrt{151}$

Project duration = $155 \pm 3\sqrt{151}$

13. Find the critical path, standard deviation. variance of critical path, and project duration.

Activity	t_0	t_m	t_p	T_E
1–2	1	2	3	2
1–3	5	6	7	6
1–4	3	5	7	5
2–5	5	7	9	7
3–5	2	4	6	4
5–6	4	5	6	5
4–7	4	6	8	6
6–7	2	3	4	3

Table 4.11

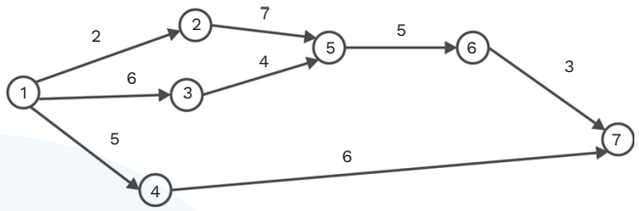


Fig. 4.39

Critical path: 1 – 3 – 5 – 6 – 7 = 18

$\sigma_{cp} = \sqrt{V_{c_P}} = \sqrt{\left(\frac{7-5}{6}\right)^2 + \left(\frac{6-2}{6}\right)^2 + \left(\frac{6-4}{6}\right)^2 + \left(\frac{4-2}{6}\right)^2}$

$= 0.87$

Project duration $= 18 \pm 3 \times 0.87$

14. Find the critical path, standard deviation. variance of critical path, and project duration.

Activity	Predecessor	t_0	t_p	t_m	T_E
A	–	1	5	3	3
B	–	2	4	3	3
C	–	3	5	4	4
D	A	2	10	9	8
E	C	4	6	5	5
F	B, D, E	5	3	6	7
G	A	2	6	4	4
H	G, F	0	6	3	3

Table 4.12

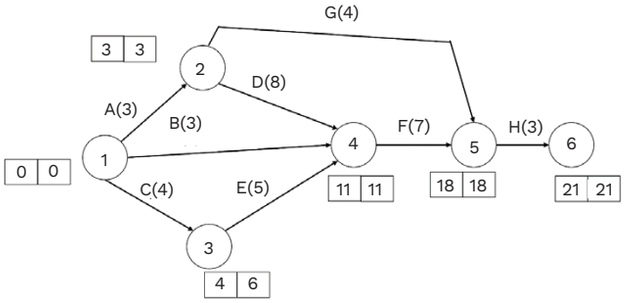


Fig. 4.40

Critical path: 1 – 2 – 4 – 5 – 6 = 21

$$\sigma_{cp} = \sqrt{V_{c_P}} = \sqrt{\left(\frac{5-1}{6}\right)^2 + \left(\frac{10-2}{6}\right)^2 + \left(\frac{13-5}{6}\right)^2 + \left(\frac{6-0}{6}\right)^2}$$

$= 2.23$

Project duration = $21 \pm 3 \times 2.23$

15. Find the critical path, standard deviation. variance of critical path, and project duration.

Activity	$t_E = t_m$	t_0, t_m, t_p
1–2	2	1, 2, 3
1–3	6	(5, 6, 7) (4, 6, 8)
1–4	5	
2–5	7	
3–5	4	(2, 4, 6) (3, 4, 5)
5–6	5	(4, 5, 6) (2, 5, 5)
4–7	6	
6–7	3	(2, 3, 4)

Table 4.13

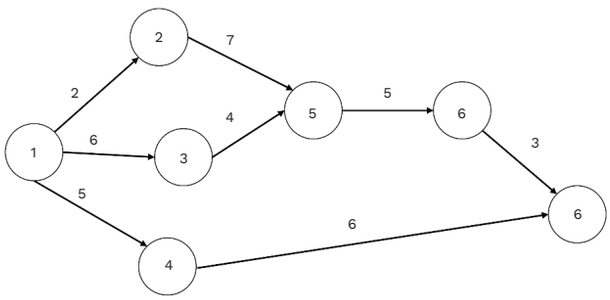


Fig. 4.41

Critical path: 1 – 3 – 4 – 5 – 6 – 7 = 18

$$\sigma = \sqrt{V_{c_P}} = \sqrt{\left(\frac{7-5}{6}\right)^2 + \left(\frac{5-3}{6}\right)^2 + \left(\frac{8-2}{6}\right)^2 + \left(\frac{4-2}{6}\right)^2}$$

$= 1.13$

Project duration $= 18 \pm 3 \times 1.13$

16. Find the relation between total float, free float, independent float, and interference float of the activity.

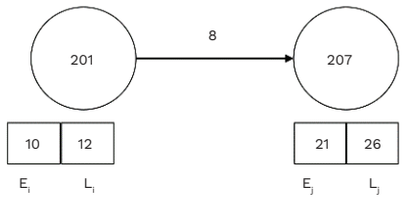


Fig. 4.42

Total float (TF) = (Lj – Ei) – Dij

(26 – 10) – 8 = 8

Free float (FF) = Ej – Ei – Dij

21 – 10 – 8 = 3

Independent float (IF) = Ej – Li – Dij

21 – 12 – 8 = 1

Interference float (InF) = Total float – free float

8 – 3 = 5

Total float > interference float > free float >

17. What is the proportional variation and 'range' of project duration for the network as shown with the indicated probabilistic a (optimistics time), m (most likely time), and b (pessimistic time) durations of the respective activities?

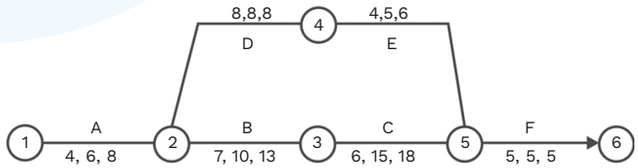


Fig. 4.43

(A) $\frac{1}{15}, 28$ to 42

(B) $\frac{1}{15}, 30$ to 45

(C) $\frac{1}{12}, 28$ to 42

(D) $\frac{1}{12}, 30$ to 45

Solution: (A)

Activity	T_E	σ
A	6	2/3
B	10	1
C	14	2
D	B	0
E	5	1/3
F	5	0

Table 4.14

Critical path: A – B – C – F = 35

$$\sigma_{cp} = \sqrt{v_{c_p}} = \sqrt{\left(\frac{2}{3}\right)^2 + (1)^2 + (2)^2} = 2.33$$

Project duration = $35 \pm 3 \times 2.33$

28 – 42

Proportional variation = Standard deviation/ project duration = 2.33/35

18. Activity A has a pessimistic (P) estimate of 36 days, a most likely (ML) estimate of 21 days, and an optimistic (O) estimate of 6 days. What is the probability that activity A will be completed in 16 to 26 days?

(A) 55.70 per cent
(B) 68.26 per cent
(C) 95.46 per cent
(D) 99.73 per cent

Solution: (B)

$$T_e = \frac{T_0 + 4T_m + T_p}{6}$$

$$T_e = \frac{6 + 4 \times 21 + 36}{6}$$

T_e = 21 days

$$\sigma = \frac{T_p - T_0}{6}$$

$$\sigma = \frac{36 - 6}{6} = 5 \text{ days}$$

21 ± 5

19. The optimistic and pessimistic time estimates for the PERT network of a project are shown in the given figure. What will be the expected duration of the project?

(A) 14 days
(B) 17 days
(C) 17.83 days
(D) 18.67 days

Solution: (C)

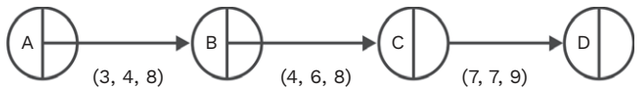


Fig. 4.44

$$T_e = \frac{3 + 4 \times 4 + 8}{6} = 4.5$$

$$T_e = \frac{4 + 4 \times 6 + 8}{6} = 6$$

$$T_e = \frac{7 + 4 \times 7 + 9}{6} = 7.33$$

4.5 + 6 + 7.33 = 17.83

Comparison between PERT and CPM

PERT	CPM
It is an event-oriented network.	It is an activity-oriented network.
It is associated with probabilistic activities.	It is associated with deterministic activities.
It is based on a three-time estimate to complete an activity and is used where the time required to complete various activities is not certain.	It is based on a single time to complete an activity and is used for repetitive jobs where one has prior experience in handling similar projects.
It usually does not consider cost analysis and is mainly used for research and development projects.	It gives importance to cost analysis and crashing is done to minimise the cost of the CPM project. It is mainly used for construction projects.

Table 4.15

Advantages of Network Method over Bar Chart and Milestone Chart

- The relationship between the project's activities and events are clearly apparent.
- The project can be considered as a whole, with all of its sub-activities being clearly connected. It helps in controlling the project.
- The network method is advantageous for extremely complex projects with a high volume of activities.
- It denotes the time interval between two activities during which a project may be rescheduled.
- Since it accounts for time uncertainty, it is also applicable to research and development projects.
- It serves as the foundation for all planning and forecasting, assisting management in determining the optimal allocation of resources to meet time and cost objectives.
- It increases visibility and empower the management to manage 'one-of-a-kind' initiatives.
- It assists the management in evaluating alternatives by addressing issues, such as how time delays will affect project completion, where slack exists between elements, and which elements are critical to meeting the project's completion date.
- It establishes a framework for acquiring facts necessary for decision-making.
- It employs a technique known as time network analysis to ascertain manpower, material, and capital requirements, as well as, to monitor progress.
- It establishes the fundamental framework for data reporting. They demonstrate the interdependence of activities.
- It enables 'what if' scenarios to be explored. It assists with risk analysis during scheduling.

Practice Exercise

1. The amount of time that an activity can be delayed from its early start without delaying the project end date is called _____.
(A) negative float
(B) free float
(C) total float
(D) independent float

2. Which of the following statements regarding the critical path method is not true?
(A) It modifies the schedule to account for limited resources.
(B) The first step is to use conservative estimates for activity durations.
(C) Duration buffers are added to the critical path.
(D) It focuses on managing buffer activity durations.

3. Scheduling helps in _____.
(A) planning for the project
(B) conducting financial control of the project
(C) preparing the estimate for the project
(D) carrying out the project in an orderly and effective manner

4. To control the schedule, a project manager is re-analysing the project to predict project duration. She does this by analysing the sequence of activities with the least amount of scheduling flexibility. What technique is she using?
(A) Critical path method
(B) Flow chart
(C) Precedence diagramming
(D) Work breakdown structure

5. The WBS for the project represents _____.
(A) all tangible items that must be delivered to the client
(B) all work that must be completed for the project
(C) the work that must be performed by the project team
(D) all activities of the project

6. Consider the following statements in the work breakdown structure.
I. It is a graphical representation of an entire programme.

II. The top–down approach to planning is adopted.
III. The down–top approach to planning is adopted.
IV. It is suitable for complex projects.

Which of these statements is/are correct?

(A) I only
(B) II and IV
(C) III and IV
(D) IV only

7. Consider the following statements.

I. In the work breakdown structure, top–down approach is adopted.
II. Bar-chart depicts interdependencies between activities.
III. Controlling can be better achieved in the milestone chart.

Which of these statements is/are correct?

(A) I and III only
(B) I and II only
(C) II and III only
(D) I, II, and III

8. Most statistical simulations of budgets, schedules, and resource allocations use which one of the following approaches?

(A) Programme Evaluation and Review Technique
(B) Decision tree analysis
(C) Present value analysis
(D) Monte Carlo analysis

9. A network-based technique that allows for branching and looping is _____.

(A) Programme Evaluation and Review Technique
(B) Graphical Evaluation and Review Technique
(C) Project Data Management
(D) Application Data Management

10. A project manager is in the middle of the executing process of a large construction project when he discovers that the time needed to complete the project would be longer than the time available. What is the best thing to do?

(A) Cut product scope
(B) Meet with the management and inform that the deadline cannot be met
(C) Work overtime
(D) Determine the options for schedule compression and present the recommended option before the management

11. How are project milestone durations decided?

(A) It depends on the type and nature of the project.
(B) It is decided by the sponsor.
(C) The manager and the team decide this duration.
(D) It is zero because it is only a checkpoint on the project.

12. Milestone schedules are primarily meant for use by the _____.

(A) vendors working on a project
(B) executive management
(C) project team members
(D) all stakeholders in the project

13. Consider the following statements concerning PERT.

I. It consists of activities with uncertain time phases.
II. This evolved from the Gantt chart.
III. Total slack along the critical path is not zero.
IV. There can be more than one critical path in the PERT network.
V. It is similar to an electrical network.

Which of these statements is/are correct?

(A) I and II
(B) II and IV
(C) IV and I
(D) III and II

14. What does lag mean?

(A) The amount of time an activity can be delayed without delaying the project finish date
(B) The amount of time an activity can be delayed without delaying the early start date of its successor
(C) Waiting time for an activity
(D) The product of a forward and backward pass

15. Which of these is one of the main reasons for creating a schedule for a project?
(A) For the team to know when to plan vacations during the project life cycle
(B) To know the end date for the project and to fix the schedule baseline
(C) To know the critical path items to assign resources
(D) To hire the vendors for the project in a planned manner

16. While developing a project schedule, you want to define a distribution of probable results for each schedule activity and use that distribution to calculate another distribution of probable results for the entire project. Which is the most common technique to use?
(A) PERT
(B) Monte Carlo analysis
(C) Linear programming
(D) Concurrent engineering

17. Unlike bar charts, what do milestone charts show?
(A) Scheduled start or completion date of major deliverables, and key external interfaces
(B) Activity start and end dates for critical tasks
(C) Expected duration for the critical path
(D) Dependencies between complementary projects

18. Project managers should pay attention to critical and sub-critical activities while evaluating project time performance. One way to do this is to analyse 10 sub-critical paths in the order of ascending float. This approach is part of _____.
(A) variance analysis
(B) simulation
(C) earned value management
(D) trend analysis

19. The shortest time necessary to complete all activities in a network is called the _____.
(A) activity duration length
(B) critical path
(C) maximum slack path
(D) compression path

20. Several types of float are found in project networks. A float that is used by a particular activity and does 'not' affect the float in later activities is called _____ float.
(A) extra
(B) free
(C) total
(D) expected

21. In PERT analysis, the time estimates of activities and probability of their occurrence follow _____ distribution.
(A) beta
(B) gamma
(C) normal
(D) Poisson's

22. The work defined at the lowest level of the Work Breakdown Structure to estimate and manage time, risk, quality, and cost is called _____.
(A) requirement
(B) contract
(C) work package or activity
(D) cost account

23. The WBS for the project represents _____.
(A) all tangible items that must be delivered to the client
(B) all work that must be completed on the project
(C) the work that must be performed by the project team
(D) all activities of the project

24. The client changes the product specification. This will add only two weeks to the critical path. Which of the following is the best thing for the project manager to do?
(A) Compress the schedule to recover the two weeks.
(B) Cut scope to recover the two weeks.
(C) Analyse how the project will be impacted and relay that to the client before taking any action.
(D) Refuse to change the plans.

25. Consider the following statements.

I. A dummy activity is artificially introduced into a network when necessary.
II. A dummy activity consumes some time.
III. A dummy activity is represented by a dotted arrow.
IV. A dummy activity must necessarily be introduced in every network.

Which of the above statements are correct?

(A) I, II, and III
(B) I and III
(C) II, III, and IV
(D) I and II

26. Match List-I (activity type) with List-II (represented by) and select the correct answer.

List-I

A. Artificially introduced
B. Critical
C. Non-critical type
D. Dangler

List-II

I. A single thick arrow
II. A single arrow
III. An arrow emerging from an event and ending into an event, which is not a finishing event, and yet there is no emerging arrow from that event
IV. A dotted arrow

Codes:

	A	B	C	D
(A)	IV	I	II	III
(B)	II	III	IV	I
(C)	IV	III	II	I
(D)	II	I	IV	III

27. What is the time by which the completion of an activity may be delayed without affecting the start of subsequent activities called?

(A) Total Float
(B) Interfering float
(C) Independent Float
(D) Free Float

28. Slack time is associated with _____.

(A) a real activity
(B) an event
(C) both event and real activity
(D) dummy activity

29. The work breakdown structure 'control points' for the management of a project are the _____.

(A) milestones
(B) work packages
(C) activities
(D) constraints

30. Which of the following cannot be generally validated using a work breakdown structure?

(A) Schedule control
(B) Cost control
(C) Quality control
(D) Risk management

31. The area under a beta-distribution curve is divided into two halves by vertical ordinate through _____ time.

(A) expected
(B) optimistic
(C) most likely
(D) pessimistic

32. Consider the following statements of the network.

I. Only a one-time estimate is required for each activity.
II. Three-time estimates are required for each activity.
III. Time and cost are both controlling factors.
IV. It is built-up of an event-oriented diagram.

Which of the above statements are correctly applied to the CPM network?

(A) I and III
(B) I and II
(C) II and IV
(D) III and IV

33. In the PERT analysis, which one of the following is followed by the time estimates of activities and the probability of their occurrence?

(A) Normal distribution

(B) Poisson's distribution
(C) Beta distribution
(D) Binomial distribution

34. What does a higher standard deviation imply in cost analysis?
(A) Higher uncertainty
(B) Lower uncertainty
(C) Nothing to do with uncertainty
(D) Extra costs

35. What does the critical path in PERT represent?
I. The shortest path for the earliest completion of the project
II. The longest path of the network from the initial to the final event
Select the correct answer using the code(s) given below.
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

36. Which of the following is the correct sequence to analyse a project for implementation?
(A) Time-cost study, network, WBS, scheduling with resource allocation
(B) Network, time-cost study, scheduling with resource allocation, WBS
(C) WBS, network, scheduling with resource allocation, time-cost study
(D) WBS, time-cost study, network, scheduling with resource allocation

37. List the following processes in correct sequence, from the earliest to the latest, in project implementation planning.
I. Project duration
II. Resource histogram
III. Standardised input/performance for each activity, including alternatives
IV. Work Breakdown Structure
V. The optimisation of resource(s), considering the constraints
VI. Activities and relationship between them
Select the correct answer using the codes given below.
(A) II, I, III, V, VI, and IV
(B) II, VI, III, V, I, and IV
(C) IV, I, III, V, VI, and II
(D) IV, VI, III, V, I, and II

38. Assertion (A): Critical path can pass through dummy arrows also.
Reason (R): Necessary additional nodes are introduced to ensure the uniqueness of I–j notation for activities.
(A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

39. Which aspects should be considered in planning for a project?
(A) Past, present, and future
(B) Design considerations
(C) Financial considerations
(D) Manpower availability

40. Which of these is the main objective of planning while executing a project?
(A) Most economically and on time
(B) Only timely completion
(C) Only the most economically
(D) Economically and safely

41. Planning is important as it is advantageous to _____.
(A) contractors
(B) architects and engineers
(C) clients
(D) All of the above

42. Which of these is one of the main disadvantages of bar chart?
(A) Progress of the work cannot be monitored.
(B) It does not show the interdependence of activities.
(C) The time schedule is not shown properly.
(D) The financial aspect is not shown.

43. Work Breakdown Structure helps in _____.
(A) identifying the activities
(B) identifying the functional elements of a project and their interrelationship
(C) breaking the project into several elements
(D) breaking the project into sub-systems

44. Precedence networks are best suited when _____.
(A) there are a large number of overlapping activities
(B) there are many critical activities
(C) the project is to be executed efficiently
(D) A-O-A networks are to be avoided

45. The method, usually adopted in arriving at the duration of the project is _____.
(A) Graphical Evaluation and Review Technique (GERT)
(B) Critical Path Method (CPM)
(C) Programme Evaluation and Review Technique (PERT)
(D) Decision networks

46. Which of these networks takes care of scheduling implications of alternative jobs that are considered before the final choice is made?
(A) CPM
(B) PERT
(C) GERT
(D) Decision CPM

47. The independent float affects _____.
(A) only the succeeding activities
(B) only the preceding activities
(C) only a particular activity involved
(D) both the succeeding and preceding activities

48. Interfering float for an activity is the difference between its _____.
(A) total and independent float
(B) free and independent float
(C) total and free float
(D) None of the above

49. The software package used for the development of networks is _____.
(A) Primavera project planning software
(B) Successfully planned software
(C) Harvard Total Project Manager (HTPM)
(D) All of the above

50. A serious limitation of interdependence between various activities is generally observed in _____.
(A) bar charts
(B) milestone charts
(C) network analysis
(D) job layouts

Answer Key

1. (C)	**2.** (B)	**3.** (D)	**4.** (A)	**5.** (B)	**6.** (B)	**7.** (A)	**8.** (D)	**9.** (B)	**10.** (D)
11. (D)	**12.** (B)	**13.** (B)	**14.** (C)	**15.** (B)	**16.** (B)	**17.** (A)	**18.** (A)	**19.** (B)	**20.** (B)
21. (A)	**22.** (C)	**23.** (B)	**24.** (C)	**25.** (B)	**26.** (A)	**27.** (D)	**28.** (B)	**29.** (B)	**30.** (C)
31. (A)	**32.** (A)	**33.** (B)	**34.** (A)	**35.** (B)	**36.** (C)	**37.** (D)	**38.** (B)	**39.** (A)	**40.** (A)
41. (D)	**42.** (B)	**43.** (B)	**44.** (A)	**45.** (B)	**46.** (D)	**47.** (C)	**48.** (C)	**49.** (D)	**50.** (A)

5 Project Monitoring, Control Evaluation, and Termination

Project Monitoring and Control System

- Project monitoring is the process of collecting, recording, and reporting information concerning project performance that the project manager and others wish to know, whereas control uses the monitored data and information to bring actual performance into agreement with the plan. Although the data gathered is primarily used for control, it can also be used for auditing and improving quality.
- A control system is a structure that provides the project management with information regarding deviations from the project from what was planned, as well as, recommendations for corrective actions required to correct the deviations.
- Control systems are intended to monitor the characteristics of cost, time, and quality. In other words, control systems aid in the identification of instances involving cost and time overruns, and quality parameter deviations.
- As projects are inherently dynamic, project control systems must be more advanced than those designed for functional departments. The precision of the control system is contingent upon the nature and complexity of the projects, as well as, the administrative skills of the project participants.
- Control systems must be simplified to tables, graphs, or charts to provide a visual representation of events.
- The control system should promptly indicate deviations (in terms of time, cost, and quality) from the plan. It must be able to foresee or predict deviations so that corrective action may be implemented on time.
- The control system must be designed with the active participation of all senior executives of the project team so that it can meet the desired needs.

Tools Used for Project Control

- Work Breakdown Structure
- Bar chart or milestone chart
- Line of Balance (LOB) chart
- Programme–progress chart
- Critical ratio
- Earned Value Analysis (EVA)
- Network plans (precedence diagrams, CPM, or PERT networks)
- Graphical Evaluation Review Technique (GERT)
- Network simulation
- Outsourcing method
- Monthly status reports
- Review meetings

Earned Value Analysis (EVA)

- Earned Value Analysis (EVA) is a project management technique that objectively tracks the physical accomplishment of work.
- EVA can be used to show the past and present performance of a project, and predict its future performance through the use of statistical techniques. It integrates the scope, schedule, and cost of the project.
- EVA answers many questions for stakeholders involved in the project related to its performance. It is an early warning programme or project management tool that enables the managers to identify and control problems before they become insurmountable.
- The only way to exercise effective control over a project is to conduct periodic and systematic performance analysis. Modern performance analysis (also known as Earned Value Analysis) provides an analytical framework for project control by estimating cost variance, schedule variance, cost performance index, schedule performance index, estimated cost performance index, etc.
- To arrive at the above parameters, a project needs the understanding of the following terms.

Planned Value (PV) or Budgeted Cost of Work Scheduled (BCWS)

It refers to the sum of budgets for all work packages scheduled to be accomplished within a given period.

Earned Value (EV) or Budgeted Cost of Work Performed (BCWP)

It implies the budgeted cost of work performed in a given period or the budgeted level of effort expended.

Actual Cost (AC) or Actual Cost of Work Performed (ACWP)

It is the amount of money (and effort) spent in completing a work in a given period.

Cost Variance (CV)

It is the difference between the actual expenditure incurred in the project till the date of review to the value of work accomplished for the expenses incurred.

CV = BCWP – ACWP

Cost Performance Index

A Cost Performance Index of less than 1.00 indicates that the project is spending more than the scheduled cost.

CP Index = BCWP/ACWP

Schedule Variance (SV)

- It is the difference between the value of work that has been completed and the value of the work scheduled to have been completed.

  SV = BCWP – BCWS

- A negative SV indicates that the project is running behind schedule. The above formula gives the schedule variance in cost terms.

Schedule Performance (SP) Index

- This parameter gives the schedule variance in the form of an index.

  SP Index = BCWP/BCWS

- An SP Index of less than 1.00 indicates that the progress of work performed is less than the schedule.

Time Variance (TV)

It gives the time lag at the date of review by indicating the time equivalent of the value of work, which has to be completed to stay on schedule.

TV = (Date of review) – (Date on which BCWP = BCWS)

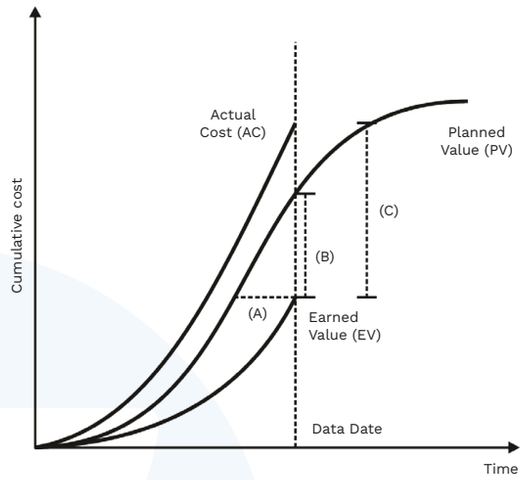


Fig. 5.1

A = Delay in units of time
B = Schedule Variance (SV)
C = Cost Variance (CV)

Critical Ratio

- A critical ratio (CR) indicates to a manager when a task or process is becoming unacceptable. Typically, when the ratio drops below one, then the project is unacceptable.
- This indicator combines both the cost performance index and the schedule performance index to represent project status. This indicator takes into consideration cost and schedule trade-off.
- As critical ratios are easy to determine, they provide a valuable measure of the overall health of a particular project.
- The calculation of the critical ratio for project tasks is the product of a schedule ratio and a cost ratio.

  Critical Ratio (CR) = EV/AC × EV/PV

- Taking the product of these two ratios gives an overall measure that includes performance, cost, and schedule.

CR = (actual progress/scheduled progress)
× (budgeted cost/actual cost)
CR < 1 means poor project performance
CR = 1 means project performance is on target
CR > 1 means good project performance

Line of Balance Techniques

- This is an improvement over milestone charts. The Line of Balance (LOB) activity schedule chart shows a graphical plan of work execution in a linear or 'S' curve shape. These graphs, representing the work cycles, are referred to as 'cyclographs' or 'cyclograms. The time unit (day or week) in cyclographs is represented along the horizontal axis, while the vertical axis shows the number of similar work units of the project. The time unit reflected on the horizontal axis can further be divided into months (or weeks), after assessing the working days available in each month.
- LOB is a planning technique for repetitive work. In this technique, the required resources for each stage of operation are found so that the subsequent stages of the activities are not interfered with and the target output can be achieved. The technique can be applied in repetitive works involving construction projects, such as mass housing development, high-rise buildings, tunnels, etc.

Graphical Evaluation Review Technique

- Precedence networks and decision CPM networks are 'deterministic' in nature. This means they do not involve probabilistic elements, whereas Graphical Evaluation Review Technique (GERT) is based on a probabilistic approach.
- Many researchers have proposed and expanded more general networking systems to expand the modelling capabilities of PERT and CPM. Most of these make use of probabilistic nodes. These differ considerably from ordinary project network nodes (i.e., events). In the latter case, every node is eventually reached or released when all jobs leading into the node have been completed. Similarly, all jobs proceeding from the node may start after the node is released, and all of them must eventually be completed.
- But probabilistic networks show that certain nodes may be released when one or more (but not necessarily all) jobs leading into them are completed. These networks also show that there is a possibility of probabilistic exits from the node. This implies that only one of several proceeding jobs is taken. Thus, jobs themselves may be probabilistic, occurring with some probability, which may be less than 1.

Network Simulation

- For each activity, PERT considers three probable time estimates and a specified distribution for arriving at the estimated time of completion. In actuality, however, the distribution of time estimates is not constant and will fluctuate depending on the circumstances.
- Network simulation models allow for any activity-time distribution to be assumed. The model replicates the project duration distribution for a given collection of activity time distributions. The output of the network simulation model is the project duration distribution, whereas the input to the network simulation model is the activity-time distribution of the various project activities.
- Since a network simulation model may take any sort of activity-time distribution, it reflects real-life scenarios and allows the assignment of any actual activity-time distribution, thereby, allowing for a more accurate assessment of project completion time.

Outsourcing Method

- Every project consists of different type of tasks, each needing unique talents and resources. An organisation embarking on a new project may not have all necessary resources and capabilities at hand. Either the organisation can develop the necessary resources and abilities internally, or it can outsource the project responsibilities, for which the available internal resources and skills are insufficient.

- Outside agencies, with specialised knowledge in the field, can perform the job more efficiently when they are contracted to do so.
- Recruiting specialists and training existing people to perform specialised jobs will incur expenses. The expense of recruitment and/or training will increase the overheads of the organisations, which may no longer be necessary once the project has been completed. When a company decides to outsource specific duties, it can only utilise the services of the external agency when necessary. This technique of paying for what you need and when you need it will substantially reduce the costs.
- Mobilising and training personnel with specialised skills could take a considerable amount of time, which will have an impact on the duration of the project. Consequently, outsourcing non-essential tasks can aid in the timely completion of projects. This will provide greater value to the organisation, if the project personnel are liberated to focus on their main responsibilities.
- Consequently, smart and well-managed outsourcing will increase the competitive edge of businesses.

Review Meetings

- Apart from the use of control tools like charts, reports, analytical evaluation, etc., review meetings at frequent intervals will help achieve proper control of project activities. Review meetings can be arranged for the overall review of projects by the top management, and also for specific reviews like technical, or design, or progress reviews, etc., by the middle management.
- Since projects consist of multidisciplinary activities, coordination and control of the many roles can be attained with direct engagement of the top management, which can be facilitated through talks at the review meetings.
- Due to the direct participation of all important individuals, review sessions will assist in resolving unresolved concerns.
- Technical deficiencies in the project can be explored, and alternate courses of action can be determined in a review meeting. Unresolved topics may be the subject of consensus in such a meeting.
- Since review meetings provide an opportunity for contact among important individuals, it is simple and expedient to identify the possible issues and take appropriate preventive measures.
- Since the execution of projects requires the active participation of individuals from various disciplines, the synergy necessary to move the project in proper direction can be accomplished by discussing the concerns openly and inspiring the participants to achieve the project's objectives.

Project Management Information System

- Project Management Information System (PMIS) is a set of communication methods used by the project team to share plans and results of project activities.
- Computer-based PMIS are most commonly used and allow regular updating of schedules to provide a basis for management action.
- Bar charts and milestone charts produced by planning software are a convenient tool for continuous monitoring and updating of plans.

Resource Allocation

- A project manager often faces resource limitations. Either there could be a delay in the arrival of material resources, or there could be a shortage of human resources, or there could be an excess of human resources at some point.
- It is the project manager's responsibility to plan and distribute resources to the various activities to maximise resource utilisation.
- A resource is a physical variable that is necessary to complete an activity. It may be material, manpower, machinery, space, cash, or time.
- Resource allocation refers to the decision on the allocation of resources to each activity. Due to limited resources or limited project length, it is necessary to allocate the available resources in a manner that generates a demand

that is more or less constant throughout the project's duration.

- A resource usage profile or resource histogram is a diagram that illustrates the fluctuation of resource requirements over time.
- Resource allocation may be achieved by the following two processes.
  - Resource levelling
  - Resource smoothening

Resource Levelling

- In resource levelling, the resources are seen as limitless, although the project duration remains unchanged or is fixed. Resource levelling strives to reduce peak demands and smoothen the output period assignments without increasing the overall project duration.
- Certain circumstances necessitate a project to be finished by a certain date. The management determines the deadline for completing the project for a variety of reasons.
- A project manager regularly encounters a mismatch between the available and required resources. This means that during peak periods, when there are excess resources available on certain days and a scarcity of resources on other days, this can be accommodated by recruiting workers on a temporary basis. As the need for staff varies on a daily basis, the project manager must prepare accordingly so that the most efficient use of available personnel is achieved.
- Therefore, the purpose of the project manager in such a scenario is to level as much as possible the demand for resources over the duration of the project's execution, keeping in mind that the defined project completion time does not exceed. The defined project completion time may be the minimum possible time by which the project may be completed, as determined by defining the critical route, or an even shorter timeframe, as determined by the management to meet other needs.

Ideal Resource Profile

- While project scheduling is done, it is assumed that human resources will be available as and when required, but it is rarely possible.
- Employment and utilisation are mostly stable and increases at a constant rate.

Methods of Resource Levelling

Trial and Error Method

- The Earliest Start Time (EST) schedule is depicted in the form of a squared network, and resource scheduling is aggregated for different time periods and plotted graphically.
- Activities are moved forward or backward on the time scale in order to provide a more steady level of engagement for both individual and group skills. This may require withdrawing resources from activities with free floats and extending their durations, or cannibalising a portion of the total float to achieve a constant or almost constant rate of engagement.

Trigger Level Method

- The objective of this method is to smooth the peak demand for resources by delaying 'slack' work beyond the peak demand time.
- This technique, also referred to as the multishop method, was developed in a naval shipyard.
- The EST schedule is translated into a manpower loading chart, depicting each department's daily manpower needs.
- The trigger level is set one unit below the peak demand, and slack jobs are rescheduled outside of the peak period so that the peak demand does not exceed the trigger level.

Comprehensive Method

- Systematic, iterative, and exhaustive computer programmes are now available for handling resource-levelling issues. We begin with the EST schedule—determine the resource loading chart for the initial schedule, and then calculate the sum of squares.
- In the next phase, we utilise all free floats from the activities, beginning with the longest free float. For each progressive schedule, we compute the sum of squares and compare it with the sum of squares of the initial or previous schedule to confirm that we are progressing in the correct direction.

Resource Smoothening

- It refers to a situation, wherein the resources available are limited and the project duration may be modified. Activities are rescheduled to reduce the peak resource demand so that it does not exceed the resource limit. There should never be fewer available resources than the minimum required for any project activity. First, available floats are utilised, and then, if necessary, the time of particular activities is lengthened or shortened in accordance with the resource requirements.
- Hence, an attempt has been made to explain a simple set of heuristics (thumb rules), which when applied will give reasonably good solutions.

Steps Involved in Resource Smoothening

- List out the resources required for the completion of various activities and identify the ones that are considered important.
- Prepare the resource profiles by carrying out the resource aggregation exercise. The cumulative resource requirements for each unit are plotted in the form of histograms.
- Periods of peak and low demand are identified. Efforts are made to lower the peaks and fill the gaps.
- As there is no constraint on the availability of resources, the demand needs to be made as uniform as possible.
- This can be achieved by altering the start and finishing dates of the non-critical activities as these activities will have a certain amount of float.

Project Cost Control

Time-Cost Trade-off

- Time and cost are the two most essential resources for a project manager. The task of a project manager is to strike a balance between these two resources, given their respective limitations. The appropriate balance between time and cost is referred to as the Time-Cost Trade-off, and it may be attained by analysing the availability and demand for these two resources for the current project.
- As each project is unique, the same time-cost trade-off rule cannot be applied uniformly to all.

Cost Model Analysis

- In cost model analysis, time is related to cost, and the objective is to develop an optimum time-cost relationship.
- The overall project duration can be reduced by reducing the duration of only the critical activities in the project network. The duration of such activities may be reduced in two ways.
  - By deploying more resources for the early completion of such activities
  - By relaxing the technical specifications for such activities
- In cost model analysis, it is assumed that the time of a project can be shortened by allocating more resources to critical activities without compromising on quality standards.
- There are two time and cost estimates for each task in cost model analysis: 'normal estimate' and 'crash estimate'. In a standard estimate, the emphasis is on cost, with minimal cost connected to the shortest time. The 'crash estimate' consists of the bare minimum time and cost required to complete the project. Here, the emphasis is on time.

Project Cost

The components of the total cost are as follows.

Indirect Project Cost

- Indirect costs are expenses that cannot be allocated clearly to individual activities of a project but are evaluated as a whole. The indirect cost consists of administrative and establishment expenses, overhead, supervision, expenditures on a central shop organisation, loss of income, loss of profit, etc.
- Considering just overhead and supervision expenses, indirect costs rise proportionally to duration. It is depicted as a straight line with a slope equal to the daily overhead.
- However, when there is a loss in profits owing to inability to satisfy a demand or penalty for a delay, a corresponding cost increase must

be added to the cost of overheads, resulting in the curve. This type of loss is known as an outage loss.

- The overall indirect cost curve will, therefore, be curved. But for the sake of computation, it will be assumed to be linear.

Direct Project Cost

- It is the cost, which is directly dependent on the number of resources involved in the completion of activities. It includes labour, materials, plants, machinery, etc.
- To get the same work done in less time, one needs to increase the amount of labour, equipment, and time-saving materials, that too, at extra charges, which simply means an increase in direct cost.
- The project has the highest direct cost corresponding to the crash duration, and has a normal cost corresponding to the normal duration.
- Normal time (tn): It is the standard time that an estimator would usually allow for an activity.
- Crash time (tc): It is the minimum possible time, in which an activity can be completed by employing extra resources. Crash time is the time beyond which the activity cannot be shortened by any amount of increase in resources.
- Normal cost (Cn): This is the direct cost required to complete the activity in a normal time duration.
- Crash cost (Cc): This is the direct cost corresponding to the completion of the activity within crash time.
- The straight line or segmented approximation of the direct cost curve helps in carrying out the project cost analysis. In such an analysis, the cost slope is used.

Cost Slope

- It is the slope of the direct cost curve, approximated as a straight line. It is defined as follows.

$$\text{Cost price} = \frac{(\text{Crash cost} - \text{normal cost})}{(\text{Normal time} - \text{crash time})}$$

or

$$C_s = \frac{(C_c - C_n)}{(t_n - t_c)} = \frac{\Delta_c}{\Delta_t}$$

- The segmented approximation of the cost curve, having multiple cost slopes, is more accurate, but the calculations involved are more. Generally, a single cost slope is assumed.

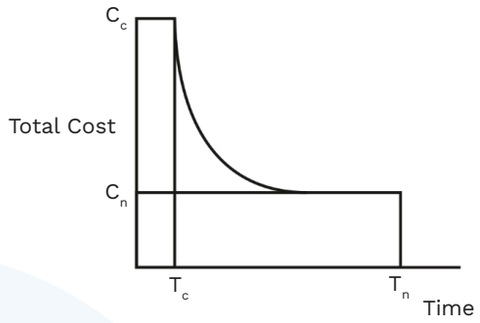


Fig. 5.2

Total Project Cost and Optimum Duration

- The total project cost is the sum of both direct and indirect cost.
- The minimum total cost is obtained at a duration known as the optimum duration. The corresponding cost is known as the minimum cost. If the project duration is increased, the total cost will also increase, while if the project duration is decreased to the crash value, the project cost will be the highest.

Steps in Crashing

Step 1: Estimate the project duration and find the critical path.
Step 2: Find the cost slope of all activities.
Step 3: The critical activity having the minimum cost slope is crashed in the first stage. In the next stage, crashing will involve activity having the second lowest cost slope in the critical path.
Step 4: The total cost of the project at this stage is calculated.
Step 5: Steps 3 and 4 are repeated till all activities of the project are crashed along the critical paths, and the corresponding time is the crash time of the project.
Step 6: It is to be noted that only critical activities should be crashed.

Example:

The management of the company is interested in crashing the following project by spending an additional amount not exceeding ₹ 2000. Suggest how this can be accomplished.

Activity	Predecessor	Normal Time (Weeks)	Crash Time (Weeks)	Normal Cost (₹)	Crash Cost (₹)	Δt	$\frac{\Delta_c}{\Delta_t}$
A	-	7	6	15000	18000	1	3000
B	A	12	9	11000	14000	3	1000
C	A	22	21	18500	19000	1	500
D	B	11	10	8000	9000	1	1000
E	C, D	6	5	4000	4500	1	500

Table 5.1

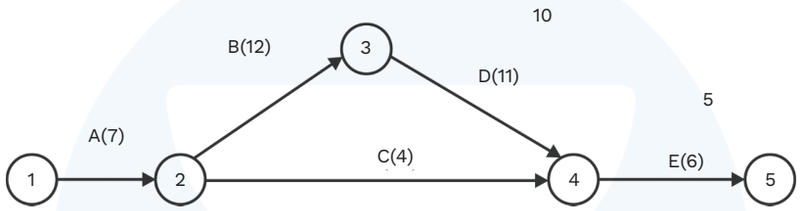


Fig. 5.3

Critical path: 1–2–3–4–5 = 36 weeks

- As per the procedure, we have to crash the activity where the cost slope is the lowest. The E (3–4) activity is crashed by one week. So, the amount utilised to crash by one week is ₹ 500. Now, the critical path is 35 weeks.

Activity	Δt	$\frac{\Delta_c}{\Delta_t}$
1–2	1	3000
2–3	3	1000
3–4	1	500
4–5	1	1000

Table 5.2

- Next D (4–5) activity is crashed by one week and the amount utilised to crash is ₹ 1000. Now, the critical path is 34 weeks.
- The critical path changes; it is 1–2–4–5 = 34 weeks.

Activity	Δt	$\frac{\Delta_c}{\Delta_t}$
1–2	1	3000
2–4	3	500
4–5	1(0)	1000

Table 5.3

- Now C (2–4) activity has to be crashed and the amount utilised to crash is ₹ 500. The critical path changed to 33 weeks.
- With ₹ 2000 allocated by the company, the critical path can crash from 36 to 33 weeks.

Project Evaluation and Termination

Project Evaluation

- To evaluate implies to assess. Therefore, project evaluation is an assessment of the project during its implementation. It is an interim assessment at significant milestones. The project will be

directed along the proper path, if it is evaluated at important implementation stages.

- Evaluation of a project is an attempt to establish whether the overall status and progress of the project are satisfactory relative to what was originally envisaged and whether the project's objectives are being met. The ultimate goal of project evaluation is to improve project planning and execution in every way.

Project Evaluation Objectives

- One might thing of having delivered the outputs. But this does not mean that the project is complete because unless the project has been closed with planned procedures, its value will be null or void.
- Project closing ensures that there are no further obligations in the project.
- Ensuring project output can be used.
- Hearing about good and bad points is equally important so that success can be repeated and failures can be avoided in future.
- Releasing resources like staff, equipment, etc., is important to start a project.
- It is not good that we deliver outputs to our customers without closing the contracts with our vendors.
- Once the required products are ready to be delivered to the customers, there is no meaning in keeping vendors' contracts alive.
- Contract closing involves verifying that the specified work has been completed and contract records have been updated.
- A contract record is an important document. It includes the contract and other relevant documents, such as progress reports, financial records, invoices, and payment records.
- These documents are kept in a contract file, which should be part of the complete project file.

Evaluation Methods

- Smaller projects may be evaluated by direct observation and by getting first-hand information from field staff, specialists, and others connected with the project's implementation.
- Evaluation based on periodic written reports is done for major projects. The written reports may be supplemented with charts, graphs, and illustrations to add value to the reports.

Post Project Evaluation (Post Audit)

- A post project evaluation is often referred to as post audit, or post completion audit. This is the post completion analysis of a project.
- At the time of project appraisal, estimates are established for the project's cost and duration. After the project is completed, a project audit is conducted to evaluate the actual project cost and time required to execute it.
- While project appraisal is the prediction of the future, post project evaluation is an analysis of the past.
- A project is, ideally, deemed to be successful if it is finished on time, within budget, and with the level of performance anticipated.
- In addition to evaluating the variances in the project cost and the completion time, post project evaluation of public projects also evaluates the actual societal cost–benefit aspects and the degree to which the project's objectives are met.

Advantages of Post Audit

- The knowledge gathered throughout the implementation of the project is of considerable value to all parties involved. During the implementation of the project, the project owner and the project management team gain information that they can share with others or apply in future endeavours.
- By conducting a post project evaluation, a financial institution that has supported the project can identify faults in its project appraisal and the control mechanism that should have been implemented for effective project monitoring.

Phases of Post Audit

A post audit is carried out in two phases, viz., immediately after the completion of the project and after the lapse of some time (say two to three years) since its completion. The main aim of post project evaluation, whether conducted immediately after the completion of the project or after the lapse of some time, is to study the

pitfalls and grey areas, and identifying appropriate corrective measures.

Immediately after the completion of the project, objectives of such a type of audit are as follows.

- Analysing the difference between the actual project cost and the estimated project cost
- Analysing the difference between the actual time taken for project completion and the estimated time
- Identifying areas that have contributed to the variances in project cost and time
- Identifying the reasons for such variances and classifying them as avoidable or unavoidable
- Analysing the steps that could have been taken to avoid the avoidable variances
- Conducting an in-depth analysis of the factors that could have caused the unavoidable variances and examining the possibilities of their elimination in future projects by adopting a suitable methodology

The objectives of audit that is conducted two to three years after the project is completed are as follows.

- Analysing whether the product goals and objectives have been achieved
- Determining whether the project produces products or services of acceptable quality
- Determining whether the estimated output is achieved
- Determining whether the product or service is accepted by the market
- Determining whether the production volume achieved is in proportion with the planned market share

Types of Post Audit

Technical Evaluation (Technical Audit)

- Technical evaluation refers to the evaluation of the quality and quantity of output, as well as, the production's running expenses, etc. There is a contrast between what was given in the feasibility or detailed project report and what was accomplished with regard to these factors.
- Regarding rated output, certain projects are anticipated to provide the estimated rated output upon commissioning. Such projects can be examined for output variances, if any, immediately following commissioning. Regarding many other initiatives, the output will not be steady for a while. These are instances, in which the output is dependent on the objectives of a post audit that can be summed up as the quality of raw materials, the competence of the staff operating the plant, etc., based on the aforementioned factors. The operation of the plant is to be researched by altering these parameters and observing the output until the production reaches its maximum and is stable. After all inputs have been synced and production has stabilised, can the output be evaluated.
- Other areas of interest include the examination of utilities consumed by a plant, such as electricity, fuel, water, steam, consumables, spares, etc., in addition to the evaluation of actual versus rated output. These factors are evaluated and compared to what was estimated or predicted at the time of the project's review. The technical evaluation also entails assessing the output's quality and determining if the specified quality requirements are met.

Financial Evaluation (Financial Audit)

- At the time of project evaluation, financial projections and estimates of the project's costs under various headings and running expenses, such as maintenance expenses, profitability projections, cash flow and fund flow projections, sources and uses of funds for the project, etc., are made.
- The objective of financial evaluation is to determine whether the actual project cost, operational costs, profitability, cash or fund flows, etc., correspond with the estimations and projections provided at the time of appraisal.
- While the assessment of actual project costs can be done immediately after the completion of the project, other financial metrics, such as running costs, profits, etc., cannot be evaluated until the project has been in operation for at least two to three years as the record of actual operating costs, profits, etc., would be available only after the project is in operation for some time.

Economic Evaluation (Economic Audit)

- Economic evaluation is the most challenging as it incorporates numerous subjective variables that are difficult to quantify.
- The evaluator should have an 'eye' for spotting the project's societal costs and benefits. As public sector and community development initiatives are carried out with social objective (in addition to financial and other objectives) in mind, economic evaluation is more applicable to these types of endeavours.
- A project in the public sector would have been implemented to attain specific social benefits. A Social Cost-Benefit Analysis (SCBA) is performed during the project's appraisal phase to quantify the social costs and benefits, despite their subjectivity. The SCBA audit of the project should examine if the anticipated social benefits are achieved and whether the project has had any negative effects, which were not anticipated initially. Such unanticipated negative outcomes tend to increase the social cost of the endeavour. Post project evaluation from a social cost-benefit perspective is at its infancy, despite the fact that SCBA at the appraisal stage has developed into a well-structured system in recent years.

Practice Exercise

1. Consider the following statements regarding updating the network.

I. For a short duration project, updating is done frequently.
II. For a large duration project, the frequency of updating is decreased as the project nears completion.
III. Updating is caused by overestimated or underestimated times of activities.
IV. The outbreak of natural calamity necessitates updating.

Which of these statements are correct?

(A) I, II and III
(B) II, III and IV
(C) I, III and IV
(D) I, II, and IV

2. Of the following tools and techniques, which one is not used for schedule control?

(A) Fast tracking
(B) What if scenario analysis
(C) Three-point estimates
(D) Adjusting leads and lags

3. Your customer wants the project to be completed six months earlier than planned. You believe you can meet this target by overlapping the project activities. The approach you plan to use is known as _____.

(A) critical chain
(B) fast tracking
(C) levelling
(D) crashing

4. What is the process of incorporating changes and rescheduling or replanning called?

(A) Resource allocation
(B) Resource smoothing
(C) Resource levelling
(D) Updating

5. Total project cost versus time curve is/an _____.

(A) S-shaped curve
(B) parabola
(C) U-shaped curve
(D) straight line

6. In the cost optimisation procedure, the cost slope for each activity can be estimated by which of these formulas?

(A) $\frac{\text{Crash cost} - \text{normal cost}}{\text{Crash time} - \text{normal time}}$

(B) $\frac{\text{Crash time} - \text{normal cost}}{\text{Crash cost} - \text{normal cost}}$

(C) $\frac{\text{Normal time} - \text{normal cost}}{\text{Crash cost} - \text{normal cost}}$

(D) $\frac{\text{Crash cost} - \text{normal cost}}{\text{Normal time} - \text{crash time}}$

7. The normal duration and normal cost of an activity are 10 days and ₹ 350, respectively. The cost slope is ₹ 75 per day. If the crash duration is eight days, then what is the crash cost of the activity?
(A) ₹ 400
(B) ₹ 500
(C) ₹ 600
(D) None of the above

8. In the time-cost optimisation of a project, the project can be crashed by expending.
(A) All activities on the critical path
(B) Critical activities having minimum cost slope
(C) Activities on subcritical path
(D) Critical activities having maximum cost slope

9. Match List-I (Cost) with List-II (Features) and select the correct answer using the codes given below the lists.

List-I
A. Optimal cost
B. Overhead cost
C. Direct cost
D. Indirect cost

List-II
I. Activity related
II. Developed by crashing process
III. Project related
IV. Contained in or contributed exclusively to the related product

Codes:

	A	B	C	D
(A)	IV	III	II	I
(B)	II	I	IV	III
(C)	IV	I	II	III
(D)	II	III	IV	I

10. In big construction projects. The per cent cost of construction equipment in terms of the total project cost is about _____.
(A) 10–20
(B) 15–30
(C) 45–50
(D) 5–10

11. The objectives of cost control is to _____.
(A) locate the areas of inefficient functioning
(B) analyse the basis for preparing estimates
(C) conduct an estimation of profit or loss
(D) All of the above

12. The stage of the work at which cost control is not affected is at _____.
(A) planning
(B) designing
(C) construction
(D) None of the above

13. The time beyond which the direct cost will not be reduced with the increase in time is _____ time.
(A) normal
(B) crash
(C) optimistic
(D) optimum

14. The direct cost of a project with respect to normal time is _____.
(A) minimum
(B) maximum
(C) zero
(D) infinite

15. The reduction in project time normally results in _____.
(A) increasing the direct cost and decreasing the indirect cost
(B) decreasing the direct cost and increasing the indirect cost
(C) increasing both the direct and indirect costs
(D) decreasing both the direct and indirect costs

16. Economic saving of project time would result from crashing _____.
(A) cheapest critical activity
(B) cheapest non-critical activity
(C) costliest critical activity
(D) costliest non-critical activity

17. In a construction project, the cost slope of an activity is an indication of _____.
(A) extra time needed
(B) extra cost needed
(C) reduction of the duration of critical activity
(D) crashing of an activity

18. Which of the following are examples of indirect costs?
(A) Salaries of corporate executives
(B) Salaries of full-time project staff
(C) Overhead costs
(D) Both (A) and (C)

19. Which of the following is not a commonly used technique for schedule compression?
(A) Resource reduction
(B) Reducing the scope
(C) Fast tracking activities
(D) Overtime

20. Overall cost estimates must be allocated to individual activities to establish the cost performance baseline. In an ideal situation, a project manager would prefer to prepare estimates _____.
(A) before the budget is complete
(B) after the budget is approved by the management
(C) using a parametric estimating technique and model specific to that project type
(D) using a bottom–up estimating technique

21. The key to effective cost control is _____.
(A) using the earned value to forecast project status
(B) focusing on projected expenditures and actively networking with key stakeholders to ensure that funds will be available as requested
(C) informing stakeholders of the project's cost status
(D) managing the approved cost baseline and any changes to it

22. What is the significant purpose of monitoring a project through its implementation phase?
(A) To fix responsibility for delays
(B) To retail the project with control over cost overrun
(C) To retail the project with minimum time overrun
(D) To retail the project with optimal time and cost overrun

23. Which of the following is not an input to the execution process?
(A) Enterprise environmental factor
(B) Organisational process assets
(C) Project management plan
(D) Accepted deliverables

24. Which of the following is an output of the project execution process?
I. Deliverables
II. Project documents update
III. Project change requests
IV. Project management plan
(A) All of the above
(B) I, II, and III
(C) II, III, and IV
(D) II and III only

25. Which is not a tool for the project execution process?
I. Expert Judgement
II. Project management information system
III. Meetings
(A) All of the above
(B) I, II, and III
(C) I and III only
(D) None of the above

26. The maximum amount of budget is expended during which process?
(A) Initiation
(B) Planning
(C) Execution
(D) Closing

27. Which of these processes acts as an umbrella for all other processes?
(A) Initiation
(B) Planning
(C) Execution
(D) Monitoring and control

28. The project execution phase is done with the aim of which of the following?
I. To make necessary changes in a product
II. To make action into operation
III. To change the project charter
IV. To achieve deliverables

(A) All of the above
(B) I, II, and III
(C) I, II, and IV
(D) II and IV only

29. Which of the following is correct?
I. Project execution is the most difficult part of a project, where real risks are involved.
II. Stakeholders are responsible for project execution.
(A) Both I and II
(B) I only
(C) II only
(D) None of the above

30. Regarding project management the various monitoring and control tools are _____.
I. WBS
II. PERT or CPM
III. Line of Balance
IV. Milestone chart
(A) All of the above
(B) I, II, and III
(C) II, III, and IV
(D) III and IV only

31. In project management, various schedule compression techniques include _____.
I. Line of Balance
II. Crashing
III. Fast tracking
IV. Bar chart
(A) All of the above
(B) I, II, and III
(C) II and III only
(D) II and IV only

32. Cost variance in a project is the difference between
(A) Earned value and planned value
(B) Earned value and actual cost
(C) Planned value and actual cost
(D) Cost variance and cost performance index

33. Which among the following is not a cost estimating tool for exiting (old) product?
I. Parametric
II. Analogous
III. Bottom-up
IV. Monte Carlo
(A) All of the above
(B) I and II only
(C) III and IV only
(D) I only

34. Which of the following is correct regarding crashing?
I. It reduces the project duration.
II. It increases the cost.
III. It increases the project duration.
IV. It reduces the cost.
(A) I and II
(B) II and III
(C) I and IV
(D) IV only

35. Which of the following is correct?
I. Resource smoothing is done after resource levelling.
II. Resource levelling may change the project duration
(A) Both I and II
(B) None of the above
(C) I Only
(D) II Only

36. What are the different software used in a project information system?
I. Yojna
II. Scada
III. Microsoft Project
IV. Prism
(A) All of the above
(B) I, II, and III
(C) II, III, and IV
(D) III and IV only

37. Resource allocation is done to:
I. optimise the resource utilisation.
II. generate a uniform demand for resources.
Which of these statements is/are correct?
(A) Both I and II
(B) I only
(C) II only
(D) None of the above

38. In the precedence diagram the relation is from one activity to another by _____.
(A) start to start
(B) start to finish
(C) finish to start
(D) finish to finish

39. If the actual cost is more than the estimated cost, then _____.
(A) cost performance index CPI < 1
(B) being the schedule
(C) CPI = 1
(D) cannot be predicted

40. In a project, poor execution may be the effect of which of the following?
I. Poor leadership
II. No conflict management system
III. Demotivated workers
IV. Multitasking
(A) All of the above
(B) I, II, and III
(C) I, III, and IV
(D) II and IV only

41. A positive cost variance means the project is _____.
(A) over budget
(B) under budget
(C) equal to budget
(D) cannot be predicted

42. A positive schedule variance indicates the project is _____.
(A) ahead of schedule
(B) behind the schedule
(C) on schedule
(D) cannot be predicted

Answer Key

1. (A)	**2.** (C)	**3.** (B)	**4.** (D)	**5.** (C)	**6.** (D)	**7.** (B)	**8.** (B)	**9.** (B)	**10.** (B)
11. (D)	**12.** (A)	**13.** (A)	**14.** (A)	**15.** (A)	**16.** (C)	**17.** (B)	**18.** (D)	**19.** (A)	**20.** (A)
21. (D)	**22.** (C)	**23.** (D)	**24.** (A)	**25.** (C)	**26.** (C)	**27.** (C)	**28.** (A)	**29.** (A)	**30.** (A)
31. (C)	**32.** (B)	**33.** (C)	**34.** (A)	**35.** (C)	**36.** (A)	**37.** (A)	**38.** (B)	**39.** (A)	**40.** (C)
41. (B)	**42.** (A)								

6 Project Risk Analysis

Risk

- A risk is an uncertain event or condition that has a positive or negative effect on at least one of the objectives of a project. A risk may occur due to one or more causes.
- Generally, the word 'risk' reflects a negative image only. But it may have a positive or negative effect on the project. Hence, it may be classified into positive and negative risk.

Positive Risk

A positive risk is a condition that has a positive impact on any of the project objectives.

Negative Risk

- A negative risk is a condition that harms the project objectives. Every risk is made of the following two components.
- Likelihood (probability): It is the extent to which the risk event is likely to occur.
- Impact (consequence): It is the effect that risk will have on the project, if it occurs.
- Project risk management is the art and science of identifying, assigning, and responding to risk throughout the life of a project in the best interests of meeting objectives.
- Hence, risk management is concerned with the outcome of future events, whose exact outcome is unknown, and the development of strategies to deal with such uncertainties over a range of possible outcomes.
- Risk management helps the project sponsor, project team, and especially the project manager to take appropriate action towards negative risks by minimising their adverse impacts on the scope, cost, schedule (and quality or result), and positive risks concerning the project by maximising opportunities to improve its objectives with lower costs and shorter schedules, thereby, enhancing its overall scope and quality.
- The project team should review project documents about risk management to get an idea about the organisation's and sponsor's ability to risk tolerance or risk threshold.
- Risk tolerance: It gives an idea about an organisation's or individual's sensitivity towards the various risks. High tolerance means people are willing to take high risks, whereas low tolerance means people are not willing to take a high risk unless the benefits of taking such a risk outweigh the fear of risk.
- Risk threshold: It is the amount of risk that an organisation or individual is willing to accept. It is usually a definite figure.
- Many a time, risk management is neglected. However, it has the potential to increase project's performance by contributing to the successful selection of a suitable project, determination of the project scope, and development of realistic estimates. Therefore, risk management might be compared to an investment or insurance policy in a project.
- To successfully carry out risk management, one must first identify the probable sources of risks that are likely to arise during the implementation of a project, and then, evaluate their probability and specific remedies. After that, one may go on to the next step of risk management.

Risk versus Uncertainty

Risk is defined as the variance in return on investment. If the probabilities of the probable outcomes of a particular situation are known, it may be deduced that the problem is risky or the task is risky. Alternatively, if the probabilities of an issue's various outcomes are unknown, it may be concluded that the problem has an element of uncertainty, i.e., the outcome cannot be predicted.

Types of Project Risks

Operational Risk

- A decrease in operational efficiency may cause reduced profitability, or even a loss for the project at times.
- In some projects, the effects of decreased operational efficiency may or may not be high. Similarly, the possibility of variation in efficiency may vary from project to project.
- Breakdown of machines, demand, and supply of resources and products; shortfall of goods and services; and lack of efficient logistics and inventory may lead to inefficiency of production. Operational leverage is the measure of operational efficiency.

Market Risk

Although every project undergoes market potential analysis and demand forecasting, there are chances of variations due to various foreseen or unforeseen reasons, which may adversely affect the market potential or sales of products or services to be rendered by the planned project. This is a user acceptance risk.

Economic Risk

- The global or national economy is always in different phases and no project can shield itself from ongoing economic changes
- All projects are prone to risk for the economy. A recession in the economy reduces the purchasing power of the customers, and thus, can affect the sales or revenues of the project. It can also lead to reduced demand and selling prices.
- The inflation rate varies with changes in the economic situation, which also affects the prospects of a project.

Financial Risk

- A shift in the cost of financing or interest rates will have an impact on the potential outcomes of a project. Current interest rates have a direct bearing on the predicted returns.
- Financial risk varies by the project's debt-equity ratio and leverage.

Technological Risk

- The failure of the selected technology is one of the biggest risks to any project. Moreover, a project may face the problem of upgradation of technology.
- Many projects have failed due to their inability to cope with technological innovations. This particular source of risk is vital in the current scenario as technology is changing and evolving at a fast pace.

Commercial Risk

There is always a risk of a customer or debtor becoming insolvent. Although the company manages and keeps provision for such debts, at times when the customer is a major one and the amount outstanding with him/her is large, it may affect the functioning of the project.

Quality Risk

Today's world is quality conscious and any degraded product may lead to rejection and the brand may lose its value in the long run, if it is not able to maintain the required quality standards and meet the customer's expectations.

Legal or Regulatory Risk

New tariffs or obstacles to imports or exports might emerge as a result of political changes in a country's legal or regulatory framework, posing a threat to international trade.

International Risk

- If international prices with additional or transaction costs (transportation costs and duties) become less than domestic prices, there is an import threat.
- Any change in foreign exchange rates also affects projects with international exposure.
- A devaluated Indian currency may adversely affect a project, which is dependent on imports of goods or services. Similarly, the devaluation of foreign exchange may adversely affect export-oriented projects.
- Risk of war, imposition of export or import bans, and changes in the political scenario of a country may also lead to risk in a project involving international business.

- The indicators used to measure risk are range, standard deviation, variance, coefficient of variation (CV), semi variance, mean absolute deviation, etc.
- Range: It is the simplest to calculate. The range of a distribution is the difference between the highest and the lowest values.

$$\sigma = \left[\sum_{P_i} \left(X_1 - \bar{X}\right)^2\right]^{\frac{1}{2}}$$

Where, σ is the standard deviation, p_i is the probability associated with its value, X_i is the value of the product, $\bar{x}$ is the expected value/ average/weighted average.

Coefficient of variation (CV) = Standard deviation/expected value

Risk Assessment Methods

Although there is a wide variety of mathematical methods available for risk assessment, the following is a list of straightforward tools that are useful for evaluating projects of a smaller or medium scale.

- Sensitivity analysis
- Scenario analysis
- Monte-Carlo technique (simulation analysis)
- Break-even analysis
- Decision tree analysis
- Game theory

Sensitivity Analysis

Sensitivity analysis determines the effect on the whole project by changing one of its risk variables, such as delay in design or cost of materials.

- The NPV, IRR, or any other indicator of a project is analysed for variability due to change in any one variable at a time, keeping the other factors constant.
- The pessimistic, optimistic, and expected forecasts are made for NPV. Based on it due to a change in a variable, the change in NPV is calculated.
- The output of sensitivity analysis is generally shown with the help of a tornado diagram, which is useful for comparing the relative importance and impact of variables that have a high degree of uncertainty to those that are more stable.

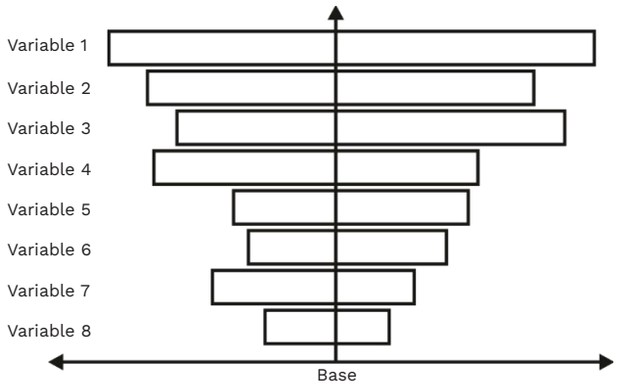


Fig. 6.1

Scenario Analysis

- Three scenarios that are commonly considered are optimistic, pessimistic, and most likely scenario.
- This method is an improvement over sensitivity analysis as it considers variation in several variables together, which are more realistic.

Simulation Analysis (Monte Carlo Method)

- Sensitivity analysis indicates the variability of NPV, IRR, or any other indicator due to variation of any one parameter.
- The decision-maker would also like to know the likelihood of other such occurrences. This information can be generated by simulation analysis.

Steps Involved in Simulation Analysis

Step 1: It involves modelling the project. The modelling of the project will reveal how NPV is related to other parameters or exogenous variables.

Step 2: Parameter values and exogenous variable distribution probabilities are required.

Step 3: Choose a value at random from each exogenous variable's probability distribution.

Step 4: Find NPV that pertains to the values of exogenous variables that are produced arbitrarily.

Step 5: Till a significant number of simulated NPV is achieved, repeat steps 3 and 4.

Step 6: Create a graph showing the NPV frequency distribution.

Break-Even Analysis

- Break-even analysis starts with dividing the costs into two broad heads, namely fixed and variable costs.
- Fixed costs: Certain expenditures are inherent to every undertaking and are constant in character. These expenses do not alter, no matter howsoever the higher or lesser a product is produced.
- Variable costs: These are expenses that shift proportionally with production volume. As it varies with production volume, the price of raw materials is considered a variable expense.
- It may be noted that at the break-even point, Sales realisation (revenue) = Fixed cost
  + variable cost
- Contribution: The contribution is the total sales revenue minus the variable costs. The word 'contribution' is used to describe the amount by which the difference between the selling price and variable cost helps to cover fixed expenses and generate a profit.

Decision Tree Analysis

- It is assumed that a firm's project will be implemented following a set plan. In reality, firms have some flexibility in the execution of a project. When such flexibility exists, decision tree analysis is a useful tool for valuing project flexibility.
- When future consequences are unknown, a decision tree may be used as a visual aid to help choose the most appropriate way to proceed.

Steps in Decision Tree Analysis

- Identifying the problem and alternatives
- Delineating the decision tree (exhibiting the anatomy of the decision situation; it is in a diagrammatic format)
- Specifying the probabilities and monetary outcomes
- Evaluating various decision alternatives

Risk Management or Managing the Risk

- The method of choosing how to undertake and carry out risk management activities for a project is known as the planning stage of risk management. The main output of risk management planning is the risk management plan.
- Some of the necessary questions that need to be addressed in any risk management plan are as follows.
  - Why is it important to take/not take this risk about the project objectives?
  - What are the various types and sources of risks?
  - How are risks going to be mitigated?
  - Who are the individuals responsible for implementing the risk management plan?
  - When will the milestones associated with the mitigation approach occur?

7 Tender and Contract

Project Contract

- A legally binding agreement is known as a contract. This is a transaction involving the exchange of commitments between two or more individuals or entities.
- According to the Indian Contract Act of 1872, Section 2(b), the definition of a contract is "an agreement that is enforceable by law". The Indian Contract Act provides a definition that is broken down into two main elements.
  - A contract is an agreement.
  - The agreement must be enforceable by law.
- For a contract to be considered legally binding, it must fulfil many requirements that are referred to together as the 'essentials of a valid contract'. Even while an agreement must always be present before there can be a contract, not all agreements are contracts. Contracts are legally binding agreements, and the law must be able to uphold them.
- An individual, an organisation, or a corporation that accepts the responsibility of carrying out the terms of a contract is known as a contractor.

Fundamental Elements of a Contract

- A legally binding contract has to be both an offer and a reaction to that offer, for example, an offer is made when a public notice requests bids for the execution of work on a contract basis. An offer is accepted when a contractor submits his/her tender papers in response to the public notice and in line with the terms and conditions that have been stated.
- For the contract to be valid, all parties involved need to have the intention of forming a contractual binding. The parties' consent must be free and informed.
- A genuine consideration between the parties to the contract is necessary. All contracting parties must be of legal age and capacity. People who lack the legal ability to engage in contracts include children and the mentally ill.
- The contract's terms cannot be illegal or unethical in any way. To the best of knowledge, the contract should not be null and void under any applicable legislation in the country.
- The provisions of contract must be able to be carried out. The purpose of the contract must be clear and there must be no room for ambiguity or different interpretations of its terms.
- A contract has to be in accordance with all necessary legal procedures.

Type of Contracts

There are two types of contracts. They are as follows.

- Turnkey contracts
- Non-turnkey contracts

Turnkey Contracts

- Up to the point that the project is ready to start, the contractor is incharge of taking care of all elements of the project's implementation. Under the terms of the turnkey contract, the contractor is accountable for the completion of the entire project. It is almost as if the owner does not get involved in the process until the very end when the project is finished and he/she starts cranking out the goods at the factory.
- Because the contractor of a turnkey project is responsible for starting from scratch, the work covered by this contract includes design, engineering, construction, structural work, supply and instalation of equipment and machinery, supply and erection of spares, testing, and commissioning. Since the entire project is handled by one company, a performance guarantee is standard in turnkey contracts.
- Turnkey contracts might be for the entire project or a specific subset of it. When a sub-unit is an autonomous part of the whole project and its performance can be measured separately from that of other sub-units, a turnkey contract is signed for that sub-unit.

Non-turnkey Contracts

- When the project's scale is small, the promoters have relevant experience, and the organisation's project team is competent, non-turnkey contracts are the most appropriate.
- In a non-turnkey set-up, each part of the project is broken down into smaller tasks. What must be delivered to contractors in terms of packaging is clear. Contractors might be given responsibility for a single task or a series of related tasks. The internal project management team, however, will be incharge of directing and executing the project as a whole. A few examples of non-turnkey agreements are shown as follows.
    - Lump sum contract is a one-time, set price service agreement. The contractor, hereby, undertakes to do the work for the stated fee. When the owner hires a contractor, he or she gives detailed instructions for the job. The benefits and drawbacks of a fixed price contract as they apply to this agreement are as follows.
        - Before work on the project even begins, the owner is given an estimate of how much it will cost.
        - It saves the owner and the contractor from having to worry about a large number of intricacies and paperwork.
        - The contractor is given the chance to implement value engineering via the usage of this contract, which is combined with the design contract method of delivery. The contractor is given unrestricted authority to carry out the task.
        - This contract is only as reliable as the information included in the contract itself. If there are inaccuracies in the contract document, the contract will need to be restructured; as a result, the owner will assume an increased level of risk. Thus, it is challenging to take into account any modification or specification.
        - While building a project, the contractor may be placed in a precarious position if unanticipated dangers occur.
        - In a lump sum contract, the owner may pay the whole amount upon completion or a portion of the total cost at various stages of the project's completion. For example, suppose a storage vessel contract was issued with a certain sum of money for the work to be done, an initial 10 per cent will be compensated upon the installation of the substructure, 60 per cent with the execution of staging, and 85 per cent when the vessel is ready. There will be no final payment until the vessel is fully functional. When hiring labour, it is a common practice to agree on a single price all at once.
    - Unit price contract: This contract structure bases payment on a predetermined quantity of labour performed. There are indeed a few distinct forms of this agreement.
    - Bill of quantities contract: The company owner is the one who draws up the plan, determines how much work is needed, and writes the specifications under this sort of agreement. The contractor submits a quotation depending on the overall price per unit of all building materials.
        - The unit cost of the job may include the contractor's overhead, profit, and other expenses. It is not uncommon for a contractor to provide two distinct quotes: one for the actual job and another for the profit overhead. A bill of quantities is a document that specifies the agreed upon total amount of work to be done. Contracts of this type are flexible enough to provide for nominal changes in the figures. This is a fixed design project. Thus, no change to the plan is expected. Changes to the design are permissible during construction but will not affect the per unit cost.
        - This is the standard technique of building for government funded infrastructure projects of this scale.

This contract structure provides the owner with a favourable offer. The drawbacks of this approach include the following.

- The owner must measure the quantity of work performed in the field, which necessitates one's presence on the job site.
- The ultimate cost of the building will not be known until after all dust settles. The owner is in a sticky situation, if the predicted results significantly differ from the actual state of affairs. Unbalanced bids refer to incorrect amounts. Unethical behaviour includes grossly asymmetric bids.

- Schedule of rate contract: Often, it is impossible to predict to predict the number of labiyrs required to complete the work. Agreements are reached based on the per-unit cost of the work being done. Due to the potential difficulty in accurately identifying each item, contracts often contain additional things that are being performed. It is not guaranteed that all of the required materials will be used in the building process. Typical uses include sub-surface work, flood mitigation, and highway construction. Similar to the invoice quality agreement, this kind of agreement offers both benefits and drawbacks. There are multiple variations of the rate contract schedule, some include charging a profit on top of the price per unit.
- Cost-plus contract: Here, the contractor gets paid not only for the actual costs of the project but also for additional benefits, such as overhead, profit, etc. Upper and lower limits cost limitations are often imposed on contracts, as in a specified maximum price contract. The contractor is accountable for additional costs incurred over this threshold. Incentives for completing the project ahead of schedule are often added to the existing contracts. The engineering and building phases may overlap extensively, which is a major benefit of this type of contract. Because of this, one can continue with the project. This kind of agreement is the most appropriate for projects, where precisely defining the work to be performed before finalising the deal would be challenging.
- Item rate contract: It is an agreement, in which the price of goods or services is determined according to a predetermined timetable or list of items. A contractor completes tasks on an hourly or per item basis. The department then provides a bill of quantities, and the contractor will have to offer a rate quotation for each line item on the invoice. The remuneration for the contractor is based on the amount of work performed. The contractor gets compensated in accordance with the exact dimensions of the different parts of the project that one has finished. Almost all engineering projects for government agencies, including rail and road, are conducted under an item rate contract. It functions well for jobs that can be broken down into discrete components, the amounts of which can be reliably calculated.

Project Tender

A proposal to perform a defined job or provide a specified item at a specific rate within a given timeframe under certain terms of the agreement is referred to as a tender. It is made on paper by the tenderer (the individual who provides the tender).

Tender Notice or Notice Inviting Tender (NIT)

- Public notices, including tender notice, are often posted on major news websites. This is a common practice among government agencies and other public sector organisations. The tender notice contains the following information.
  - The company or department making the request for proposals, as well as, the official's designation and department, should be included.

- It also includes the specifics of the job, i.e., where it will be performed, how much is it expected to pay, and how much money is needed to pay for bid packets and blueprints.
- The closing timestamp for the receipt of bids, as well as, the location and accessibility of the forms, are all provided. The kind of requirements, as well as, the anticipated end date is also included.
- It also includes the value of the Earnest Money Deposit (EMD) that must be made, as well as, the manner through which it should be paid. The value of the security deposit is to be provided by the victorious bidder. Date, time, and place of opening tenders is also covered.

Tender Documents

- The purpose of tender papers is to keep potential bidders aware of the basic and particular criteria that apply to tenderers.
- Typically, a letter of invitation to the tenderers; a specimen tender form; general instructions to the tenders; details of civil or structural work, along with a complete set of civil or structural drawings; details and specifications of machinery or equipment to be supplied, if any; a proposed contract agreement; arbitration authority who will make a decision in the event of a dispute; a time schedule for the completion of the work; and the amount of earnest money to be deposited are some of the components that constitute tender documents.

Pre-qualification of Tenderers

- Tenderers are selected in advance only for big and complicated projects. To guarantee that only eligible bidders submit proposals, pre-qualification is used. To assess whether a potential bidder can take part in the tenders, the information written next is gathered from them.
- Pre-qualification of tenderers analyses the money one has to complete the work, and also the machinery and equipment, skilled/semi-skilled/unskilled employees, etc., at one's disposal.
- Pre-qualification analyses the track of prospective tenderers leading similar initiatives. Only those tenderers or bidders, who are judged qualified, are allowed to submit tenders or bids.

Earnest Money Deposit

- Every bidder is required to provide this amount as deposit when one submits a bid. The EMD amount is typically between 1 and 3 per cent of the tender value. Those who did not win the contract get their EMD back once it is finalised. To prevent the chosen bidder from backing out of the deal, his EMD is held in reserve. If the winning bidder cancels or backs out of the contract, his/her EMD will not be returned to the bidder. Bids may be withdrawn at any time before acceptance. Once a particular bid has been selected and communicated, and the winning bidder withdraws his/her offer after transmission, he/she forfeits the EMD.
- The EMD is often payable as a pay order or demand draft from a nationalised or listed bank. When the EMD sum is substantial, EMD is also accepted as a bank guarantee from a nationalised or scheduled bank.

Security Deposit

The amount of the security deposit is due when the contract is signed by the winning bidder. In most cases, it amounts to around 15 per cent of the overall contract price. This is a security deposit to guarantee that the contractor (successful bidder) would finish the job as agreed upon in the contract. The contractor's security deposit shall be withheld in the event of non-compliance with any terms of the contract. Contractors often get their security deposits back within six months of a project's completion, following a period of inspection known as the 'defects liability period'.

Acceptance of Tenders or Bids

- On the designated day, tenders and EMDs will be unfurled from their sealed envelopes. If the EMD envelopes are found to be in order, then

the tender paper envelopes may be opened. Typically, bids are opened in the presence of the bidders.

- All qualified proposals will be reviewed and scored by a committee of industry professionals. Tender review committee members will include department heads (legal, finance, operations, buying, etc.), as well as, invited experts in the area. The following criteria must be met before selecting a bid.
  - How stable the tenderer's finances are
  - Whether the tenderer has the necessary technological resources
  - The bidder's track record of completing comparable projects in the past
  - General and detailed descriptions of the work being bid on, as well as, variations from those descriptions, if any
- In addition to the above, the drafted contract and its provisions must be reviewed and understood before a bid is accepted. To put it simply, the lowest bidder typically wins.
- Any bid that meets the above requirements and is not the lowest may be considered, if there are compelling reasons for doing so.

Letter of Intent

This letter is to notify the undersigned that their proposal has been selected. The Letter of Intent (LOI) is released before a contract is signed. The successful bidder must communicate any preconditions within the allotted timeframe. A contract must be signed once an LOI is issued by the project's promoters and accepted by the winning bidder. To finalise the business, the parties should accept the binding contract in addition to the LOI as it is not a legally binding document.

Implementation of the Deed of Contract

An agreement between parties might be verbal or written. However, if the law requires a written contract, registration, and/or attestation, then those requirements must be met. If these procedures are not followed, the contract will be unenforceable.

Clause of Sanction for Non-performance Regarding Guaranteed Parameters

The contract has a provision addressing liquidated damages, and it also includes a clause discussing penalties for failure to perform guarantee criteria. Both of these clauses may be found in the contract. If any of the requirements of the guarantee parameters are not satisfied, the provision states that the price of the contract will be decreased by a certain percentage, as mentioned in the clause. To provide sufficient compensation for the loss experienced as a result of non-performance, the percentage of the deduction that will be withheld has to be carefully assessed in close collaboration with the contractors and consultants. There should be no effect of the deduction percentage on either the customers or contractors.

Various Forms of Compensation for Contractors

There are different modes of payment to contractors

- First and final bill
- Running account bill

First and Final Bill

After the contractor has finished all work, there will be just one payment due to him/her. This type of remuneration is appropriate for relatively little and inconsequential labour.

Running Account Bill

- A running account is a bill that is issued regularly to collect payment for services rendered or goods delivered in the past. The contractor gets paid in stages as the work progresses, with the full payment due upon project completion. When the invoice is generated, it reflects the amount of work completed up to that point in time and is generated at the rates specified in the contract. Approximately 90 per cent of the payment is made, reserving 10 per cent. As a result, there will be several bills at various stages and a single final bill.
- An account is considered closed and a final amount is sent after all work has been completed.
- When measurements cannot be obtained for various reasons, but the work has advanced far enough, an advance payment may be granted to the running account.

Project Cost Estimate

Objectives of Cost Estimate

- To create an estimate of anticipated costs for individual activities on the project from beginning to end
- To create a budget for the project by aggregating the estimates for individual activities
- To obtain funding for the project based on this budget
- To use this budget as a management and control tool (both underestimation and overestimation are bad for the project)

Methods of Cost Estimate

There are four methods to estimate durations and costs related to project activities. They are as follows.

- Analogous estimate (top–down estimate): Estimates are created by using the past experience(s).
- Bottom–up estimate: Individual items are estimated separately by creating a work breakdown structure (WBS) and then rolled up for a project total done on projects that are new.
- Parametric estimate: Parameters are established for creating an estimate. These estimates are also created on the basis of past experience(s).
- Computerised/Monte Carlo estimate: Software is used to estimate individual items and then rolled up for a project total.

Top–Down and Bottom–Up Estimates

- Estimates at a high level are frequently referred to as top–down estimates, whereas estimates at a detailed level are referred to as bottom–up estimates. The British standard method permits a project manager to estimate in various degrees of detail. The estimate may be at a higher level (top–down) for the entire project, or it may be more precise (bottom–up) at the work package level. At the lowest level, costs can be further broken down into many cost categories.
- Using standard per-square-metre rates, it is possible to quickly and easily calculate a rough estimate for a housing construction project. A bottom–up estimate for the same project would include all individual contracts, including those for the builder, plumber, electrician, materials, plant rental, etc. The bottom–up estimate should be more precise, but it will take longer to construct and entail more estimating costs.

Types of Cost Estimates

Conceptual Cost Estimate

- It is also known as a rough, preliminary, feasibility, or analogy estimate. It is, generally, considered to be the basis for all cost estimates.
- Usually, it is done by the owner. It is based on cost per unit item. It is done when the construction details available are less than 30 per cent and may have an error of 10–20 per cent.
- There are three methods for a conceptual cost estimate.
  - Unit cost method
  - Factor method
  - Simulation method
- Unit cost method: Unit cost is, generally, prepared using previous project data. Hence, an estimator must have the cost per unit for the previously same type of completed project. It is also called weighted unit cost.

$$\text{Unit cost} = \frac{A + 4B + C}{6}$$

Where, A is the minimum unit cost of the previous project, B is the average unit cost of the previous project, and C is the maximum unit cost of the previous project.

- The following adjustments are needed in unit cost method.
  - Adjustment of time: The adjustment between two projects represents the relative inflation and deflation of cost, material rate, and interest rate.

$$\text{Adjustment cost} = \left(\frac{\text{Cost index for the current year}}{\text{Cost index for the year of a sample project}}\right) \times \text{cost of the previous project}$$

- Adjustment of location: Tender prices vary according to the region of the country. Similarly, the use of cost information from a previous project to forecast the cost of a proposed project will not be reliable unless an adjustment is made proportional to represent the difference in cost between the locations of two projects.

  The construction cost of the project in city A

  $$= \left(\frac{\text{Cost index for city A}}{\text{Cost index for previous project city}} \right) \times \text{cost of the previous project}$$

- Adjustment of size: Adjustment should also be made for the difference in the size of the two projects. The adjustment is a simple ratio of the size of the proposed project to the size of a similar project for which cost data are available.
- Combined adjustment:

Proposed cost = Previous cost × time adjustment × location adjustment × size adjustment

Semi-detailed Cost Estimate

Usually, a semi-detailed cost estimate is done by an engineer to prepare tender notice. It is suitable when 30–80 per cent of construction documents are available. It may have a 5–10 per cent error.

Detailed Cost Estimate

- Here, the quantity of an item is multiplied by the cost of that item. It is suitable when 80–100 per cent drawings and specifications are available. It may have a 2–4 per cent error. It is the most accurate approach.
- Individual, or separate, or short and long wall, and centre line methods are the methods of detailed cost estimates.

Project Sickness

- An industrial firm (being a business registered for not less than five years) with cumulative losses after a financial year equivalent to or surpassing its whole network is defined as 'sick' by the Sick Industrial Companies (Special provisions) Act, 1985, as modified in 1993.
- All reserves credited out of the profits and share premium account, excluding reserves out of re-evaluation of assets, write back of depreciation provisions, and reserves created out of amalgamations, are considered 'free reserves' to calculate the 'net worth' (which is the sum of paid-up capital and free reserves) under Section 3(1) (ga) of the Amendment Act, 1993.
- Companies owned by the state or federal government (51 per cent or more) are exempted from the Act's requirements. Additional exceptions are made for ancillary and small-scale industrial units.
- Within 60 days of finalising the audited accounts for the financial year at the end of which the firm has become ill, its board of directors is obligated by law to disclose the sickness to the Board for Industrial and Financial Reconstruction (BIFR). According to BIFR's guidelines, this report must follow a certain structure. A corporation is required by law to notify any firm's sickness to the BIFR. The firm's health may be reported by anybody with a stake in the matter to the Business Insolvency and Financial Restructuring Board.
- The Sick Industrial (Special provisions) Act, 1985, mandates that before a case may be submitted to the BIFR, the BIFR registrar must ensure that all facts related to the case comply with the Act. In such an instance, the BIFR will schedule a hearing and notify the parties involved. The rehabilitation of a sick unit requires the cooperation of several different organisations working in tandem. By working together, the BIFR can achieve its goals. Representatives of the ill firm providing the information, as well as, representatives of commercial banks, central or state governments, trade unions, etc., are all invited to attend the hearing and investigation undertaken by the BIFR under Section 16 of the Act.
- Once a firm has been identified as ailing, the BIFR may provide it with the opportunity to strengthen its infrastructure and resume

operations without external funding. If the BIFR concludes that the firm cannot become financially viable without significant external investment or within an acceptable timeframe, it will order the operating agency to put together an appropriate revival package to bring the company back from the brink of bankruptcy.

- Additional financial assistance or reliefs, concessions, sacrifices, etc., (e.g., waiving of penal or compound interest charged, waiving of interest in part or full, and a waiver from sales tax) are included in the revival package as finalised by the BIFR. These will be circulated to the concerned agencies for their consent to be received within 60 days. Once all participating agencies in the revitalisation programme have given their stamp of approval, the revitalisation scheme's suggestions will be legally obligatory for all parties.

Causes of Sickness

- Prevention is better than cure. This stresses the need for researching the potential causes of occupational illness to implement preventive measures. Similar to how people might be born with sickness or acquire some during their lifetime, industries too may face setbacks during the execution phase or acquire illness through their existence. Like humans may get ill in one of two ways, either by being born with a certain disease or by acquiring one during development, an industry can have difficulties during the implementation phase or develop illness over its lifetime.
- Internal and external causes lead to sickness. Causes that originate within a company, yet are ultimately within the purview of the management, are considered as internal. In-house illnesses may be prevented, if the management is perceptive enough to identify and address the underlying causes. Factors outside of an organisation over which the management has little influence or control are known as external causes. The goals and activities of the government, the failure of the monsoon, which has a negative impact on agriculture and allied sectors, the rise of powerful competitors, etc., are examples of external variables. Although illness might originate from either inside or beyond the organisation, the leadership may be able to reorganise the business, develop effective initiatives, and address external factors that are having an outsized impact on productivity and morale of those involved.

White Elephants

- The term 'white elephant' is used to describe an item whose upkeep is excessive in relation to its worth. When discussing investments, the term is used to describe an asset, property, or company that has such high operating and maintenance costs that making a profit is next to impossible. White elephants are often ill liquid assets, indicating that they cannot be quickly or readily traded for anything else without incurring a heavy loss to the seller.
- Project overruns have become a severe economic concern in India, as shown by the quarterly reports of the Ministry of Project Implementation, Government of India. Such massive overruns not only erode financial outlays on development projects but also thwart progress in industrial and infrastructural development.
- In the context of reducing the cost and time overruns, the Ministry of Project Implementation and the Committee on Public Undertakings (COPU) have suggested several measures. Some of the reasons for cost overruns are as follows.
  - Delay in supply
  - Delay in decision-making
  - Political interference
  - Complex procedure
  - Administrative reasons and a no objection certificate (NOC) required from various departments
- As the adage goes, 'a problem well-formulated is half solved', so the project team's delays can also be eliminated, if the root reasons are identified. Overruns in both time and money are often the result of poor planning, organisation, and monitoring; therefore, it is crucial to do the former and utilise scientific

techniques to forecast the latter's needs. Cost overruns may be kept to minimum by selecting the right project manager, organisation, or team, and delegating power proportionally to responsibilities. The timely completion of a project will be ensured by a suitable information management system for project management and effective coordination with all parties involved.

- Following is a list of the other factors that contribute to finishing the project on time.
  - Extensive preparation, a solid plan for tracking progress, and the allocation of resources based on an estimate of how far the project should be are the hallmarks of effective planning and implementation strategy.
  - Incorporating risk-free practises into the contract-drafting process is important. It is critical for the project team to consider financial incentives in order to choose a workable technological solution.
- For effective deployment, it is important to list technical parameters and designs, encourage community involvement in planning and execution, and practise decentralised decision-making.
  - The project manager should stay in place at least until the plant's commissioning phase. Workers and their managers must get requisite training.
  - Foreseeing oversights and errors and readying the organisation to deal with the impending disaster is also important. Also, errors in management should be kept to a minimum.
  - Maintaining contact with suppliers and sub-contractors to find out where materials are in the construction process at any given moment is crucial. Incorrect communication may lead to missed transportation constraints.
  - The project team's creative mindset and expertise, together with a solid project management database, are also an important parameter.
  - Conservation of natural resources and prevention of pollution are essential goals. The objectives of the project should be defined precisely. It is important to clearly describe the team and its sub-units.
  - Task and milestone charts, audit reports, and minutes from meetings between contractors and government entities may all be easily understood with the use of transparent financial cost estimates.

Cost Overrun

To put it another way, a cost overrun is when actual costs for a project end up being more than the initial projections. Time overruns (project delays) almost always lead to budget overruns for the reasons listed as follows.

- The impact of inflation on the cost of project components may rise, if completion is delayed.
- When implementation is postponed for longer than anticipated, it implies workers will be needed at a higher total cost of labour.

Credit Rating

- Credit rating is an evaluation of the financial trustworthiness of a company or an individual, particularly concerning meeting some obligation.
- There are three credit rating agencies in India. They are as follows.
  - Credit Rating and Information Services of India Ltd (CRISIL)
  - Investment Information and Credit Rating Agency of India Ltd (ICRA)
  - Credit Analysis and Research Ltd (CARE)

Capital Cost

It is the investment required for acquiring the asset. Thus, if the cost of a machine is ₹ 15 lakh, its capital cost is ₹ 15 lakh. Suppose, the machine is used for 15 years, this means that the capital cost of ₹ 15 lakh is spent during these 15 years.

Uniform Annual Equivalent of Capital Cost

- When acquiring a machine with a physical life of, say, 5 years, it should be determined whether the machine can be operated profitably during its whole physical life. The capital cost of the machine is depleted in 5

years if it is used until the end of its physical lifespan. As the capital cost is spent over 5 years, the capital cost's Uniform Annual Equivalent (UAE) can be used to facilitate calculations. Using the following relationship, the UAE of the capital cost is calculated after 5 years of machine use. (This connection links the cost of capital to the UAE's present value.)

Capital cost

$$= \frac{x}{(1+r)} + \frac{x}{(1+r)^2} + \frac{x}{(1+r)^3} + \ldots + \frac{x}{(1+r)^5}$$

- If the machine is used only for 3 years and then replaced by a new machine, the UAE of capital cost can be arrived at by equating the capital cost to the present value of its UAE.

i.e., $$\frac{x}{(1+r)} + \frac{x}{(1+r)^2} + \frac{x}{(1+r)^3}$$

- So, as the period for which the asset is put into use increases, the UAE of capital cost will decrease and vice-versa.

Operating and Maintenance Cost

Operating and maintenance cost is the cost required to repair, maintain, and operate an asset. As the life of an asset increases, its repair and maintenance requirement also increases. Hence, the operating and maintenance costs to be incurred every year increase with the age of the asset.

Depreciation of Assets

- In accounting or accountancy, the idea of depreciation is used to describe the gradual decline in the value of a company's asset over time. This is accomplished by allocating the asset's or item's cost over its duration of usage. In addition to recognising asset-related costs in the balance sheet of the firm, one of the primary motivations for employing this idea is to minimise the company's taxable income or net income. This is accomplished by deducting the cost of assets used during a specific time period from a company's total during that duration. This expense is known as depreciated cost, and the asset is known as a depreciable asset. When an asset is placed into operation for business purposes, depreciation expense is incurred.
- Calculation of depreciation expense or value of an asset requires four inputs.
  - The purchase value of the asset
  - Salvage value (remaining or residual value) of the asset at the end of its useful life
  - The estimated useful life of the asset
  - The method used to calculate depreciation

Types of Depreciation

Depreciation is commonly classified into three types.

- Physical depreciation
- Functional depreciation
- Contingent depreciation

Physical Depreciation

When an asset deteriorates in a way that compromises its physical condition, that process is known as physical depreciation. Because of this, the asset cannot perform as well as it could. Wear and tear from repeated uses (abrasion, shocks, vibration, impact, etc.) and degradation from exposure to the environment (pipe corrosion, chemical decomposition, etc.) are the primary causes of physical wear and tear.

Functional Depreciation

'Obsolescence' is shorthand for 'functional depreciation.' It is defined as decline in the value of an asset as a result of factors, such as shift in taste or style, inability to keep up with the rising demand, or need for a replacement because of the advent of a more efficient and cost-effective alternative.

Contingent Depreciation

It is due to accidents (due to negligence), disease (pollution of water, parasites), and diminution of supply (natural gas, electricity water, etc.).

Depreciation	Obsolescence
• This is the physical loss in the value of a property due to wear and tear, and decay. • It depends on its original condition, quality of maintenance, and mode of use. • It varies with age. • There are several methods for calculating the amount of depreciation.	• This is the functional loss in the value of a property due to changes in design, structure, fashion, utility, demand, etc. • It depends on technological advancement, art, etc. • It is not dependent on age. • No method is available for the calculation of obsolescence.

Salvage Value (Resale Value)

When a piece of property has served its purpose and cannot be used again, its residual value is calculated. It is safe to assume that if a thing has salvage value, it serves some purpose beyond its current state. Salvage value increases when a building has been properly cared for.

Scrap Value

When an item is no longer usable for any purpose other than as scrap, its market value is determined by its scrap value. The item is thought to have little practical use.

Book Value

Book worth is the value of an asset as recorded in the books on a given year. This is calculated as the original cost less the cumulative depreciation up to that point.

Methods for Calculating Depreciation

- There are several methods of calculating depreciation. Some of them are simple and used regularly, whereas some are relatively uncommon. The following notations have been used.
- C_i = Initial cost of an asset at zero time (this will include the cost of asset + transportation charge + installation and other charges spent initially)

  C_s = Salvage value (or scrape value) to be estimated at the end of the utility period

  n = Life of the asset

  B_m = Book value at the end of the period 'm'

Straight Line Method

In this method, the property is assumed to lose value by a constant amount every year. At the end of the life, the salvage value (or scrap value) is left.

$$D_m = \frac{C_i - C_s}{n}$$

$$D_m = D_1 = D_2 = D_3 = D_n$$

$$B_m = C_1 - \left(\frac{C_i - C_s}{n}\right)$$

Declining Balance Method (Constant Percentage Method)

In this method, the property is assumed to lose value annually at a constant percentage of its book value. Let FDB be the fixed percentage.

D_1 = depreciation for first year = $C_i \times$ FBD

B_1 = Book value at the end of first year

$= C_i - C_i \times \text{FBD} = C_i (1 - \text{FBD})$

D_2 = Depreciation for second year = Book value at beginning of second year × FBD

$= C_i (1 - \text{FBD})\ \text{FBD}$

(Book value at beginning of second = Book value at the end of first year) depreciation for the second year.

$B_2 = C_i (1 - \text{FBD}) - C_i (1 - \text{FBD})\ \text{FBD}$

$= C_i (1 - \text{FBD})^2$

Similarly, $B_3 = C_i (1 - \text{FBD})^3$

Book value at the end of n years = $C_i (1 - \text{FBD})^n$

$C_s = C_i (1 - \text{FBD})^n$

$$\text{FBD} = 1 - \left(\frac{C_s}{C_i}\right)^{1/n}$$

Double Declining Balance Method

In this method also, the property is assumed to lose value annually by a fixed factor of the book value.

FDDB = Fixed factor for double-declining balance method

FDDB is taken as double the straight line.

$$\text{Rate, i.e., FDDB} = \frac{2}{n}$$

The process of calculation of depreciation and book value for each year is the same as adopted in the declining balance method.

Sum of the Years Digit Method

- In common with the declining balance technique and the double declining balance method, this approach is an example of an accelerated form of depreciation method, in which it is assumed that an asset's value would decline over time.
- In this method, the digits corresponding to the number of each year of life are listed in reverse order. The sum of these digits is then determined. The depreciation factor for any year is the reverse digit for the year divided by the sum of the digits, and depreciation for a year is the product of the depreciation factor of that year and the amount ($C_i - C_s$). The general expression for the annual depreciation for any year (m) when the life is *n* years is expressed as

$$D_m = (C_i - C_s)\left[\frac{n-m+1}{\frac{n(n+1)}{2}}\right]$$

Sinking Fund Method

- A sinking fund refers to the creation of a fund, wherein money is accumulated by setting aside a certain amount periodically in such a way that the fund, along with the interest accrued, gets accumulated so that there remains a pre-determined value at the end of a specified period of time.
- The sinking fund depreciation model assumes that the value of an asset decreases at an increasing rate. An equal amount (*D*) is assumed to be deposited into a sinking fund at the end of each year of the asset's life. The sinking fund is ordinarily compounded annually. At the end of the estimated life, the amount accumulated equals the total depreciation of the asset ($C_i - C_s$). It is to be clearly understood that the depreciation amount is the sum of two components. The first component is the amount deposited into the sinking fund and the second is the amount of interest earned from the accumulated value of the sinking fund at the beginning of the particular year.
- Let '*I*' be the rate of interest. The first component of depreciation, which is to be deposited equally at the end of each year.

$$= (C_i - C_s)\left[\frac{A}{F}, i, n\right]$$

$$D = (C_i - C_s)\left[\frac{i}{(1+i)^n - 1}\right]$$

Combining both the factors, the depreciation for *m*th year by this method is enumerated as follows.

$D_m = D[1 + i]^{m-1}$

$m = 1,2,3 \ldots$

$D_1 = D$

$D_2 = D(1 + i)$

$D_3 = D(1 + i)^2$

$D_m = \ldots$

Solved Examples

1. Anirudh requires, say, ₹10 lakh after nine years from now. He prepared to invest a fixed sum of money regularly every year. If the money so invested is expected to earn interest at the rate of, say, 12 per cent p.a., what will be the annual instalments that need to be invested? The annual instalment is called the 'sinking fund'. This can be arrived at as under.

Expected Maturity value at the end of 9 years = ₹ 100000

Interest rate = 12 per cent p.a.

Let the annual instalment be invested = A

Maturity value (MV) = $A\left[\frac{\{(1+i)\}^n - 1}{i}\right]$

$$A = \frac{MV.i}{\left[(1+i)^n - 1\right]}$$

Substituting the given values, A

$$= \left[\frac{100000 \times 0.12}{(1+0.12)^9 - 1}\right]$$

$$A = \left[\frac{120000}{(2.7731-1)}\right]$$

Annual sinking fund to be invested = ₹ 67678

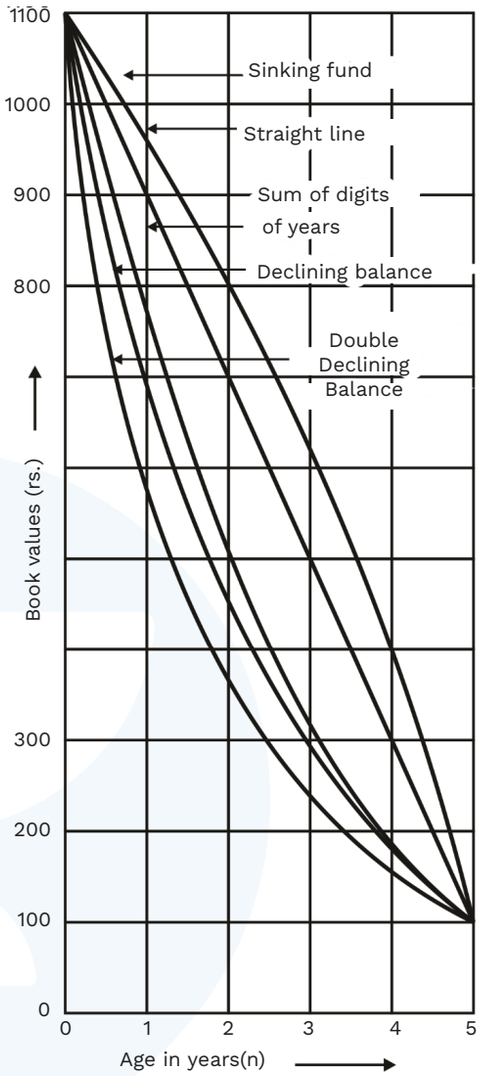


Fig. 7.1 Sinking Fund Method

Depletion

- The term 'Depletion' is used to describe the diminishing value of the natural resources that are consumed in the production of goods and services. The phrase is usually associated with the extraction of coal from a mine, timber from a forest, stone from a quarry, oil from a reservoir, etc. When these natural resources are sold or mined, the reserve and value of the property drop.
- However, there is a fundamental distinction between how amounts received through depletion and depreciation must be handled. When a piece of property has depreciated to its fullest, it is typically replaced by a similar piece of property. In case of mineral or other natural resource depletion, it is impossible to replenish them. Once gold or oil has been extracted from a mine or well, it cannot be restored. In the actual functioning of many natural resource businesses, the depletion fund may be used to acquire new properties, such as new mines, to ensure the enterprise's continuation.

Depletion for a year

$$= \left(\frac{\text{Cost of property}}{\text{Number of units in the property}} \right) \times \text{unit sold during the year}$$

Capital and Operating Costs on Projects

- Capital expenditure (or CAPEX) refers to the fixed, one-time expenses incurred to acquire assets for a business, which are often incurred to achieve future benefits. This is an investment in the organisation. Examples include purchasing real estate, equipment, and patent rights. Depreciation applies to capital expenditures.
- Operating expenses (or OPEX) are the funds a business spends daily to maintain its operations. Examples are salary, insurance, taxes, research and development, travel expenses, licence fees. Fixed expenses and variable costs are the two most common types of OPEX. Generally, OPEX includes the cost of leased equipment rather than purchased equipment.

UAE of Operating and Maintenance Cost

As the age of a machine increases, the annual operating and maintenance costs will also increase. The Uniform Annual Equivalent (UAE) of operating and maintenance costs can be arrived at as under.

Let, x_1 = Operating and maintenance cost for the first year

x_2 = Operating and maintenance cost for the second year and so on...

Present value of operating and maintenance cost

$$\frac{x_1}{(1+r)} + \frac{x_2}{(1+r)^2} + \frac{x_3}{(1+r)^3} + \ldots$$

Let UAE (operating and maintenance cost) = x

Present value of UAE of operating and maintenance cost

$$= \frac{x}{(1+r)} + \frac{x}{(1+r)^2} + \frac{x}{(1+r)^3} + \ldots$$

Enterprise Resource Planning

- Enterprise refers to an endeavour, in which several individuals cooperate to achieve a shared objective. Men, materials, money, time, and everything else needed to operate and used by a business are all examples of resources. The Enterprise Resource Planning (ERP) system is the software that facilitates the ERP.
- The ERP software allows for streamlined communication and data sharing across departments due to its unified architecture and central data repository.

ABC Analysis

- ABC analysis is based on the principle of vital few and trivial many. ABC analysis segregates the material items that are consumed in a production process/project into three groups: A-items, B-items, and C-items.
  - **A-items:** We could observe that about 70–75 per cent of the total money is spent on a few items of materials that account for only about 5–10 per cent of the total items of material consumed. Such materials that form part of only 5–10 per cent of the total items of material consumed, but account for a major chunk of the money spent are called A-items. Cement and steel are the A-items and the inventory of these items is to be closely monitored for controlling the cost of the project.
  - **B-items:** These goods typically account for 10–15 per cent of the overall number of items and 10–15 per cent of the total expenditure on the materials. As opposed to A-items, B-items do not require stringent oversight. B-items include steel rods, asbestos sheets, and other materials used in the construction of an industrial shed.
  - **C-items:** These are those items that normally form about 70–80 per cent of the total number of items but account for only about 5–10 per cent of the total money spent. In the construction of an

industrial shed, for example, sand, bricks, paint, wood, glass sheets, exhaust fans, binding wires, etc., will be C-items.

Value Analysis and Value Engineering

- The Society of American Value Engineering (SAVE) defines value analysis as the systematic application of recognised techniques, which identify the function of a product or service, establish a monetary value for that function and provide the necessary function at the lowest overall cost. Value engineering is also defined as the systematic, analytical approach of identifying unnecessary costs in construction. Unnecessary costs are those which do not provide any enhanced quality, appearance, utility, or improved life. It is estimated that by adopting value engineering techniques, savings up to 5–10 per cent on large construction projects can be achieved.
- For understanding value analysis and value engineering, we shall first define the term 'Value.'

  Value = Performance + Cost

  = What does a function do? + How much does it cost?
- Value analysis and value engineering are used interchangeably. Though both value analysis and value engineering attempt to identify unnecessary costs to eliminate them, there exists a subtle difference between the two. The difference between the two lies in the time and stage at which the techniques are applied.
- Value analysis applies to an existing project, whereas value engineering applies the same principles to a new project. The purpose of value analysis is to increase the worth of an existing product. It is a method that identifies and reduces the unnecessary costs already incurred in a product. Value engineering is utilised during the design phase to omit unnecessary product features. Value engineering is a preventive process. Value engineering, when carried out at the initial stage of a product, can reduce the cost of production substantially.
- Value analysis reduces the cost of a product. It helps employees to have a better understanding of the jobs that they do in the organisation.
- Employees become more cost conscious and give value-oriented ideas. It creates a conducive climate for innovation.
- Implementing changes becomes easier as employees become aware of the benefits of value analysis and do not offer any resistance to changes that are carried out for the benefit of the organisation.
- The following situations in an organisation indicate the immediate need for carrying out value analysis.
  - There is a continuous decline in the sales of the company's products.
  - The rate of return on investment shows a falling trend. Competitors offer their products at a relatively lower price to the customers. The company defaults in meeting delivery dates.
  - Competitors come up with new and improved designs for the products. Processing and tooling costs go up steeply.
- Value can be measured using value index. Value index (VI) is the ratio of function worth to function cost.

$$VI = \frac{FW}{FC} \leq 1 = \frac{\text{Aggregate value in given year}}{\text{Aggregate value in base year}}$$

  Where, FW is the function worth and FC is the function cost.
- VI can take a maximum value of 1.00. When the value of VI is less, it indicates that there is scope for value improvement.

Value Gap

- The difference between cost and worth indicates the scope for possible value improvement. When the worth is much below the cost, it indicates that there is a greater scope for value addition. The difference between function cost and function worth is known as value gap (VG).

  VG = Function cost – function worth
- A higher value of the value gap indicates a greater scope for value improvement.

Life Cycle Cost

- It is also known as the 'Cradle to Grave Cost'. It is the cost of an item from its acquisition till it is abandoned after the completion of its useful life period.
- The components of Life Cycle Cost (LCC) are cost of acquisition, cost of operation, cost of repairs and maintenance, cost of replacement of parts/components, and salvage value.

  LCC = Cost of [aquisition + operation + repair and maintenance + replacement of components] – salvage value
- In assessing the Value Index, if LCC is used

  LOC = (Present value of all cost incurred) – (present value of salvage value)

Future Value of a Series of Cash Flow (Future Value of Annuity)

- A series of payments of a fixed sum at regular intervals over a period of time is called annuity. We come across investments of this nature in recurring deposits, pension/ provident funds, life insurance policies, etc., wherein the investor makes investments of constant amounts at regular time intervals, over a period of time.
- The maturity value of Annuity (i.e., an annual series of investments) is given by the following relationship:

$$\text{Future value} = \text{Maturity value} = A\left[\frac{(1+i)^n - 1}{i}\right]$$

Where, A is the annuity amount (the constant amount invested every year).

I = Interest rate, and n = Number of years

Solved Examples

2. Calculate the compound value (i.e., maturity value) of ₹ 1000 invested every year for 10 years, at an interest rate of 12 per cent p.a.

Solution:

$$\text{Maturity value} = A\left[\frac{\{1+i\}^n - 1}{i}\right]$$

$$\text{Maturity value} = 1000\left[\frac{\{1+0.12\}^{10} - 1}{12}\right]$$

$$= 1000\left[\frac{3.1058 - 1}{0.12}\right]$$

$$= ₹\,17548$$

Present Value of Money (The Concept of Discounting)

We have seen that the future value of a given sum of money is calculated by the process of compounding. Similarly, the present value is calculated through the process of discounting. Discounting refers to the procedure of calculating the present value of future cash flows by applying suitable interest factor, thus, adjusting for the time value of money. The technique of discounting is the exact opposite of the technique of compounding.

Present Value of Future Money Receivable after '*n*' Years

- The maturity value of ₹ P at an interest rate of 'i per cent' per annum, compounded annually for a period of 'n' years is given by:

  Maturity value (MV) = $P(1+i)^n$
- When the future value (maturity value) is known, the present value can be found from the above formula. Accordingly,

$$P = \frac{MV}{(1+i)^n}$$

Solved Examples

3. An investor wants to get a sum of ₹ 400000 after 5 years. The prevailing rate of interest for deposits is 11 per cent p.a. and the interest rate is expected to remain stable. Interest is compounded annually. How much money should the investor invest now?

Solution:

$$P = \frac{MV}{(1+i)^n}$$

Substituting the given data, $P = \frac{400000}{(1+0.11)^5}$

$$P = 400000 \left[\frac{1}{(1+0.11)^5}\right]$$

$= 400000 \times 0.5935$

$= ₹\ 237400$

Present Value of Perpetual Annuity (Perpetuity)

- When the Periodical payments are received at annual intervals perpetually, the present value of such future cash flows is called the present value of the perpetuity

Let A = Amount paid or received at the end of every year

i = Interest rate per annum

Present Value of perpetual annuity = A (1/i)

Solved Examples

4. If a sum of ₹ 1500 is receivable every year perpetually, find out the present value of the perpetual cash flow. Assume a discount rate of 16 per cent p.a.

Solution:

Present value of perpetual annuity = $A \times \left(\frac{1}{i}\right)$

$= 1500 \times \left(\frac{1}{0.16}\right) = ₹\ 9375$

Practice Exercise

1. The principal function of a warranty is to _____.
(A) assure the level of quality to be provided
(B) provide a way to assert claims for late payment
(C) provide a way to allow additional time following acceptance to correct deficiencies, without additional costs
(D) ensure that the goods purchased fit the purposes for which they are to be used

2. A contract is an agreement between _____.
(A) two parties valid in law
(B) several agencies
(C) three agencies
(D) two parties without legal binding

3. A lump sum contract is _____.
(A) commonly followed by many agencies
(B) followed only by government departments
(C) followed only in private contracts
(D) followed only by local bodies

4. Unit price contract means _____.
(A) item-wise contract
(B) K-2 form of contract
(C) same as lump sum contract
(D) Both (A) and (B)

5. A turn-key contract means _____.
(A) to design and build
(B) to build the structure only
(C) to perform all functions from inception to completion of the construction
(D) the same as the unit price contract

6. A ripple effect in the construction of a facility caused due to _____.
(A) currency fluctuation
(B) ambiguity in specification and drawings
(C) the client forcing the contractor to accelerate a portion of the whole project
(D) unusual site conditions

7. EMD is _____.
(A) depositing 2 per cent of the contract value
(B) same as security deposit
(C) depositing 10 per cent of the contract value
(D) depositing only 5 per cent of the contract value

8. The security deposit is _____.
(A) refunded in the middle of a contract
(B) refunded after the maintenance period
(C) not refunded
(D) refunded as soon as the construction is over

9. An arbitrator mediates between _____.
(A) two aggrieved parties
(B) public and workers
(C) engineer and architect
(D) public and the contractor

10. A routine audit of a cost-reimbursable (CR) contract determines that over changes are being made. If the contract does not specify corrective actions, the buyer should _____.
(A) continue to make project payments
(B) halt the payment until the problem is resolved
(C) void the contract and take legal action to recover the payments
(D) change the contract to require more frequent audits

11. A project is engaged in making electronic devices. It is necessary to purchase some materials to make the printed circuit boards, all parts are commonly available from several vendors. The most likely contract that should be issued for these parts is _____ contract.
(A) unit prices
(B) firm fixed price
(C) cost-reimbursable
(D) award fee

12. With a clear contract statement of work, a seller completes work as specified, but the buyer is not pleased with the results. In such a case, the contract is considered to be _____.
(A) null and void
(B) incomplete
(C) complete
(D) waived

Answer Key

1. (A) **2.** (A) **3.** (A) **4.** (D) **5.** (C) **6.** (C) **7.** (A) **8.** (B) **9.** (A) **10.** (A)
11. (B) **12.** (C)